

$$Ax = b$$

Grass

Solutions to Linear Algebra by FIS

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Determinants

4.3 Properties of Determinants

4.3.1 Exercises

Note. How do we understand the adjugate matrix and the corresponding theorem, that

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)?$$

[Explanation by Eric Tao](#) .

Let A be an $n \times n$ matrix. The adjugate matrix $\operatorname{adj}(A)$ is the matrix that ‘does cofactor expansion on A ’. Notice that the cofactor expansion along any i th row is a linear combination of $\{C_{ij} \mid 1 \leq j \leq n\}$. So, by defining $[\operatorname{adj}(A)]_{ij} = C_{ji}$, we have that

$$A \operatorname{adj}(A) = \begin{pmatrix} \det(A) & ? & ? & \dots & ? \\ ? & \det(A) & ? & \dots & ? \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ ? & ? & ? & \dots & \det(A) \end{pmatrix}.$$

By the exercise below, all our ‘?’s are just zero. Naturally, it follows that

$$A \operatorname{adj}(A) = \det(A)I.$$

Exercise. Let A be an $n \times n$ matrix that has nonzero determinant. Explain why, for $i \neq j$,

$$A_{j1}C_{i1} + A_{j2}C_{i2} + \dots + A_{jn}C_{jn} = 0$$

Proof. This is the cofactor expansion along the i th row of the matrix $\mathcal{A}$, obtained from A , by replacing its i th row with its j th row. Since two rows are now equal, the above cofactor expansion, i.e. $\det(\mathcal{A})$, is zero. 

Question from H&K a classmate asked me about.

Exercise. The result of Example 16 suggests that perhaps the matrix

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n+1} \\ \vdots & \vdots & & \vdots \\ \frac{1}{n} & \frac{1}{n+1} & \cdots & \frac{1}{2n+1} \end{bmatrix}$$

is invertible and A^{-1} has integer entries. Can you prove that?

Exercise 4.3.25. Let c_{jk} denote the cofactor of the row j , column k entry of the matrix $A \in M_{n \times n}(\mathbb{F})$.

- (a) Prove that if B is the matrix obtained from A by replacing column k by e_j , then $\det(B) = c_{jk}$.
- (b) Show that for $1 \leq j \leq n$, we have

$$A \begin{pmatrix} c_{j1} \\ c_{j2} \\ \vdots \\ c_{jn} \end{pmatrix} = \det(A) \cdot e_j.$$

Hint: Apply Cramer's rule to $Ax = e_j$.

- (c) Deduce that if C is the $n \times n$ matrix such that $C_{ij} = c_{ji}$, then

$$AC = [\det(A)]I.$$

- (d) Show that if $\det(A) \neq 0$, then $A^{-1} = [\det(A)]^{-1}C$.

Proof.

- (a) The cofactor expansion of B along the k th column is just

$$(-1)^{j+k} \det(\tilde{B}_{jk}) = (-1)^{j+k} \det(\tilde{A}_{jk}) = c_{jk}.$$

- (b) Suppose $\det(A) \neq 0$. By Cramer's rule and (a), we have that the k th coordinate of x is

$$x_k = \frac{\det(B)}{\det(A)} = \frac{c_{jk}}{\det(A)}.$$

If $\det(A) = 0$, then note that the multiplication of the i th row of A with $(c_{j1} \ c_{j2} \ \dots \ c_{jn})^t$ is the determinant of the matrix A' whose i th and j th rows are both identical to the i th row of A . This must evaluate to zero.

Parts (c) and (d) follow easily from (b).

An alternate way to prove this result is presented in [Note 4.1](#) and the associated exercise.



Exercise 4.3.28. Let $y_1, y_2, \dots, y_n$ be linearly independent functions in C^∞ . For each $y \in C^\infty$, define $T(y) \in C^\infty$ by

$$[T(y)](t) = \det \begin{pmatrix} y(t) & y_1(t) & y_2(t) & \dots & y_n(t) \\ y'(t) & y_1'(t) & y_2'(t) & \dots & y_n'(t) \\ \vdots & \vdots & \vdots & & \vdots \\ y^{(n)}(t) & y_1^{(n)}(t) & y_2^{(n)}(t) & \dots & y_n^{(n)}(t) \end{pmatrix}.$$

The preceding determinant is called the **Wronskian** of $y, y_1, \dots, y_n$.

- (a) Prove that $T: C^\infty \rightarrow C^\infty$ is a linear transformation.
- (b) Prove that $N(T)$ contains $\text{span}(\{y_1, y_2, \dots, y_n\})$.

Proof.

- (a) Recall that $(ay + z)^{(k)}(t) = ay^{(k)}(t) + z^{(k)}(t)$ for any scalar $a \in \mathbb{C}$, and functions $y, z \in C^\infty$. Furthermore, the determinant is a n -linear function. These facts suffice to show $T(ay + z) = aT(y) + T(z)$.
- (b) For any y_i , notice that $T(y_i)$ has two identical columns, and hence must be zero. In other words, $\{y_1, y_2, \dots, y_n\} \subseteq N(T)$.



Diagonalization

5.1 Eigenvalues and Eigenvectors

5.1.1 Theorems

Theorem 5.2. Let $A \in M_{n \times n}(\mathbb{F})$. Then, a scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof. If λ is an eigenvalue of A , then $Ax = \lambda x$ for some nonzero $x \in \mathbb{F}^n$. So, $(A - \lambda I_n)x = 0$. Hence, $\det(A - \lambda I_n) = 0$.

Conversely, when $\det(A - \lambda I_n) = 0$, there exists nonzero $x \in \mathbb{F}^n$ for which $(A - \lambda I_n)x = 0$. Since $Ax = \lambda x$, λ is indeed an eigenvalue. 

Lemma. Let B be an $n \times n$ matrix whose entries are all constants, except for the first $m \leq n$ diagonal entries, each of which is some degree 1 polynomial $c - t$. Then, $\det(B)$ is of degree m .

Proof. When $n = 1$, the result is trivial. So, assume it is also true of $n = k$, and consider $n = k + 1$. We do cofactor expansion on the first row. For each $\tilde{A}_{1j}$, we can rearrange the rows and columns to obtain a matrix such that all $m - 1$ or $m - 2$ entries of degree 1 are in the first $m - 1$ or $m - 2$ diagonal entries. By assumption, every $\det(\tilde{A}_{1j})$ must now be of degree $m - 1$ (e.g. for $j = 1$) or $m - 2$. Hence, $\det(A)$ is of degree m . 

Theorem 5.3 (Exercise 24). Let $A \in M_{n \times n}(\mathbb{F})$.

- (a) The characteristic polynomial of A is a polynomial of degree n with leading coefficient $(-1)^n$.
- (b) A has at most n distinct eigenvalues.

Proof.

- (a) The case of $n = 1$ is again apparent. Therefore, we suppose this is true for $n = k$, and consider $n = k + 1$. As such, $\tilde{A}_{11}$ is a degree $n - 1$ polynomial with leading coefficient $(-1)^{n-1}$. Furthermore the preceding lemma says $\det(\tilde{A}_{1j})$ is of degree $n - 2$, for all $j \neq 1$. Hence, cofactor expansion on the first row shows that $\det(A)$ has degree n and leading coefficient $(-1)^n$.
- (b) This is now clear, from theorem 5.2 and the fact that degree n polynomials have at most n roots.



Question. Why *should* any $m \times m$ matrix A have at most m distinct eigenvalues? Is it due to the following conjecture?

Eigenvectors corresponding to different eigenvalues are linearly independent.

Proof. Let $\{\lambda_\alpha\}$ be the set of all distinct eigenvalues of A . For each α , let v_α be an eigenvector corresponding to λ_α . When $n = 1$, it is clear that $\{v_1\}$ is linearly independent. Now assume $\{v_1, v_2, \dots, v_n\}$ is linearly independent. Then, suppose for contradiction that

$$v_{n+1} = \sum_{i=1}^n a_i v_i.$$

Then,

$$\sum_{i=1}^n a_i (\lambda_{n+1} - \lambda_i) v_i = 0.$$

By assumption, $\lambda_{n+1} = \lambda_i$ for all $1 \leq i \leq n$. A contradiction; linear independence holds.

Therefore, $\{\lambda_\alpha\}$ contains at most m members because of the linear independence of eigenvectors corresponding to different eigenvalues.



Theorem 5.4. (Exercise 5.1.6) Let T be a linear operator on a vector space V , and let λ be an eigenvalue of T . A vector $v \in V$ is an eigenvector of T corresponding to λ if and only if $v \neq 0$ and $v \in N(T - \lambda I)$.

Proof. First assume $v \in V$ is an eigenvector (corresponding to λ). Then, $v \neq 0$ and

$$([T]_\beta - \lambda I)[v]_\beta = [(T - \lambda I)(v)]_\beta = 0.$$

Since the map $u \xrightarrow{\phi_\beta} [u]_\beta$ is an isomorphism, $(T - \lambda I)v = 0$.



Question. Let V be a vector space of dimension at least 2 and $\{v_1, v_2, \dots, v_n\}$ be a basis for V . Then, is the linear operator T defined by $T(v_i) = v_{i+1}$ (where $n + m := m$) an linear transformation without any eigenvectors?

Proof. No: observe that $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ is an eigenvector of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.



5.1.2 Exercises

Exercise 5.1.1. Label the following statements as true or false.

- (a) Every linear operator on an n -dimensional vector space has n distinct eigenvalues.
- (b) If a real matrix has one eigenvector, then it has an infinite number of eigenvectors.
- (c) There exists a square matrix with no eigenvectors.
- (d) Eigenvalues must be nonzero scalars
- (e) Any two eigenvectors are linearly independent.
- (f) The sum of two eigenvalues of a linear operator T is also an eigenvalue of T .
- (g) Linear operators on infinite-dimensional vectors spaces never have eigenvalues.
- (h) An $n \times n$ matrix A with entries from a field $\mathbb{F}$ is similar to a diagonal matrix if and only if there is a basis for $\mathbb{F}^n$ consisting of eigenvectors of A .
- (i) Similar matrices always have the same eigenvalues.
- (j) Similar matrices always have the same eigenvectors.
- (k) The sum of two eigenvectors of an operator T is always an eigenvector of T .

Proof.

- (a) False ✓; rotation in $\mathbb{R}^2$ has no eigenvalues as seen in page 256. An even simpler example is the zero transformation, which only has a single eigenvalue, namely zero.
- (b) True ✓. Let v be an eigenvector of A , and λ the corresponding eigenvalue. Then, for all $c \in \mathbb{R}$, $A(cv) = c\lambda v = \lambda(cv)$.
- (c) True ✓; rotation in $\mathbb{R}^2$ has no eigenvectors as seen in page 256.
- (d) False ✓, for consider the zero transformation $T_0: \mathbb{R} \rightarrow \mathbb{R}$. Then, 1 is an eigenvector with the corresponding eigenvalue 0, since $T(1) = 0 \cdot 1$.
- (e) False ✓
- (f) False ✓; consider the identity matrix.
- (g) False ✓. In fact, the zero transformation in any infinite dimensional vector space (e.g. $P(\mathbb{F})$) has the eigenvalue 0.
- (h) True ✓. Assume $A = Q^{-1}BQ$ for some diagonal matrix B and invertible matrix Q . Notice each e_i is an eigenvector of B corresponding to the eigenvalue B_{ii} . Hence, every $Q^{-1}e_i$ is an eigenvector of A (corresponding to the eigenvalue B_{ii}), for

$$A(Q^{-1}v) = (Q^{-1}BQ)(Q^{-1}v) = \lambda(Q^{-1}v).$$

Our answer to a previous question says $\{Q^{-1}e_i \mid 1 \leq i \leq n\}$ is a basis. Conversely, suppose there exists a basis $\beta = \{v_1, v_2, \dots, v_n\}$ consisting of eigenvectors of A . It follows that

$$A = Q^{-1}[A]_{\beta}Q,$$

for the diagonal matrix $[A]_\beta$ and the change of coordinate matrix Q that changes coordinates in the standard ordered basis to β -coordinates.

- (i) True ✓, see exercise 5.1.13.
 (j) False ✓, as shown in (h): Consider any matrices such that $A = Q^{-1}BQ$. So, v is an eigenvector of B iff $Q^{-1}v$ is an eigenvector of A .

A specific example. We have the similar matrices

$$\begin{pmatrix} 0 & -1 & -1 \\ 0 & 0 & -1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Notice that

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \text{but} \quad \begin{pmatrix} 0 & -1 & -1 \\ 0 & 0 & -1 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}.$$

- (k) False ✓; consider the operator T on $\mathbb{R}_2$ given by $T(e_i) = ie_i$. Then, $T(e_1 + e_2) = (1, 2)$ is not an eigenvector of T .



Exercise 5.1.3. For each of the following linear operators T on a vector space V and ordered bases β , compute $[T]_\beta$ and determine whether β is a basis consisting of eigenvectors of T .

- (c) $V = \mathbb{R}^3$, $T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 3a + 2b - 2c \\ -4a - 3b + 2c \\ -c \end{pmatrix}$, and $\beta = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\}$.
 (e) $V = P_3(\mathbb{R})$,

$$T(a + bx + cx^2 + dx^3) = -d + (-c + d)x + (a + b - 2c)x^2 + (-b + c - 3d)x^3,$$

$$\text{and } \beta = \{1, 1 + x^2, x + x^2, 1 - x + x^3\}.$$

Proof.

- (c) ✓ Notice that

$$T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = - \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad T \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \text{and} \quad T \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

Indeed, we see that β is a basis consisting of eigenvectors of T , since

$$[T]_\beta = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(e) ✗ Similarly, we see that

- $T(1) = -1 + (1 + x^2)$,
- $T(1 + x^2) = -(1 + x^2) + (1 - x + x^3)$,
- $T(x + x^2) = -(x + x^2)$,
- $T(1 - x + x^3) = (1 + x^2) - (x + x^2) - 2(1 - x + x^3)$.

Therefore, β is not a basis consisting of eigenvectors of T , as

$$[T]_{\beta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & -2 \end{pmatrix}.$$

Careless mistake: $T(1 - x + x^3) = -(1 - x + x^3)$. Other than that, the rest of the answer is fine!



Exercise 5.1.4. For each of the following matrices $A \in M_{n \times n}(\mathbb{F})$,

- (i) Determine all the eigenvalues of A .
 - (ii) For each eigenvalue λ of A , find the set of eigenvectors corresponding to λ .
 - (iii) If possible, find a basis for $\mathbb{F}^n$ consisting of eigenvectors of A .
 - (iv) If successful in finding such a basis, determine an invertible matrix Q and a diagonal matrix D such that $Q^{-1}AQ = D$.
- (a) $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ for $\mathbb{F} = \mathbb{R}$.

Proof.

- (a) ✓ We compute that the characteristic polynomial of A is $(t + 1)(t - 4)$. So, the eigenvalues of A are -1 and 4 , with corresponding eigenspaces

$$E_{-1} = \left\{ t \begin{pmatrix} 1 \\ -1 \end{pmatrix} \mid t \in \mathbb{R} \right\} \quad \text{and} \quad E_4 = \left\{ t \begin{pmatrix} 2 \\ 3 \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

Thus, a basis for $\mathbb{F}^n$ is clearly

$$\beta = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}.$$

Now, we have that

$$\frac{1}{5} \begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}.$$



Exercise 5.1.5. For each linear operator T on V , find the eigenvalues of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix.

- (i) $V = M_{2 \times 2}(\mathbb{R})$ and $T \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c & d \\ a & b \end{pmatrix}$.
- (j) $V = M_{2 \times 2}(\mathbb{R})$ and $T(A) = A^t + 2 \cdot \text{tr}(A) \cdot I_2$.

Proof.

- (i) ✓ Let the basis $\gamma := \{E^{11}, E^{21}, E^{12}, E^{22}\}$. So,

$$[T]_{\gamma} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Accordingly, we compute the characteristic polynomial to be $(t-1)^2(t+1)^2$.

i.e. the eigenvalues are 1 and -1 . Now, we notice that

1. $[T]_{\gamma} \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}^t = \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix}^t$,
2. $[T]_{\gamma} \begin{pmatrix} 0 & 0 & 1 & 1 \end{pmatrix}^t = \begin{pmatrix} 0 & 0 & 1 & 1 \end{pmatrix}^t$,
3. $[T]_{\gamma} \begin{pmatrix} 1 & -1 & 0 & 0 \end{pmatrix}^t = -\begin{pmatrix} 1 & -1 & 0 & 0 \end{pmatrix}^t$,
4. $[T]_{\gamma} \begin{pmatrix} 0 & 0 & 1 & -1 \end{pmatrix}^t = -\begin{pmatrix} 0 & 0 & 1 & -1 \end{pmatrix}^t$.

Therefore, we obtain the basis

$$\beta := \{E^{11} + E^{21}, E^{12} + E^{22}, E^{11} - E^{21}, E^{12} - E^{22}\},$$

such that

$$[T]_{\beta} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

is a diagonal matrix.

- (j) ✓ Similarly, we first notice

$$[T]_{\beta} = \begin{pmatrix} 3 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 3 \end{pmatrix}.$$

Again, we compute the characteristic polynomial. This time it is $(t-5)(t-1)^2(t+1)$. i.e. the eigenvalues are 5, 1, and -1 . Furthermore,

1. $[T]_{\beta} \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix}^t = 5 \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix}^t$,
2. $[T]_{\beta} \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix}^t = \begin{pmatrix} 1 & 0 & 0 & -1 \end{pmatrix}^t$,
3. $[T]_{\beta} \begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}^t = \begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}^t$,
4. $[T]_{\beta} \begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix}^t = -\begin{pmatrix} 0 & 1 & -1 & 0 \end{pmatrix}^t$.

As such, we have the basis

$$\alpha := \{E^{11} + E^{22}, E^{11} - E^{22}, E^{21} + E^{12}, E^{21} - E^{12}\}.$$

Indeed, the matrix representation of T in the basis α is diagonal. In fact,

$$[T]_{\alpha} = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$



Exercise 5.1.9.

- Prove that a linear operator T on a finite dimensional vector space is invertible if and only if zero is not an eigenvalue of T .
- Let T be an invertible linear operator. Prove that a scalar λ is an eigenvalue of T if and only if λ^{-1} is an eigenvalue of T^{-1} .
- State and prove results analogous to (a) and (b) for matrices.

Proof.

- Notice that T is invertible iff $\text{nullity}(T) = 0$ iff zero is not an eigenvalue of T .
- A scalar λ is an eigenvalue of T iff $T(u) = \lambda u$ for some vector u iff $T^{-1}(u) = \lambda^{-1}u$ iff λ^{-1} is an eigenvalue of T^{-1} .
- This translates easily for any matrix A by taking $T = L_A$.



Exercise 5.1.10. Prove that the eigenvalues of an upper triangular matrix M are the diagonal entries of M .

Proof. The characteristic polynomial of the $n \times n$ upper triangular matrix, M , is $(A_{11} - t)(A_{22} - t) \dots (A_{nn} - t)$. Its roots, i.e. the eigenvalues of M , are hence the diagonal entries of M .



Exercise 5.1.12. A scalar matrix is a square matrix of the form λI for some scalar λ ; that is, a scalar matrix is a diagonal matrix in which all the diagonal entries are equal.

- Prove that if a square matrix A is similar to a matrix λI , then $A = \lambda I$.
- Show that a diagonalizable matrix having only one eigenvalue is a scalar matrix.
- Prove that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not diagonalizable.

Proof. (a) Let A be similar to λI . Then for some invertible matrix Q ,

$$A = Q^{-1}(\lambda I)Q = \lambda I.$$

- (b) Suppose A is diagonalizable with only one eigenvalue, λ . Let the standard ordered basis be β . Then, $[L_A]_\gamma = \lambda I$ for some basis γ . Clearly, $[L_A]_\gamma$ is similar to $A = [L_A]_\beta$. Therefore, $A = \lambda I$ by (a).
- (c) Its characteristic polynomial has exactly one root, 1. So from (b), since it is not a scalar matrix, it can't be diagonalizable.



Exercise 5.1.13.

- (a) Prove that similar matrices have the same characteristic polynomial.
- (b) Show that the definition of the characteristic polynomial of a linear operator on a finite-dimensional vector space V is independent of the choice of basis for V .

Proof.

- (a) Let A and Q be $n \times n$ matrices, such that Q is invertible. Then,

$$\det(Q^{-1}AQ - tI_n) = \det(Q^{-1})\det(A - tI_n)\det(Q) = \det(A - tI_n).$$

Hence, the characteristic polynomials of $Q^{-1}AQ$ and A are identical.

- (b) Let $T: V \rightarrow V$ be a linear transformation, and Q be the change of coordinate matrix that changes β -coordinates into γ -coordinates, i.e. $[I_V]_\beta^\gamma$. Then, $[T]_\beta$ is similar to $[T]_\gamma$ because

$$[T]_\beta = Q^{-1}[T]_\gamma Q.$$

By (a), their characteristic polynomials are identical.



Exercise 5.1.15. For any square matrix A , prove that A and A^t have the same characteristic polynomial (and hence the same eigenvalues). Visit goo.gl/7Qss2u for a solution.

Proof. The characteristic polynomials of A and A^t are identical, since

$$\det(A - tI) = \det(A - tI)^t = \det(A^t - tI).$$



Exercise 5.1.16.

- (a) Let T be a linear operator on a vector space V , and let x be an eigenvector of T corresponding to the eigenvalue λ . For any positive integer m , prove

that x is an eigenvector of T^m corresponding to the eigenvalue λ^m .

- (b) State and prove the analogous result for matrices.

Proof.

- (a) Notice that $T^m(x) = \lambda T^{m-1}(x) = \cdots = \lambda^m T(x)$.
 (b) Let A be an $n \times n$ matrix with $Ax = \lambda x$. Then, for any positive integer m , we have $A^m x = \lambda A^{m-1} x = \cdots = \lambda^m x$.



Exercise 5.1.17. Let T be a linear operator on a finite-dimensional vector space V , and c be any scalar.

- (a) Determine the relationship between the eigenvectors of T (if any) and the eigenvalues and eigenvectors of $U = T - cI$.

Justify your answers.

- (b) Prove that T is diagonalizable if and only if U is diagonalizable.

Proof.

- (a) Notice that $U(x) = \lambda x$ iff $T(x) = (c + \lambda)x$. Hence, the eigenvectors of U and T are identical. But, the corresponding eigenvalues differ by the scalar c .
 (b) Follows immediately from (a) (using theorem 5.1).



Exercise 5.1.18. Let T be the linear operator on $M_{n \times n}(\mathbb{R})$ defined by $T(A) = A^t$.

- (a) Show that ± 1 are the only eigenvalues of T .
 (b) Describe the eigenvectors corresponding to each eigenvalue of T .
 (c) Find an ordered basis β for $M_{2 \times 2}(\mathbb{R})$ such that $[T]_\beta$ is a diagonal matrix.
 (d) Find an ordered basis β for $M_{n \times n}(\mathbb{R})$ such that $[T]_\beta$ is a diagonal matrix for $n > 2$.

Proof.

- (a) We notice that $T(A) = \lambda A$ iff $A^t = \lambda A$. Let $J \in M_{n \times n}(\mathbb{R})$ be zero everywhere except $J_{12} = -1$ and $J_{21} = -1$. It is clear that ± 1 are eigenvalues of T (when $n \geq 2$), since

$$T(I) = I \quad \text{and} \quad T(J) = -J.$$

Now consider an eigenvector A of T corresponding to the eigenvalue λ , that is nonzero in some (i, j) th entry. Then, $A = T^2(A) = \lambda^2 A$ implies $(1 - \lambda^2)A_{ij} = 0$. Indeed, λ must be ± 1 .

- (b) The eigenvectors corresponding to the eigenvalues ± 1 are symmetric and skew-symmetric, respectively.

(c) Consider

$$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}.$$

Notice that

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = 2 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and similarly,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

It is clear that β is a basis for V . Moreover, $[T]_\beta$ is the diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This is because

$$\begin{aligned} 1. \quad T \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & 2. \quad T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \\ 3. \quad T \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, & 4. \quad T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \end{aligned}$$

(d) Consider the $n \times n$ matrices A^{ij} and B^{ij} , which are zero everywhere except for $A_{ij}^{ij} = 1$ and $A_{ji}^{ij} = -1$; as well as $B_{ij}^{ij} = B_{ji}^{ij} = 1$. Then the set of all such matrices, namely

$$\beta = \{A^{ij} \mid 1 \leq i < j \leq n\} \cup \{B^{ij} \mid 1 \leq i \leq j \leq n\},$$

is a basis of purely eigenvectors for $M_{n \times n}(\mathbb{R})$.

Notice that $E^{ii} = B^{ii}$. Even if $i < j$, we see that $A^{ij} + B^{ij} = 2E^{ij}$ and $-A^{ij} + B^{ij} = 2E^{ji}$. Since β contains all E^{ij} and has n^2 members, it indeed is a basis for $M_{n \times n}(\mathbb{R})$.

Furthermore,

$$T(A^{ij}) = -A^{ij} \quad \text{and} \quad T(B^{ij}) = B^{ij}.$$

So β contains only eigenvectors, as claimed. i.e. $[T]_\beta$ is diagonal.



Exercise 5.1.19. Let $A, B \in M_{n \times n}(\mathbb{C})$.

- (a) Prove that if B is invertible, then there exists a scalar $c \in \mathbb{C}$ such that $A + cB$ is not invertible.

- (b) Find nonzero 2×2 matrices A and B such that both A and $A + cB$ are invertible for all $c \in \mathbb{C}$.

Proof.

- (a) By the fundamental theorem of algebra, there exists $-c \in \mathbb{C}$ for which $\det(AB^{-1} + cI) = 0$. Hence, $\det(A + cB) = 0$ as $\det(B) \neq 0$.
 (b) Let $A = B = I$. Then, $A^{-1} = I$ and $(A + cB)^{-1} = (c + 1)^{-1}I$.



Exercise 5.1.20. Let A be an $n \times n$ matrix with characteristic polynomial

$$f(t) = (-1)^n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0.$$

Prove that $f(0) = a_0 = \det(A)$. Deduce that A is invertible if and only if $a_0 \neq 0$.

Proof. Notice that $a_0 = f(0) = \det(A - 0 \cdot I) = \det(A)$ as claimed. So, A is invertible (iff $\det(A) \neq 0$) iff $a_0 \neq 0$.



Exercise 5.1.22.

- (a) Let T be a linear operator on a vector space V over the field $\mathbb{F}$, and let $g(t)$ be a polynomial with coefficients from $\mathbb{F}$. Prove that x is an eigenvector of T with corresponding eigenvalue λ , then $g(T)(x) = g(\lambda)x$. That is, x is an eigenvector of $g(T)$ with corresponding eigenvalue $g(\lambda)$.
 (b) State and prove a comparable result for matrices.
 (c) Verify (b) for the matrix A in exercise 5.1.4(a) with polynomial $g(t) = 2t^2 - t + 1$, eigenvector $x = (2, 3)^t$, and corresponding eigenvalue $\lambda = 4$.

Proof.

- (a) Let $g(t) = \sum_{i=0}^n a_i t^i$. Then, we see that

$$g(T)(x) = \sum_{i=0}^n a_i T^i(x) = \sum_{i=0}^n a_i \lambda^i x = g(\lambda)x.$$

- (b) Indeed, x is an eigenvector of A with corresponding eigenvalue $\lambda = 4$, because

$$Ax = \begin{pmatrix} 8 \\ 12 \end{pmatrix} = 4 \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Furthermore, (b) holds as expected, for

$$g(A) \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 14 & 10 \\ 15 & 19 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 29 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = g(4) \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$



Exercise 5.1.23. Use Exercise 5.1.22 to prove that if $f(t)$ is the characteristic polynomial of a diagonalizable linear operator T , then $f(T) = T_0$, the zero operator. (In section 5.4 we prove that this result does not depend on the diagonalizability of T .)

Proof. Let $f(t)$ be a characteristic polynomial of a diagonalizable linear operator T . Then, there are eigenvectors v_i with corresponding eigenvalues λ_i that form a basis $\beta := \{v_1, v_2, \dots, v_n\}$, for which $[T]_\beta$ is diagonal. Since $f(T)(v_i) = 0$ for each i , we have $f(T) = T_0$. 

Exercise 5.1.25. Determine the number of distinct characteristic polynomials of matrices in $M_{2 \times 2}(\mathbb{Z}_2)$.

Proof. $\times$ There are six distinct polynomials, since there are six diagonal entries. Let $e, f \in \mathbb{Z}_2$. We see that the characteristic polynomials of

$$(a) \begin{pmatrix} 0 & e \\ f & 0 \end{pmatrix}, \quad (b) \begin{pmatrix} 1 & e \\ f & 0 \end{pmatrix}, \quad (c) \begin{pmatrix} 1 & e \\ f & 1 \end{pmatrix},$$

are, respectively,

$$(a) t^2 \text{ or } t^2 - 1, \quad (b) t(t-1) \text{ or } t(t-1) - 1, \quad (c) (1-t)^2 \text{ or } (1-t)^2 - 1.$$

The above is wrong. We see that there are only four such polynomials, because $(1-t)^2 - 1 = t^2 - 1$ and $(1-t)^2 - 1 = t^2 - 2t = t^2$. 

5.2 Diagonalizability

5.2.1 Theorems

Theorem 5.6. The characteristic polynomial of any diagonalizable linear operator on a vector space V over a field $\mathbb{F}$ splits over $\mathbb{F}$.

Proof. Let T be a diagonalizable linear operator on an n -dimensional vector space V . Then, there is a basis β for which $A = [T]_\beta$ is diagonal. As such, the characteristic polynomial of T is just

$$(-1)^n(t - A_{11})(t - A_{22}) \dots (t - A_{nn}).$$



Lemma. For any A be an $n \times n$ matrix whose first $i \leq n$ diagonal entries are λ , we have that $(\lambda - t)^i$ is a factor of the characteristic polynomial of A .

Proof. This is trivial for $n = 1$. So suppose it is true for $n = k$ and consider $n = k + 1$. For each $(A - tI)_{1j}$ with $j > 1$, shift the rows below the j th row up by one (and the j th row to the n th row). Now, the first $i - 1$ diagonal entries are $\lambda - t$. As such, every $(\lambda - t)^{i+1}$ is a factor of each c_{1j} . Hence it is also a factor of

the characteristic polynomial of A .



Theorem 5.7. Let T be a linear operator on a finite-dimensional vector space V , and let λ be an eigenvalue of T having multiplicity m . Then, $1 \leq \dim(E_\lambda) \leq m$.

Proof. Let $n := \dim(V)$, $\beta := \{v_1, v_2, \dots, v_i\}$ be a basis for E_λ , and $\gamma := \{v_1, v_2, \dots, v_n\}$ be an extension of β to a basis for V . By the above lemma, $(\lambda - t)^i$ is a factor of characteristic polynomial of $[T]_\gamma$.



Theorem 5.8. Let T be a linear operator on a finite-dimensional vector space such that the characteristic polynomial splits. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be the distinct eigenvalues of T . Then,

- (a) T is diagonalizable if and only if the multiplicity of λ_i is equal to $\dim(E_{\lambda_i})$ for all i .
- (b) If T is diagonalizable and β_i is an ordered basis for E_{λ_i} for each i , then $\beta = \beta_1 \cup \beta_2 \cup \dots \cup \beta_k$ is an ordered basis for V consisting of eigenvectors of T .

Proof. Let m_b be the multiplicity of λ_b and assume T is diagonalizable. So $[T]_\beta$ is diagonal for some basis

$$\beta := \{v_{ab} \mid 1 \leq a \leq k, 1 \leq b \leq m_b\}.$$

Suppose that for some $T(x) = \lambda_i x$ and scalars c_{ab} , we have

$$x = \sum_{a,b} c_{ab} v_{ab}.$$

By theorem 5.5, $x = \sum_b c_{ib} v_{ib}$. That is, $\{v_{ib} \mid 1 \leq b \leq m_i\}$ spans E_{λ_i} . By theorem 5.7, $m_i = \dim(E_{\lambda_i})$.

Conversely, consider when $\dim(E_{\lambda_b}) = m_b$ for all b . Since the characteristic polynomial of T splits, $\sum m_b = \dim(V)$. Furthermore, there is a basis β_b for each E_{λ_b} . Thus by theorem 5.5, β must be a basis for V for which $[T]_\beta$ is diagonal.



Theorem 5.9. Let $W_1, W_2, \dots, W_k$ be subspaces of a finite-dimensional vector space V . The following conditions are equivalent.

- (a) $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$.
- (b) $V = \sum_{i=1}^k W_i$ and, for any vectors $v_1, v_2, \dots, v_k$ such that $v_i \in W_i$ ($1 \leq i \leq k$), if $v_1 + v_2 + \dots + v_k = 0$, then $v_i = 0$ for all i .
- (c) Each vector $v \in V$ can be uniquely written as $v = v_1 + v_2 + \dots + v_k$, where $v_i \in W_i$.
- (d) If γ_i is an ordered basis for W_i ($1 \leq i \leq k$), then $\gamma_1 \cup \gamma_2 \cup \dots \cup \gamma_k$ is an ordered basis for V .
- (e) For each $i = 1, 2, \dots, k$, there exists an ordered basis γ_i for W_i such that $\gamma_1 \cup \gamma_2 \cup \dots \cup \gamma_k$ is an ordered basis for V .

Proof.

- Assume (b) does not hold, i.e. there exists k nonzero vectors $v_i \in W_i$ for which $v_1 + v_2 + \cdots + v_k = 0$. Then, $v_1 \in W_2 + W_3 + \cdots + W_k$. Thus (a) is false.
- Now suppose (b) holds and

$$v_1 + v_2 + \cdots + v_k = u_1 + u_2 + \cdots + v_k.$$

for some $v_i, u_i \in W_i$. Therefore, $u_i - v_i = 0$ for all i . Accordingly, (c) holds.

- Consider when (c) holds and let γ_i be an ordered basis for W_i . To avoid a contradiction with the uniqueness asserted by (c), linear independence of γ must hold. Clearly, (d) is true.
- Presume (d) is valid. It is a straightforward task to find ordered bases for W_i . So, (e) follows from (d).

(Either use the result that every vectors space has a basis. Or, if we desire a Choiceless proof, for each i we pick $u_j \in W_i - \text{span}\{u_1, u_2, \dots, u_{j-1}\}$. This procedure must terminate at $j = n := \dim(W_i)$. Then, we take $\gamma_i = \{u_1, u_2, \dots, u_n\}$.)

- Finally, when (e) is true, (a) follows trivially.



Theorem 5.10. A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if V is the direct sum of the eigenspaces of T

Proof. If T is diagonalizable, theorem 5.8 tells us the sum of all eigenspaces is V . Moreover, theorem 5.5 guarantees condition (b) of the preceding theorem.

Conversely, consider when V is the direct sum of the eigenspaces of T . Hence the sum of the dimensions of all eigenspaces must be $\dim(V)$. From theorem 5.7, we deduce that the dimension of each eigenspace is identical to the multiplicity of the corresponding eigenvalue.



5.2.2 Exercises

Exercise 5.2.1. Label the following statements as true or false.

- Any linear operator on an n -dimensional vector space that has fewer than n distinct eigenvalues is not diagonalizable.
- Two distinct eigenvectors corresponding to the same eigenvalue are always linearly dependent.
- If λ is an eigenvalue of a linear operator T , then each vector in E_λ is an eigenvector of T .
- If λ_1 and λ_2 are distinct eigenvalues of linear operator T , then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.
- Let $A \in M_{n \times n}(\mathbb{F})$ and $\beta = \{v_1, v_2, \dots, v_n\}$ be an ordered basis for $\mathbb{F}^n$

consisting of eigenvectors of A . If Q is the $n \times n$ matrix whose j th column is v_j ($1 \leq j \leq n$), then $Q^{-1}AQ$ is a diagonal matrix.

- (f) A linear operator T on a finite-dimensional vector space is diagonalizable if and only if the multiplicity of each eigenvalue λ equals the dimension of E_λ .
- (g) Every diagonalizable linear operator on a nonzero vector space has at least one eigenvalue.

The following two items relate to the optional subsection on direct sums

- (h) If a vector space is the direct sum of subspaces $W_1, W_2, \dots, W_k$, then $W_i \cap W_j = \{0\}$ for $i \neq j$.
- (i) If

$$V = \sum_{i=1}^k W_i \quad \text{and} \quad W_i \cap W_j = \{0\} \quad \text{for } i \neq j,$$

then $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$.

Proof.

- (a) False ✓.
- (b) False ✓. This is true iff the dimension of the eigenspace corresponding to that eigenvalue is one.
- (c) True ✓ by definition.
- (d) True ✓ from theorem 5.10.
- (e) True ✓. In this case, $Q = [I]_\beta^\alpha$ where α is the standard ordered basis of $\mathbb{F}^n$. So, $[L_A]_\beta = Q^{-1}AQ$ is a diagonal matrix.
- (f) True ✓, see theorem 5.8.
- (g) True ✓, by definition.
- (h) True ✓, follows trivially from definition.
- (i) False ✓. Consider the vector space $\mathbb{R}^3$ and the subspaces $W_1 = \text{span}\{(0, 0, 1)\}$, $W_2 = \text{span}\{(1, 0, 0)\}$, $W_3 = \text{span}\{(1, 1, 0)\}$, and $W_4 = \text{span}\{(1, 1, 1)\}$. Then, $V = W_1 + W_2 + W_3 + W_4$ and pairwise disjointness must hold for the W_i 's. But the sum of the dimensions of the W_i 's is $4 > 3 = \dim(\mathbb{R}^3)$.

Alternatively, notice $(1, 1, 1) - (1, 1, 0) = (0, 0, 1) \in W_1 \cap (W_2 + W_3 + W_4)$.



Exercise 5.2.7. For

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}),$$

find an expression for A^n , where n is an arbitrary positive integer.

Proof. Computing the characteristic polynomial of A gives $(t+1)(t-4)$. Notice that $A(-2 \ 1)^t = -(-2 \ 1)^t$ and $A(1 \ 1)^t = 5(1 \ 1)^t$. Hence, $\beta := \{(-2 \ 1)^t, (1 \ 1)^t\}$ is a

basis for $\mathbb{R}^2$. As such,

$$\begin{pmatrix} -1 & 0 \\ 0 & 5 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} A^n &= \frac{1}{3} \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} (-1)^n & 0 \\ 0 & 5^n \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2(-1)^n + 5^n & -2(-1)^n + 2(5^n) \\ -(-1)^n + 5^n & (-1)^n + 2(5^n) \end{pmatrix}. \end{aligned}$$



Exercise 5.2.9. Let T be a linear operator on a finite-dimensional vector space V , and suppose there exists an ordered basis β for V such that $[T]_\beta$ is an upper triangular matrix.

- (a) Prove that the characteristic polynomial for T splits.
- (b) State and prove an analogous result for matrices.

The converse of (a) is treated in exercise 5.2.12(b).

Proof. Notice that the characteristic polynomial of A is

$$(B_{11} - t)(B_{22} - t) \cdots (B_{nn} - t),$$

for some upper triangular matrix B which is similar to A .



Exercise 5.2.11. Let A be an $n \times n$ matrix that is similar to an upper triangular matrix and has the distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ with corresponding multiplicities $m_1, m_2, \dots, m_k$. Prove the following statements.

- (a) $\text{tr}(A) = \sum_{i=1}^k m_i \lambda_i$
- (b) $\det(A) = (\lambda_1)^{m_1} (\lambda_2)^{m_2} \cdots (\lambda_k)^{m_k}$.

Proof. This is clear from A being upper triangularizable.



Exercise 5.2.12.

- (a) Prove that if $A \in M_{n \times n}(\mathbb{F})$ and the characteristic polynomial of A splits, then A is similar to an upper triangular matrix. (This proves the converse of exercise 5.2.9(b).)
- (b) Prove the converse of exercise 5.2.9(a).

Visit <https://goo.gl/gJSjRU> for a solution.

Proof.

(a) We claim that, for each $0 \leq m \leq n$, there is a basis

$$\beta = \{v_1, v_2, \dots, v_m, e_{m+1}, e_{m+2}, \dots, e_n\}$$

such that $([L_A]_\beta)_{ij} = 0$, for $1 \leq i \leq j \leq m$.

The case that $m = 1$ is clear. So, suppose this is true for m and consider $m + 1$.



Question. Are there upper triangular matrices that are non-diagonalizable?

Proof. Yes, consider the upper triangular matrix $\begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}$. Its only eigenvalue is 1 and the corresponding eigenspace is of dimension 1, as $\begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}$ is of rank 1.



Question. Is an upper triangular matrix $A \in M_{n \times n}(\mathbb{F})$ always similar to a lower triangular matrix?

Proof. Yes; for the ordered bases $\beta = \{e_1, e_2, \dots, e_n\}$ and $\gamma = \{e_n, e_{n-1}, \dots, e_1\}$, we have the lower triangular matrix

$$[L_A]_\gamma = [I]_\beta^\gamma A [I]_\gamma^\beta$$

whose i th column is $(A_{n, n+1-i} \ A_{n-1, n+1-i} \ \dots \ A_{1, n+1-i})^t$.



Question. Is an upper triangular matrix $A \in M_{n \times n}(\mathbb{F})$ always similar to its transpose?

Question. Let $A \in M_{n \times n}(\mathbb{F})$ and the linear operator $G: M_{n \times n}(\mathbb{F}) \rightarrow M_{n \times n}(\mathbb{F})$ be bijective. Is A always similar to $G(A)$? What if we remove the condition that G is linear?

Question. Let $A \in M_{n \times n}(\mathbb{F})$. Is cA similar to it for each scalar $c \in \mathbb{F}$?

Exercise 5.2.13. Let T be an invertible linear operator on a finite-dimensional vector space V .

- Recall that for any eigenvalue λ of T , λ^{-1} is an eigenvalue of T^{-1} (exercise 9 of section 5.1). Prove that the eigenspace of T corresponding to λ is the same as the eigenspace of T^{-1} corresponding to λ^{-1} .
- Prove that if T is diagonalizable, then T^{-1} is diagonalizable.

Proof.

- It was shown in exercise 5.1.9, that $u \in V$ is an eigenvector of T iff it is an eigenvector of T^{-1} . Thus, the desired result follows trivially.

- (b) This is immediate from (a).



Exercise 5.2.14. Let $A \in M_{n \times n}(\mathbb{F})$. Recall from exercise 5.1.15 that A and A^t have the same characteristic polynomial and hence share the same eigenvalues with the same multiplicities. For any eigenvalue λ of A and A^t , let E_λ and E'_λ denote the corresponding eigenspaces for A and A^t , respectively.

- (a) Show by way of example that for a given common eigenvalue, these two eigenspaces need not be the same.
 (b) Prove that for any eigenvalue λ , $\dim(E_\lambda) = \dim(E'_\lambda)$.
 (c) Prove that if A is diagonalizable, then A^t is also diagonalizable.

Proof.

- (a) Consider the field $\mathbb{R}$, and the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The eigenspace for A is

$$\left\{ t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mid t \in \mathbb{R} \right\},$$

while for A^t , it is

$$\left\{ t \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

- (b) This is immediate from $\text{rank}(A - \lambda I) = \text{rank}(A - \lambda I)^t = \text{rank}(A^t - \lambda I)$.
 (c) This is trivial since we know (b) is true.



Exercise 5.2.16. Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

be the coefficient matrix of the system of differential equations

$$\begin{aligned} x'_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ x'_2 &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ &\vdots \\ x'_n &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{aligned}$$

Suppose that A diagonalizable and that the distinct eigenvalues of A are $\lambda_1, \lambda_2, \dots, \lambda_k$. Prove that a differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}^n$ is a solution to

the system if and only if x is of the form

$$x(t) = e^{\lambda_1 t} z_1 + e^{\lambda_2 t} z_2 + \cdots + e^{\lambda_k t} z_k,$$

where $z_i \in E_{\lambda_i}$ for $i = 1, 2, \dots, k$. Use this result to prove that the set of solutions to the system is an n -dimensional real vector space.

Proof. Let β be the standard ordered basis and γ a basis of eigenvectors

$$v_1, v_2, \dots, v_{m_1}, v_{m_1+1}, \dots, v_n.$$

Thus, for $Q = [I]_{\gamma}^{\beta}$, we have $D := Q^{-1}AQ$. By exercise 5.2.17, the system $Ax = x'$ is equivalent to $D(Q^{-1}x) = (Q^{-1}x)'$. Accordingly, let $Q^{-1}x = (y_1 \ y_2 \ \dots \ y_n)^t$ so $y_i = c_i e^{\lambda_i t}$. Therefore, for any scalars $c_i \in \mathbb{C}$,

$$x(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$$

is a solution.

(It is clear that the solution space is $\mathbb{R}^n$. Furthermore, the z_j 's can obviously be obtained by taking $z_j = \sum_{i=m_j}^{m_{j+1}-1} c_i v_i$.) 

Exercise 5.2.17. Let $C \in M_{m \times n}(\mathbb{R})$, and let Y be an $n \times p$ matrix of all differentiable functions. Prove $(CY)' = CY'$, where $(Y')_{ij} = Y'_{ij}$ for all i, j .

Proof. This is clear by basic rules of differentiation, since

$$((CY)')_{ij} = \left(\sum_{k=1}^n C_{ik} Y_{kj} \right)' = \sum_{k=1}^n C_{ik} Y'_{kj} = C(Y')_{ij} = (C(Y'))_{ij}.$$



Exercise 5.2.19.

- (a) Prove that if T and U are simultaneously diagonalizable operators, then T and U commute (i.e., $TU = UT$).
- (b) Show that if A and B are simultaneously diagonalizable matrices, then A and B commute.

The converses of (a) and (b) are established in exercise 5.4.25.

Proof. Let β be a basis for V , such that $A := [T]_{\beta}$ and $B := [U]_{\beta}$ are diagonal. Notice that

$$(AB)_{ii} = ([TU]_{\beta})_{ij} = A_{ii}B_{ii} = B_{ii}A_{ii} = ([UT]_{\beta})_{ij} = (BA)_{ii}.$$

Since AB and BA are zero everywhere else, $AB = BA$. Recall that

$$H = \phi_\beta^{-1} \circ L_{[H]_\beta} \circ \phi_\beta,$$

for any linear transformation $H: V \rightarrow V$. As such, $TU = UT$. 

Exercise 5.2.20. Let T be a diagonalizable linear operator on a finite-dimensional vector space, and let m be any positive integer. Prove that T and T^m are simultaneously diagonalizable.

Proof. Let D , A , and Q be matrices such that D is diagonal and

$$D = Q^{-1}AQ.$$

Therefore,

$$A^n = QD^nQ^{-1}.$$

It is thus apparent that T^m is also diagonalizable. 

5.4 Invariant Subspaces and The Cayley-Hamilton-Frobenius Theorem

5.4.1 Theorems

Theorem 5.20. Let T be a linear operator on a finite-dimensional vector space V , and let W be a T -invariant subspace of V . Then, the characteristic polynomial of T_W divides the characteristic polynomial of T .

Hint. Block matrices.

Proof. Let $\beta := \{v_1, v_2, \dots, v_n\}$ be a basis for W , and $\gamma := \{v_1, v_2, \dots, v_m\}$ an extension of β to a basis for V . Further define the $n \times (m - n)$ matrix A and $(m - n) \times (m - n)$ matrix B by $A_{ij} := ([T]_\gamma)_{i, n+j}$ and $B_{ij} := ([T]_\gamma)_{n+i, n+j}$. We see that

$$[T]_\gamma - tI_m = \begin{pmatrix} [T_W]_\beta & A \\ O & B \end{pmatrix} - tI_m = \begin{pmatrix} [T_W]_\beta - tI_n & A \\ O & B - tI_{m-n} \end{pmatrix}.$$

So, the characteristic polynomial of T is

$$\det([T_W]_\beta - tI_n) \det(B - tI_{m-n}).$$

(exercise 21 of section 4.3) 

Theorem 5.21. Let T be a linear operator on a finite-dimensional vector space V , and let W denote the T -cyclic subspace of V generated by a nonzero $v \in V$. Let

$k = \dim(W)$. Then

- (a) $\{v, T(v), T^2(v), \dots, T^{k-1}(v)\}$ is a basis for W .
- (b) If $a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = 0$, then the characteristic polynomial of T_W is $f(t) = (-1)^k (a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k)$.

Proof.

- (a) Let n be the least natural number, such that $T^n(v)$ is a linear combination of $\beta := \{v, T(v), T^2(v), \dots, T^{n-1}(v)\}$. Clearly, β is a basis for W , so $k = n$.
- (b) We see that

$$[T_W]_\beta = \left(\begin{array}{cccc|c} 0 & 0 & \cdots & 0 & -a_0 \\ & & & & -a_1 \\ & & & & \vdots \\ & & & & -a_{k-1} \end{array} \right)$$

So, for each $0 \leq n \leq k$, let the $n \times n$ matrix

$$A_n := \left(\begin{array}{cccc|c} -t & 0 & 0 & \cdots & -a_n \\ 1 & -t & 0 & \cdots & -a_{n+1} \\ 0 & 1 & -t & \cdots & -a_{n+2} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{k-1} - t \end{array} \right)$$

Now, by cofactor expansion along the first row, the characteristic polynomial $f(t)$ of T_W is

$$\begin{aligned} \det(A_0) &= -t \det(A_1) + (-1)^{1+k} (-a_0) \det(I_{k-1}) \\ &= -t [-t \det(A_2) + (-1)^k (-a_1)] + (-1)^k a_0 \\ &= (-t)^2 [-t \det(A_3) + (-1)^{k-1} (-a_2)] + (-1)^k (a_0 + a_1 t) \\ &\vdots \\ &= (-t)^{k-2} (-t \det(A_k) + (-1)^{1+2} a_{k-2}) \\ &\quad + (-1)^k (a_0 + a_1 t + \dots + a_{k-3} t^{k-3}) \\ &= (-1)^k (a_0 + a_1 t + \dots + a_{k-1} t^{k-1} + t^k). \end{aligned}$$



Question. Is $T^k(v)$ always equal to v ? When does equality hold?

Proof. No, consider the zero transformation $T_0: \mathbb{R} \rightarrow \mathbb{R}$. Then, $T_0^2(1) = 0$ despite

$$\{1, T_0(1), T_0^2(1), \dots\} = \{1, T_0(1)\} = \{1, 0\}.$$

Equality holds iff T_W is invertible. If $T^k(v) = v$, then $T_W^{-1} = T_W^{k-1}$, i.e. T_W is

invertible. But the converse isn't true in general (?)



Question. Let W be a T -cyclic subspace of V . Does W always contain a 'loop'? That is, is there always a subset $L := \{u, T(u), T^2(u), \dots\}$ of W whose span is T -invariant?

Theorem 5.22 (Cayley-Hamilton-Frobenius). Let T be a linear operator on a finite dimensional vector space V , and let $f(t)$ be the characteristic polynomial of T . Then $f(T) = T_0$, the zero transformation. That is, T "satisfies" its characteristic equation.

Proof. ✓ Pick a vector $v \in V$ and let W be the T -cyclic subspace of V generated by v . By the preceding theorem 5.21, the characteristic polynomial $g(t)$ of T_W is such that $g(T)(v) = 0$. Now, $f(T)(v) = 0$ follows from theorem 5.20. Consequently, $f(T) = T_0$.



A longer proof can be found in my VSCode comments. But I find the level of brevity here to be adequate. That aside, *the following result is from the book.*

Corollary (Cayley-Hamilton-Frobenius Theorem for Matrices). Let A be an $n \times n$ matrix, and let $f(t)$ be the characteristic polynomial of A . Then, $f(A) = O$, the $n \times n$ zero matrix.

Proof. By the Cayley-Hamilton-Frobenius theorem, $L_{f(A)} = f(L_A) = L_O$. So, $f(A) = O$.



Question. Is it always true that $L_A = L_B$ implies $A = B$?

Proof. Yes, let $T: V \rightarrow W$ be a linear transformation and fix some (ordered) bases β and γ of V and W , respectively. Notice that the matrix representation of T in the ordered bases β and γ is unique (because the representation of any vector, with respect to a fixed basis, is unique).



Question. Let T be a linear operator, on a finite dimensional vector space V , whose characteristic polynomial is f . Suppose the polynomial g is such that $g(T) = T_0$. Then, must f divide g ?

Theorem 5.23. Let T be a linear operator on a finite-dimensional vector space V , and suppose that $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$, where W_i is a T -invariant subspace of V for each i ($1 \leq i \leq k$). Suppose that $f_i(t)$ is the characteristic polynomial of T_{W_i} ($1 \leq i \leq k$). Then $f_1(t)f_2(t) \dots f_k(t)$ is the characteristic polynomial of T .

Proof. ✓ Let the characteristic polynomial of T be F , and pick a basis β_i for each

i. If $V = W_1 \oplus W_2$ (i.e. $k = 2$), then

$$[T]_{\beta_1 \cup \beta_2} = \begin{pmatrix} [T_{W_1}]_{\beta_1} & O \\ O & [T_{W_2}]_{\beta_2} \end{pmatrix}.$$

So, $F(t) = f_1(t)f_2(t)$ is clear. Assume the result holds for $k = n$, and now, consider $k = n + 1$. Then, $V = W_1 \oplus (W_2 \oplus \cdots \oplus W_k)$ is apparent. As such, it follows from the above result for $k = 2$, that

$$f(t) = f_1(t)(f_2(t)f_3(t) \cdots f_k(t)) = f_1(t)f_2(t) \cdots f_k(t).$$



Alternatively, for the sake of brevity we may write the following, which I prefer.

Proof. Let the characteristic polynomial of T be F , and pick a basis β_i for each i . By invariance, it holds for $\gamma := \beta_1 \cup \beta_2 \cup \cdots \cup \beta_k$, that

$$[T]_{\gamma} = \begin{pmatrix} [T_{W_1}]_{\beta_1} & O & \cdots & O \\ O & [T_{W_2}]_{\beta_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & [T_{W_k}]_{\beta_k} \end{pmatrix}.$$

Consequently, it is clear that

$$f(t) = f_1(t)f_2(t) \cdots f_k(t).$$



5.4.2 Exercises

Exercise 5.4.1. Label the following statements as true or false.

- There exists a linear operator T with no T -invariant subspace.
- If T is a linear operator on a finite-dimensional vector space V , the the characteristic polynomial of T_W divides the characteristic polynomial of T .
- Let T be a linear operator on a finite-dimensional vector space V , and let v and w be in V . If W is the T -cyclic subspace generated by v , W' is the T -cyclic subspace generated by w , and $W = W'$, then $v = w$.
- If T is a linear operator on a finite-dimensional vector space V , then for any $v \in V$ the T -cyclic subspace generated by v is the same as the T -cyclic subspace generated by $T(v)$.
- Let T be a linear operator on an n -dimensional vector space. Then there exists a polynomial $g(t)$ of degree n such that $g(T) = T_0$.
- Any polynomial of degree n with leading coefficient $(-1)^n$ is the characteristic

polynomial of some linear operator.

- (g) If T is a linear operator on a finite-dimensional vector space V , and if V is the direct sum of k T -invariant subspaces, then there is an ordered basis β for V such that $[T]_\beta$ is a direct sum of k matrices.

Proof.

- (a) ✓ False. The zero vector space is always T -invariant.
 (b) ✓ True, see theorem 5.20.
 (c) ✓ False. Let the linear operator T on $\mathbb{R}$ be defined by $T(x) = -x$. Fix $v = 1$, and $w = -1$. Then, $W = W' = \{-1, 1\}$ even though $1 \neq -1$.
 (d) ✓ False. Consider the linear operator T on $\mathbb{R}$ defined by $T(x) = x + 1$. Then, the T -cyclic subspaces generated by 0 and 1 are $\mathbb{Z}_0^+$ and $\mathbb{Z}^+$, respectively.
 (e) ✓ True. One such g is the characteristic polynomial of T .
 (f) ✗ False. Any linear operator on the zero vector space must have characteristic polynomial 0.

Oh I misunderstood the exercise. The intended interpretation is probably “Any polynomial g of degree n with leading coefficient $(-1)^n$ (and coefficients from $\mathbb{F}$) is the characteristic polynomial f of some linear operator on *some* vector space (on $\mathbb{F}$).”

It is clear for $n = 1$ that such a linear operator T on $\mathbb{F}$ exists. So, suppose this is true for $n = k$, and consider $n = k + 1$.

- (g) ✓ True by theorem 5.24.



Exercise 5.4.2. For each of the following linear operators T on the vector space V , determine whether the given subspace W is a T -invariant subspace of V .

- (d) $V = C([0, 1])$, $T(f(t)) = \left[\int_0^1 dx \right] t$, and

$$W = \{f \in V \mid f(t) = at + b \text{ for some } a \text{ and } b\}.$$

Proof.

- (d) Notice that $T(at + b) = \left[at^2/2 + bt \right]_0^1 \cdot t = (a/2 + b)t \in W$. Hence, W is indeed a T -invariant subspace of V .



Exercise 5.4.6. For each linear operator T on a vector space V , find an ordered basis for the T -cyclic subspace generated by the vector z .

- (a) $V = \mathbb{R}^4$, $T(a, b, c, d) = (a + b, b - c, a + c, a + d)$, and $z = e_1$.
 (c) $V = M_{2 \times 2}(\mathbb{R})$, $T(A) = A^t$, and $z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Proof.

(a) ✓ We see that

$$\begin{aligned}
 e_1 &= (1, 0, 0, 0) \\
 T(e_1) &= (1, 0, 1, 1) \\
 T^2(e_1) &= (1, -1, 2, 2) \\
 T^3(e_1) &= (0, -3, 3, 3) = 3[T^2(e_1) - T(e_1)] \\
 T^4(e_1) &= (-3, -6, 3, 3) = 6T^2(e_1) - 9T(e_1) \\
 T^5(e_1) &= (-9, -9, 0, 0) = 9(T^2(e_1) - 2T(e_1)) \\
 T^6(e_1) &= (-18, -9, -9, -9) = 9(T^2(e_1) - 2T(e_1)) - 9T(e_1) \\
 T^7(e_1) &= (-27, 0, -27, -27) = -27T(e_1).
 \end{aligned}$$

So, the T -cyclic subspace generated by the vector $z = e_1$ is

$$\text{span}\{T^i(e_1) \mid 0 \leq i \leq 7\}.$$

Moreover, $\{e_1, T(e_1), T^2(e_1)\}$ is a basis for it.

(b) ✓ It is clear that $\{z\}$ is a basis for the T -cyclic subspace generated by z .



Exercise 5.4.17. Let A be an $n \times n$ matrix. Prove that

$$\dim(\text{span}(\{I_n, A, A^2, \dots\})) \leq n.$$

Proof. By the **Cayley-Hamilton-Frobenius** theorem, the characteristic polynomial $f(x)$ of A is such that $f(A) = O$. As such,

$$\dim(\text{span}(I_n, A, A^2, \dots, A^n)) \leq n.$$

Furthermore, $A^{n+m}f(A) = O$ for each $m \in \mathbb{N}$. Hence, it follows from simple induction, that $A^{n+m} \in \text{span}(A, \dots, A^n)$. Clearly,

$$\dim(\text{span}(I_n, A, A^2, \dots)) \leq n.$$



Example. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and notice $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. So, since

$$A^2 - A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix},$$

we see that $\{A, A^2\}$ is a basis for

$$\text{span}(\{I_n, A, A^2, \dots\}).$$

Hence, it must have dimension 2.

Example. Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Similarly, we observe that

$$A^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}.$$

Furthermore,

$$A - 2A^2 + A^3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad -3A + 5A^2 - 2A^3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore, it is clear that $\{A, A^2, A^3\}$ is a basis for

$$\text{span}(\{I_n, A, A^2, \dots\}).$$

i.e. the above vector space is of dimension 3.

Exercise. The above seems to suggest an interesting pattern arises with $n \times n$ upper triangular matrices A_n whose entries are all 1's. Can you find this pattern?

Exercise. Find a basis for A_n .

Exercise 5.4.18. Let A be an $n \times n$ matrix with characteristic polynomial

$$f(t) = (-1)^n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0.$$

- (a) Prove that A is invertible if and only if $a_0 \neq 0$.
- (b) Prove that if A is invertible, then

$$A^{-1} = (-1/a_0)[(-1)^n A^{n-1} + a_{n-1} A^{n-2} + \dots + a_1 I_n].$$

- (c) Use (b) to compute A^{-1} for

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -1 \end{pmatrix}.$$

Proof.

- (a) If $a_0 = 0$, then $\det(A) = f(0) = 0$. Hence, A is not invertible. Conversely,

when $a_0 \neq 0$, by the Cayley-Hamilton-Frobenius theorem we have

$$\begin{aligned} (-1)^n A^n + a_{n-1} A^{n-1} + \cdots + a_1 A + a_0 I &= 0, \\ (-1)^n A^n + a_{n-1} A^{n-1} + \cdots + a_1 A &= -a_0 I, \\ (-1/a_0)[(-1)^n A^{n-1} + a_{n-1} A^{n-2} + \cdots + a_1 I]A &= I. \end{aligned}$$

As such,

$$A^{-1} = (-1/a_0)[(-1)^n A^{n-1} + a_{n-1} A^{n-2} + \cdots + a_1 I].$$

This also conveniently proves (b).

(c) We first compute the characteristic polynomial f of A to be

$$f(t) = (1-t)(2-t)(1+t) = t^3 - 2t^2 - t + 2.$$

Accordingly,

$$A^{-1} = (-1/2)(A^2 - 2A - I) = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1/2 & 3/2 \\ 0 & 0 & -1 \end{pmatrix}.$$



Exercise 5.4.20. Let T be a linear operator on a vector space V , and suppose that V is a T -cyclic subspace of itself. Prove that if U is a linear operator on V then $UT = TU$ if and only if $U = g(T)$ for some polynomial $g(t)$.

Proof. By theorem 5.21, a basis for V is

$$\{v, T(v), \dots, T^{r-1}(v)\}$$

for some vector $v \in V$ and $r := \dim(V)$. Suppose that $UT = TU$. We notice $U(v) := \sum_{i=1}^r a_i T^i(v)$ for some $a_i \in \mathbb{F}$. Then, a simple inductive proof shows

$$UT^j(v) = T^j U(v) = \sum_{i=1}^r a_i T^{i+j}(v).$$

The converse is trivial.



Question. Let V be a vector space of dimension n . Must there exist a linear operator T on V , such that V is a T -cyclic subspace of itself?

Proof. Choose a basis $\beta := \{v_1, v_2, \dots, v_n\}$ for V . We define the linear transformation T on V by $T(v_i) = v_{i+1}$, where $v_{n+1} := v_1$. Then, it is clear that the T -cyclic subspace of v_1 is V , since it contains β .



Definition. Let T be a linear operator on a vector space V which contains the vector w . For brevity, we define $\langle w \rangle$ to be the T -cyclic subspace of V generated by w .

Exercise 5.4.21. Let T be a linear operator on a two-dimensional vector space V . Prove that either V is a T -cyclic subspace of itself or $T = cI$ for some scalar c .

Proof. Suppose T is not a scalar multiple of the identity, and choose a basis $\{u, v\}$ for V . Then, $\text{nullity}(T) < 2$ so wlog $T(u) = au + bv \neq 0$. When a and b are both nonzero, $\langle u \rangle = V$. Otherwise, *exactly one* of a or b is nonzero, but regardless, $\langle u + v \rangle = V$. 

Initially thought about using the following method, but forsook it in favor of induction, as proving the matrix has full rank seemed unnecessarily tough. Credits to Ann for reminding me to consider Vandemonde matrices, which I have proved some relevant results about.

Exercise 5.4.23. Let T be a linear operator on a finite-dimensional vector space V , and let W be a T -invariant subspace of V . Suppose that $v_1, v_2, \dots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues. Prove that if $v_1 + v_2 + \dots + v_k$ is in W , then $v_i \in W$ for every i .

Proof. Let λ_i be the eigenvalue associated with the eigenvector v_i . Since each λ_i is distinct, exercise 4.3.22(a) implies the matrix

$$\begin{pmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{k-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \lambda_k & \lambda_k^2 & \cdots & \lambda_k^{k-1} \end{pmatrix}$$

has full rank. So, each

$$e_i = \sum_{j=1}^{k-1} c_{ij} \begin{pmatrix} \lambda_1^j & \lambda_2^j & \cdots & \lambda_k^j \end{pmatrix}^t,$$

for some scalars $c_{ij} \in \mathbb{F}$. Going back to our subspace W , this implies

$$v_i = \sum_{j=1}^{k-1} c_{ij} (\lambda_1^j v_1 + \lambda_2^j v_2 + \cdots + \lambda_k^j v_k) = \sum_{j=1}^{k-1} c_{ij} T^j(v_1 + v_2 + \cdots + v_k)$$

because $\{v_1, v_2, \dots, v_k\}$ is linearly independent. Since W contains $v_1 + v_2 + \cdots + v_k$ and is T -invariant, it contains all $T^j(v_1 + v_2 + \cdots + v_k)$. Hence, $v_i \in W$ for all i . 

An alternate phrasing.

Proof. Let λ_i be the eigenvalue associated with the eigenvector v_i . Consider how each v_i can be expressed in terms of

$$\beta := \{(v_1 + v_2 + \cdots + v_k), T(v_1 + v_2 + \cdots + v_k), \dots, T^{k-1}(v_1 + v_2 + \cdots + v_k)\}.$$

For $\gamma := \{v_1, v_2, \dots, v_k\}$, notice

$$[T^j(v_1 + v_2 + \cdots + v_k)]_\gamma = \begin{pmatrix} \lambda_1^j \\ \lambda_2^j \\ \vdots \\ \lambda_k^j \end{pmatrix}.$$

Placing these columns in a matrix gives

$$\begin{pmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{k-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \lambda_k & \lambda_k^2 & \cdots & \lambda_k^{k-1} \end{pmatrix}.$$

Notice this is a Vandermonde matrix, i.e. it has full rank. Therefore, every v_i can indeed be written in terms of $\beta \subseteq W$. 

Exercise 5.4.24. Prove that the restriction of a diagonalizable linear operator T to any nontrivial T -invariant subspace is also diagonalizable.

Proof. Let W be a T -invariant subspace. Pick a maximal linearly independent subset γ of eigenvectors of T_W , and extend it to some eigenbasis β of T . If γ fails to span W , then

$$\sum_{i,j} c_{ij} v_{ij} \in W,$$

for some nonzero scalars c_{ij} and eigenvectors $v_{ij} \in \beta - \gamma$ corresponding to the eigenvalues λ_i of T . But by the preceding exercise, $\sum_j c_{ij} v_{ij} \in W$, contradicting the maximality of γ . 

Exercise 5.4.25.

- (a) Prove the converse to exercise 5.2.19(a): If T and U are diagonalizable linear operators on a finite-dimensional vector space V such that $UT = TU$, then T and U are simultaneously diagonalizable. (See the definition in the exercises of section 5.2)
- (b) State and prove a matrix version of (a).

Proof. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of T , and let $v \in E(\lambda_i, T)$. Notice that $TU(v) = \lambda_i U(v)$. That is, $E(\lambda_i, T)$ is U -invariant. By the preceding exercise, there is an eigenbasis $\beta_i := \{w_{i1}, w_{i2}, \dots, w_{ik_i}\}$ for $U_{E(\lambda_i, T)}$. Therefore, $\bigcup_i \beta_i$ is an eigenbasis for both T and U . 

Exercise 5.4.26. Let T be a linear operator on an n -dimensional vector space V such that T has n distinct eigenvalues. Prove that V is a T -cyclic subspace of itself.

Proof. Let v_i be an eigenvector of T corresponding to the eigenvalue λ_i , for $1 \leq i \leq n$. By exercise 5.4.23, we see that $v_i \in \langle v_1 + v_2 + \cdots + v_n \rangle$ for all i . Hence, $V = \langle v_1 + v_2 + \cdots + v_n \rangle$. 

Exercise 5.4.27. Let T be a linear operator on a vector space V , and let W be a T -invariant subspace of V . Define $\bar{T}: V/W \rightarrow V/W$ by $\bar{T}(v + W) := T(v) + W$ for any $v + W \in V/W$.

- (a) Prove that $\bar{T}$ is well defined. That is, show that $\bar{T}(v + W) = \bar{T}(v' + W)$ whenever $v + W = v' + W$.
- (b) Prove that $\bar{T}$ is a linear operator on V/W .
- (c) Let $\eta: V \rightarrow V/W$ be the linear transformation defined in exercise 2.1.42 by $\eta(v) := v + W$. Show that the diagram of Figure 5.1 commutes; that is, prove that $\eta T = \bar{T}\eta$. (This exercise does not require the assumption that V is finite-dimensional.)

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \eta \downarrow & & \downarrow \eta \\ V/W & \xrightarrow{\bar{T}} & V/W \end{array}$$

Proof.

- (a) For $v + W = v' + W$, we have $v' - v \in W$. So, $T(v') \in T(v) + W$. By symmetry, $\bar{T}(v + W) = \bar{T}(v' + W)$.
- (b) Let $c \in \mathbb{F}$ and $u, v \in V$. Then,

$$\begin{aligned} \bar{T}(c(u + W) + (v + W)) &= T(cu + v) + W = cT(u) + T(v) + W \\ &= c(T(u) + W) + ((T(v) + W)) \\ &= c\bar{T}(u + W) + \bar{T}(v + W). \end{aligned}$$

- (c) Notice that

$$\eta T(v) = T(v) + W = \bar{T}(v + W) = \bar{T}\eta(v).$$

Hence, Figure 5.1 commutes.



Exercise 5.4.28. Let $f(t)$, $g(t)$, and $h(t)$ be the characteristic polynomials of T , T_W , and $\bar{T}$, respectively. Prove that $f(t) = g(t)h(t)$.

Proof. We follow the notation of theorem 5.20; define $\beta/W := \{v_{n+1} + W, v_{n+2} + W, \dots, v_m + W\}$. Since

$$\begin{aligned} \bar{T}(v_{n+j} + W) &:= T(v_{n+j}) + W = \left(\sum_{i=1}^m c_{ij} v_i \right) + W \\ &= \sum_{i=n+1}^m c_{ij} (v_i + W) = \sum_{i=1}^{m-n} B_{ij} (v_{n+1} + W), \end{aligned}$$

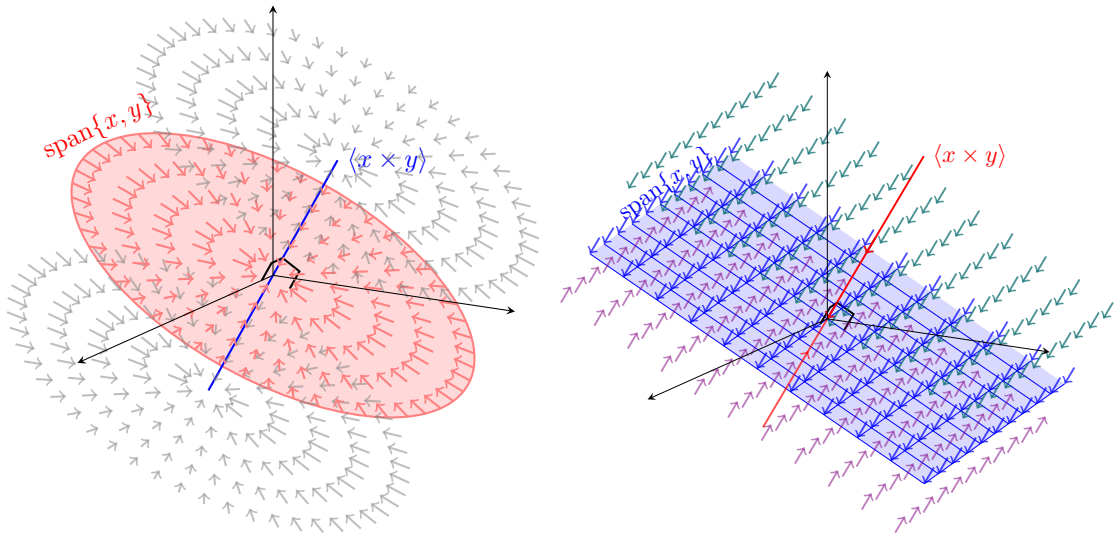
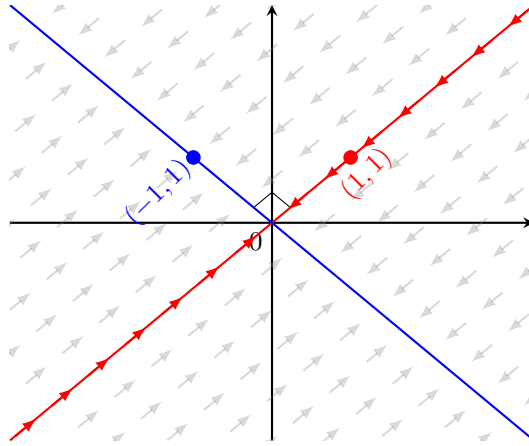
for some scalars c_{ij} , we have that $[\bar{T}]_{\beta/W} = B$. As such, $f(t) = g(t)h(t)$ follows from theorem 5.20.



Exercise 5.4.30. Prove that if both T_W and $\bar{T}$ are diagonalisable and have no common eigenvalues, then T is diagonalisable.

Proof. Assume that both T_W and $\bar{T}$ are diagonalisable. i.e. there are eigenbases $\gamma := \{v_1, v_2, \dots, v_n\}$ and $\beta/W := \{v_{n+1} + W, v_{n+2} + W, \dots, v_m + W\}$ of T_W and $\bar{T}$, respectively. Then, $\beta := \{v_1, v_2, \dots, v_m\}$ is an eigenbasis for T , since T_W and $\bar{T}$ have no common eigenvalues. 

Question. Is there a linear operator T and a T -invariant subspace W , such that both T_W and $\bar{T}$ are diagonalisable, but T itself is not?



Note. The intuition for quotient spaces.

- The quotient space V/W is obtained by squishing/collapsing W to $\{0\}$ and extending this transformation $T_0: W \rightarrow \{0\}$ to V in a way that respects the structure of V , giving us some linear $T: V \rightarrow V$ with $T_W = T_0$.
- Pick a basis β for W and an extension γ to a basis for V . Then, $V/W = \text{span}(\gamma - \beta)$. In other words, the quotient space V/W is obtained by zeroing all components of W in V .

See Figure 5.2 for an illustration.

Problem (Greub chapter 6, exercise 7). A linear transformation, T , of an n -dimensional vector space V is called *nilpotent* if $T^k = 0$ for some k . Prove that T is nilpotent if and only if the characteristic polynomial has the form

$$\chi(\lambda) = (-\lambda)^n.$$

Hint. Use problem 6 (equivalently, exercise 5.4.28).

Proof. Define $T_{n+1}: U_n/\mathcal{N}(T_n) \rightarrow U_n/\mathcal{N}(T_n)$ by

$$T_{n+1} := \begin{cases} T & \text{if } n+1 = 0, \\ \bar{T}_n & \text{otherwise,} \end{cases}$$

where $T_n: U_n \rightarrow U_n$. Clearly, $\bar{T}(v + \mathcal{N}(T)) = 0$ iff $v \in \mathcal{N}(T^2)$. Moreover, supposing $v + \cdots \in \mathcal{N}(T_{n-1})$ iff $v \in \mathcal{N}(T^n)$,

$$\begin{aligned} T_n \left(v + \sum_{i=1}^{n-1} \mathcal{N}(T_i) \right) &= T_{n-1} \left(v + \sum_{i=1}^{n-2} \mathcal{N}(T_i) \right) + \mathcal{N}(T_{n-1}) = 0 \\ \iff T(v) + \sum_{i=1}^{n-2} \mathcal{N}(T_i) &\in \mathcal{N}(T_{n-1}) \\ \iff T(v) &\in \mathcal{N}(T^n) \\ \iff v &\in \mathcal{N}(T^{n+1}). \end{aligned}$$

In other words, $v + \cdots \in \mathcal{N}(T_n)$ iff $v \in \mathcal{N}(T^{n+1})$. Since this implies $T^{k-1} = 0$, we have $\chi_{T_{k-1}} = (-t)^{\dim U_{k-1}}$:

$$\chi_T = \left(\prod_{i=1}^{k-2} \chi_{T_n|_{\mathcal{N}(T_n)}} \right) \cdot \chi_{T_{k-1}} = (-t)^{\sum_{i=1}^{k-2} \dim \mathcal{N}(T_i) + \dim U_{k-1}},$$

by exercise 5.4.28 (and where χ_T represents the characteristic polynomial of T). By theorem 5.3(a), χ_T is of exactly degree n , so $\chi_T(\lambda) = (-\lambda)^n$. The converse is a trivial application of the Cayley-Hamilton-Frobenius theorem. 

Note. The intuition for the preceding problem. Since T of a linear operator on V , we notice that $V = \mathcal{N}(T) \oplus \mathcal{R}(T)$. So, $V/\mathcal{N}(T) \cong \mathcal{R}(T)$ and $\bar{T} \cong T_{\mathcal{R}(T)}$. More generally,

$$\mathcal{N}(T_n) \cong \mathcal{N}(T^{n+1})/\mathcal{N}(T^n), \quad U_n \cong V/\mathcal{N}(T^n) \cong \mathcal{R}(T^n), \quad T_n \cong T_{\mathcal{R}(T^n)}.$$

Question. Consider the vector spaces $W \subseteq V$. Is there a canonical isomorphic embedding $T: V/W \rightarrow V$?

Proof. The goal of quotienting W from V is to ‘remove’ W from V . To ensure the removal of W is complete, it makes sense that $T[V/W]$ should be orthogonal to W , so that the vectors of $T[V/W]$ have no component parallel to W that can be further removed. So, for an inner product space V , the canonical choice for $T[V/W]$ is $W^\perp$?^a (Though there is likely no canonical choice for T itself.) For a generic vector space V with no inner product structure, a canonical choice for $T[V/W]$ probably does not exist, for the same reason. 

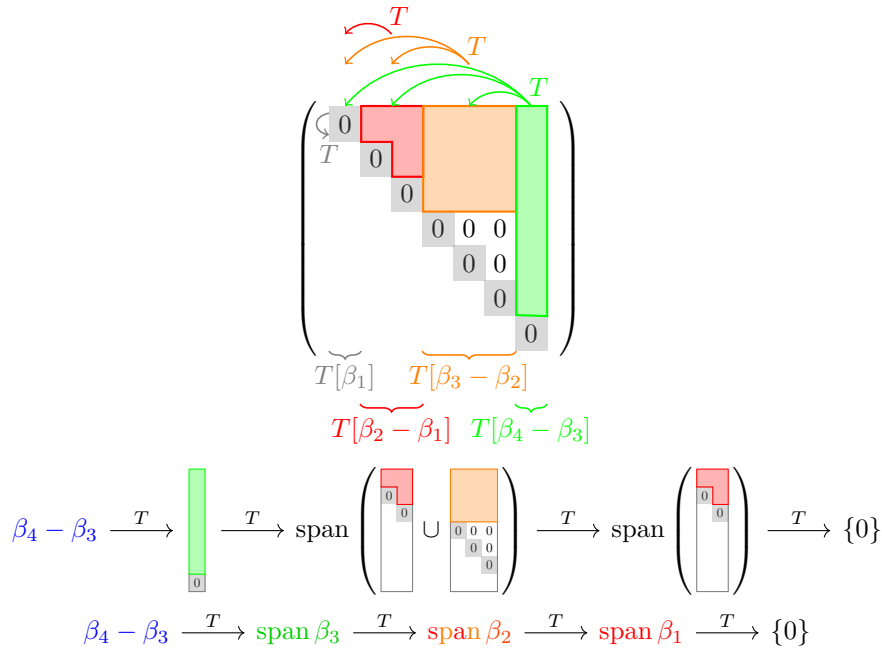
^aWe need to see prove/disprove $V = W \oplus W^\perp$.

Question. Let T be a nilpotent linear operator on an n -dimensional vector space V . Must $T^n = T_0$?

Proof. Yes. We see that $\text{rank}(T^{k+1}) < \text{rank}(T^k)$, for each $k \in \mathbb{N}$ such that $\text{rank}(T^k) > 0$ — lest $\text{rank}(T^k) = \text{rank}(T^{k+1}) = \dots > 0$ so T is not nilpotent. 

Exercise (Artin chapter 4, exercise 6.8). A linear operator T is nilpotent if some positive power T^k is zero. Prove that T is nilpotent iff there is a basis of V such that the matrix of T is upper triangular, with diagonal entries zero.

Proof. Suppose T is nilpotent and let $\beta_n := \{v_1, v_2, \dots, v_{i_n}\}$ be an extension of β_{n-1} to a basis for $N(T^n)$, where $\beta_0 = \emptyset$. Notice $T[\beta_n] \subseteq \text{span } \beta_{n-1}$. That is, $T(v_j) = \sum_{i=1}^{i_{n-1}} A_{ij} e_i$ for $j \leq i_n$ and $A := [T]_{\beta_k}$. Consider $i \geq j$; for some n we have $i_{n-1} < j \leq i_n$. Notice $i > i_{n-1}$ so $A_{ij} = 0$. The converse is trivial. 



Exercise (Artin chapter 5, exercise M.3). Let A be an $n \times n$ complex matrix. Prove that if $\text{tr } A^k = 0$ for all $k > 0$, then A is nilpotent.

Exercise 5.4.31. Let $A = \begin{pmatrix} 1 & 1 & -3 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{pmatrix}$, let $T = L_A$ and let W be the cyclic subspace of $\mathbb{R}^3$ generated by e_1 .

- (a) Use theorem 5.21 to compute the characteristic polynomial of T_W . Show that $\{e_2 + W\}$ is a basis for $\mathbb{R}^3/W$ and use this fact to compute the characteristic polynomial of $\bar{T}$.
- (b) Use the results of (a) and (b) to find the characteristic polynomial of A .

Proof.

- (a) We see that

$$\begin{pmatrix} e_1 & Ae_1 & A^2e_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 12 \\ 0 & 1 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -6 \\ 0 & 1 & 6 \\ 0 & 0 & 0 \end{pmatrix}$$

So the characteristic polynomial of T_W is $t^2 - 6t + 6$. Furthermore, $\{e_1, Ae_1\}$ and $\{e_1, Ae_1, e_2\}$ are bases for W and $\mathbb{R}^3$, respectively. Thus, $\{e_2 + W\}$ must be a basis for $\mathbb{R}^3/W$. In fact, it corresponds to the eigenvalue -1 , since

$$T(e_2) = \begin{pmatrix} 1 & 3 & 2 \end{pmatrix}^t = -e_1 + 2Ae_1 - e_2.$$

i.e. the characteristic polynomial of $\bar{T}$ is $-t - 1$.

- (b) By exercise 5.4.28, the characteristic polynomial of A is $-(t+1)(t^2 - 6t + 6)$.



Exercise 5.4.38. Let $\mathcal{C}$ be a collection of diagonalizable linear operators on a finite-dimensional vector space V . Prove that there is an ordered basis β such that $[T]_\beta$ is a diagonal matrix for all $T \in \mathcal{C}$ if and only if the operators of $\mathcal{C}$ commute under composition. (This is an extension of exercise 5.4.25.)

Hint. The result is trivial if each operator only has one eigenvalue. Otherwise, establish the general result by mathematical induction on $\dim(V)$, using the fact that V is the direct sum of the eigenspaces of some operator in $\mathcal{C}$ that has more than one eigenvalue.

Proof. Suppose this is true for $\dim(V) = 0, 1, \dots, n$ — the cases involving $\dim(V) = 0, 1$ are trivial. Now consider when $\dim(V) = n+1$. Wlog, fix a linear operator $T \in \mathcal{C}$ with two or more eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$. So, $\dim E(\lambda_i, T) \leq n$. Furthermore, $E(\lambda_i, T)$ is U -invariant, by commutativity:

$$TU(v) = UT(v) = \lambda_i U(v) \quad \text{for } v \in E(\lambda_i, T).$$

Hence, our initial supposition allows us to choose a basis $\mathcal{B}_i$ that simultaneously diagonalises all $U|_{E(\lambda_i, T)}$, for $U \in \mathcal{C}$. Now, $\bigcup \mathcal{B}_i$ simultaneously diagonalises all linear operators in $\mathcal{C}$.



Exercise 5.4.40. Let

$$A = \begin{pmatrix} 1 & 2 & \cdots & n \\ n+1 & n+2 & \cdots & 2n \\ \vdots & \vdots & & \vdots \\ n^2 - n + 1 & n^2 - n + 2 & \cdots & n^2 \end{pmatrix}.$$

Find the characteristic polynomial of A .

Exercise 5.4.41. Let $A \in M_{n \times n}(\mathbb{R})$ be defined by $A_{ij} = 1$ for all i and j . Find the characteristic polynomial of A .

Inner Product Spaces

6.1 Inner Products and Norms

Question. Preliminary questions.

- (a) Why is the weaker condition of conjugate symmetry used over symmetry? When are such inner products useful?
- (b) What is the intuition behind complex inner products?
- (c) If we allow inner products to take on complex values, then why not values in arbitrary fields?

6.1.1 Theorems

Theorem 6.1. Let V be an inner product space. Then, for $x, y, z \in V$ and $c \in \mathbb{K}$, the following statements are true.

- (a) $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$.
- (b) $\langle x, cy \rangle = \bar{c} \langle x, y \rangle$.
- (c) $\langle x, 0 \rangle = \langle 0, x \rangle = 0$.
- (d) $\langle x, x \rangle = 0$ iff $x = 0$.
- (e) If $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V$, then $y = z$.

Proof. Parts (a) to (c) are clear from the linearity in the first-coordinate and conjugate symmetry of inner products; (d) follows from positive-definiteness. Lastly, for (e), suppose that $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V$. So $\langle x, y - z \rangle = 0$ for $x = y - z$. Hence, $y - z = 0$ by (d). 

Theorem 6.2. Let V be an inner product space over $\mathbb{K}$. Then for all $x, y \in V$ and $c \in \mathbb{K}$, the following statements are true.

- (a) $\|cx\| = |c| \cdot \|x\|$.

- (b) $\|x\| = 0$ iff $x = 0$. In any case, $\|x\| \geq 0$.
- (c) (Cauchy-Schwarz Inequality) $|\langle x, y \rangle| \leq \|x\| \|y\|$.
- (d) (Triangle Inequality) $\|x + y\| \leq \|x\| + \|y\|$.

Proof. Parts (a) and (b) are trivial.

- (c) Since $y_{\parallel} := \langle y, \hat{x} \rangle \hat{x}$ and $y - y_{\parallel}$ are orthogonal, exercise 6.1.10 implies

$$|\langle x, y \rangle| = |\langle \hat{x}, y \rangle \langle x, \hat{x} \rangle| = |\langle x, y_{\parallel} \rangle| = \|x\| \|y_{\parallel}\| \leq \|x\| \|y\|.$$

- (d) By (c), $\Re \langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \|y\|$. So,

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle + 2\Re \langle x, y \rangle \leq \|x\|^2 + \|y\|^2 + 2\|x\| \|y\| = (\|x\| + \|y\|)^2.$$



6.1.2 Exercises

Exercise 6.1.1. Label the following statements as true or false.

- (a) An inner product is a scalar-valued function on the set of ordered pairs of vectors.
- (b) An inner product space must be over the field of real or complex numbers.
- (c) An inner product is linear in both components.
- (d) There is exactly one inner product on the vector space $\mathbb{R}^n$.
- (e) The triangle inequality only holds in finite-dimensional inner product spaces.
- (f) Only square matrices have a conjugate-transpose.
- (g) If x , y , and z are vectors in an inner product space such that $\langle x, y \rangle = \langle x, z \rangle$, then $y = z$.
- (h) If $\langle x, y \rangle = 0$ for all x in an inner product space, then $y = 0$.

Proof.

- (a) True ✓.
- (b) True ✓. (Why?)
- (c) False ✓. Any inner product is linear in the first component and *conjugate* linear in the second. Indeed, conjugate linearity is not equivalent to linearity; consider the standard inner product over $\mathbb{C}$. Notice $\langle 1, i \rangle = -i \neq i = i \langle 1, 1 \rangle$.
- (d) False ✓. The standard inner product is not the only inner product possible; $2\langle \cdot, \cdot \rangle$ is another.
- (e) False ✓. It holds in all inner product spaces, such as the space of all real sequences with finite nonzero entries, under the inner product $\langle \{x_n\}, \{y_n\} \rangle := \sum_{n=1}^{\infty} x_n y_n$.
- (f) False ✓. Consider the standard inner product on $\mathbb{R}$. Then, $\langle 0, 1 \rangle = \langle 0, 2 \rangle$.
- (g) No ✓: $\langle 0, 1 \rangle = \langle 0, 2 \rangle$ in $\mathbb{R}$ equipped with the standard inner product.
- (h) True ✓, since $\langle y, y \rangle = 0$.



Exercise 6.1.4.

- (a) Complete the proof in Example 5 that $\langle \cdot, \cdot \rangle$ is an inner product (the Frobenius inner product) on $M_{n \times n}(\mathbb{K})$.
- (b) Use the Frobenius inner product to compute $\|A\|$, $\|B\|$, and $\langle A, B \rangle$ for

$$A = \begin{pmatrix} 1 & 2+i \\ 3 & i \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1+i & 0 \\ i & -i \end{pmatrix}.$$

Proof.

- (a) Trivial.
- (b) Since

$$\langle A, A \rangle = \text{tr} \left(\begin{pmatrix} 1 & 3 \\ 2-i & -i \end{pmatrix} \begin{pmatrix} 1 & 2+i \\ 3 & i \end{pmatrix} \right) = \text{tr} \begin{pmatrix} 10 & 2+4i \\ 2-4i & 6 \end{pmatrix} = 16$$

and

$$\langle B, B \rangle = \text{tr} \left(\begin{pmatrix} 1-i & -i \\ 0 & i \end{pmatrix} \begin{pmatrix} 1+i & 0 \\ i & -i \end{pmatrix} \right) = \text{tr} \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} = 4,$$

we see that $\|A\| = 4$ and $\|B\| = 2$. Finally,

$$\langle A, B \rangle = \text{tr} \left(\begin{pmatrix} 1-i & -i \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 2+i \\ 3 & i \end{pmatrix} \right) = \text{tr} \begin{pmatrix} 1-4i & 4-i \\ 3i & -1 \end{pmatrix} = -4i.$$

- (b) Alternatively,

$$\langle A, A \rangle = \overline{\begin{pmatrix} 1 & 3 \end{pmatrix}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \overline{\begin{pmatrix} 2+i & i \end{pmatrix}} \begin{pmatrix} 2+i \\ i \end{pmatrix} = 10 + 6 = 16,$$

$$\langle B, B \rangle = \overline{\begin{pmatrix} 1+i & i \end{pmatrix}} \begin{pmatrix} 1+i \\ i \end{pmatrix} + \overline{\begin{pmatrix} 0 & -i \end{pmatrix}} \begin{pmatrix} 0 \\ -i \end{pmatrix} = 3 + 1 = 4,$$

$$\langle A, B \rangle = \overline{\begin{pmatrix} 1+i & i \end{pmatrix}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \overline{\begin{pmatrix} 0 & -i \end{pmatrix}} \begin{pmatrix} 2+i \\ i \end{pmatrix} = (1-4i) - 1 = -4i.$$

**Question.**

- (a) Let $A, B \in M_{n \times n}(\mathbb{K})$ and $\|\cdot, \cdot\|$ be a norm on $M_{n \times n}(\mathbb{K})$. When does it hold that $\|AB\| = \|A\|\|B\|$?
- (b) Let $(V, \|\cdot, \cdot\|, *)$ be a normed vector space with vectorial multiplication $*: V^2 \rightarrow V$. For $x, y \in V$, when does it hold that $\|x * y\| = \|x\|\|y\|$?

Question. For any $z, w \in \mathbb{C}$, we know $\overline{zw} = \bar{z} \cdot \bar{w}$. What about for $A, B \in M_{n \times n}(\mathbb{K})$? When is $\overline{AB} = \bar{A} \cdot \bar{B}$?

Proof. Always, since

$$\left(\overline{AB}\right)_{ij} := \overline{(AB)_{ij}} = \overline{\sum_{k=1}^n a_{ik}b_{kj}} = \sum_{k=1}^n \overline{a_{ik}b_{kj}} = \left(\overline{A} \cdot \overline{B}\right)_{ij}.$$



Exercise 6.1.8. Provide reasons why each of the following is not an inner product on the given vector spaces.

- (a) $\langle (a, b), (c, d) \rangle = ac - bd$ on $\mathbb{R}^2$.
- (b) $\langle A, B \rangle = \text{tr}(A + B)$ on $M_{2 \times 2}(\mathbb{R})$.
- (c) $\langle f(x), g(x) \rangle = \int_0^1 f'(t)g(t) dt$ on $P(\mathbb{R})$, where $'$ denotes differentiation.

Proof.

- (a) Notice $\langle e_2, e_2 \rangle = -1$.
- (b) Observe that $\langle -I, O \rangle = -2$.
- (c) Note that $\langle x, -1 \rangle = -1$.

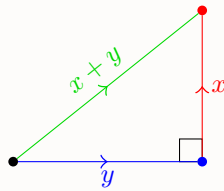


Exercise 6.1.10. Let V be an inner product space, and suppose that x and y are orthogonal vectors in V . Prove that $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. Deduce the Pythagorean theorem in $\mathbb{R}^2$. Visit goo.gl/1iTZzC for a solution.

Proof. By orthogonality, $\langle x, y \rangle = 0$ so

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, y \rangle + 2\Re\langle x, y \rangle = \|x\|^2 + \|y\|^2 + 0.$$

The Pythagorean theorem hence follows.



Question. Let V be a normed vector space, such that $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ for all orthogonal $x, y \in V$. Then, must $\|\cdot, \cdot\|$ be induced by some inner product $\langle \cdot, \cdot \rangle$ on V ?

Exercise 6.1.12. Let $\{v_1, v_2, \dots, v_k\}$ be an orthogonal set in V , and let $a_1, a_2, \dots, a_k$ be scalars. Prove that

$$\left\| \sum_{i=1}^k a_i v_i \right\|^2 = \sum_{i=1}^k |a_i|^2 \|v_i\|^2.$$

Proof. Notice that $u = \sum_{i=1}^{k-1} a_i v_i$ is orthogonal to v_k . Using exercise 10, the result hence follows from induction. 

Exercise 6.1.15.

- (a) Prove that if V is an inner product space, then $|\langle x, y \rangle| = \|x\| \|y\|$ iff one of the vectors x or y is a multiple of the other.
- (b) Derive a similar result for the equality $\|x + y\| = \|x\| + \|y\|$, and generalise it to the case of n vectors.

Proof.

- (a) In our proof of the Cauchy-Schwartz inequality, we saw that $|\langle x, y \rangle| = \|x\| \|y_{\perp}\| = \|x\| \sqrt{\|y\|^2 - \|y_{\parallel}\|^2}$. Hence, the given result is apparent.
- (b) ($\implies$) Clearly, if x and y are linearly dependent, then $\|x + y\| = \|x\| + \|y\|$.
 ($\impliedby$) Conversely, let $y_{\parallel} := \langle y, \hat{x} \rangle \hat{x}$ and $y_{\perp} := y - y_{\parallel}$:

$$\begin{aligned} \|x + y\| &= \|x\| + \|y\| \\ \sqrt{(\|x\| + \|y_{\parallel}\|)^2 + \|y_{\perp}\|^2} &= \|x\| + \sqrt{\|y_{\parallel}\|^2 + \|y_{\perp}\|^2} \text{ by orthogonality and } (\implies) \\ \|x\| \|y_{\perp}\| &= 0 \text{ by squaring and simplifying.} \end{aligned}$$

Hence, x and y are linearly independent iff $\|x + y\| = \|x\| + \|y\|$. The trivial generalisation is $\|\sum_{k=1}^n x_k\| = \sum_{k=1}^n \|x_k\|$ iff $\dim \text{span}\{x_1, x_2, \dots, x_n\} = 1$.

× see the correct statement and proof.



The following is inspired by exercise 25.

Observation. Let $\langle \cdot, \cdot \rangle$ be an inner product on V , and $f: V \rightarrow \mathbb{R}_0^+$ be such that $f(x) > 0$ if $x \neq 0$. Then, $\langle x, y \rangle_f := f(x) \langle x, y \rangle$ is an inner product.

Question. Let $[\cdot, \cdot]$ be a real inner product on the space V over $\mathbb{C}$, such that $[x, ix] = 0$ for all $x \in V$. Is $[\cdot, \cdot]$ unique up to a scalar multiple?

Exercise 6.1.17. Let T be a linear operator on an inner product space V , and suppose that $\|T(x)\| = \|x\|$ for all x . Prove that T is injective.

Proof. If $T(x) = T(y)$, then $\|x - y\| = \|T(x - y)\| = \|T(x) - T(y)\| = 0$. 

Exercise 6.1.18. Let V and W be vector spaces over $\mathbb{K}$, and $\langle \cdot, \cdot \rangle$ be an inner product on W . If $T: V \rightarrow W$ is linear, prove that $\langle x, y \rangle' = \langle T(x), T(y) \rangle$ defines an inner product on V iff T is injective.

Proof. Suppose $\langle x, y \rangle'$ is an inner product on V . Then, $\langle T(x), T(x) \rangle = \langle x, x \rangle' > 0$ for all $x \neq 0$ implies $\text{nullity}(T) = 0$. Hence, T is injective. Conversely, consider injective T . The linearity of $\langle \cdot, \cdot \rangle'$ in the first coordinate is clear from that of $\langle \cdot, \cdot \rangle$ and the linearity of T . Similarly, conjugate symmetry is inherited from $\langle \cdot, \cdot \rangle$. Finally, positive-definiteness: if $x \neq 0$, then $T(x) \neq 0$ so $\langle T(x), T(x) \rangle > 0$. Hence, $\langle \cdot, \cdot \rangle'$ is indeed an inner product on V . 

Exercise 6.1.20. Let V be an inner product space over $\mathbb{K}$. Prove the *polar identities*:

For all $x, y \in V$,

$$(a) \quad \langle x, y \rangle = \frac{\|x + y\|^2 - \|x - y\|^2}{4} \text{ if } \mathbb{K} = \mathbb{R};$$

$$(b) \quad \langle x, y \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2 \text{ if } \mathbb{K} = \mathbb{C}, \text{ where } i^2 = -1.$$

Proof.

(a) We see that

$$\begin{aligned} \frac{\|x + y\|^2 - \|x - y\|^2}{4} &= \frac{\|x\|^2 + 2\Re\langle x, y \rangle + \|y\|^2 - \|x\|^2 + 2\Re\langle x, y \rangle - \|y\|^2}{4} \\ &= \Re\langle x, y \rangle. \end{aligned}$$

The result follows from $\mathbb{K} = \mathbb{R}$.

(b) Expanding,

$$\begin{aligned} \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2 &= \frac{1}{4} \sum_{k=1}^4 i^k (\|x\|^2 + 2\Re(i^k \langle x, y \rangle) + \|y\|^2) \\ &= \frac{2i\Re(-i\langle x, y \rangle) - 2\Re(-\langle x, y \rangle) - 2i\Re(i\langle x, y \rangle) + 2\Re\langle x, y \rangle}{4} \\ &= \frac{2i\Im\langle x, y \rangle + 2\Re\langle x, y \rangle + 2i\Im\langle x, y \rangle + 2\Re\langle x, y \rangle}{4} \\ &= \langle x, y \rangle. \end{aligned}$$



Exercise 6.1.21. Let A be an $n \times n$ matrix. Define

$$A_1 := \frac{1}{2}(A + A^*) \quad \text{and} \quad A_2 := \frac{1}{2i}(A - A^*).$$

- (a) Prove that $A_1^* = A_1$, and $A_2^* = A_2$, and $A = A_1 + iA_2$. Would it be reasonable to define A_1 and A_2 to be the real and imaginary parts, respectively, of the matrix A ?
- (b) Let A be an $n \times n$ matrix. Prove that the representation in (a) is unique. That is, prove that if $A = B_1 + iB_2$, where $B_1^* = B_1$ and $B_2^* = B_2$, then $B_1 = A_1$ and $B_2 = A_2$.

Proof.

(a) Notice that

$$A_1^* = \frac{1}{2}(A + A^*)^* = \frac{1}{2}(A^* + A) = A_1,$$

$$A_2^* = \frac{1}{2i}(A - A^*)^* = -\frac{1}{2i}(A^* - A) = A_2,$$

Furthermore, $A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A_1 + iA_2$.

Defined relative to a conjugation operator $*$: $X \rightarrow X$, any real and imaginary part — x_1 and x_2 — of a ‘complex object’ $x \in X$ must have the quintessential property that $x^* = x_1 - ix_2$. Indeed, the given definitions for A_1 and A_2 (uniquely) fulfil this criterion, for the conjugate transpose $*$. Contrary to expectations from complex arithmetic, the entries of A_1 and A_2 need not be real (consider $A = iE^{12}$ as an example). From (b), we see that this is inevitable. But this poses no issue: we can simply treat Hermitian and skew-Hermitian matrices as the matrix-equivalents of real and imaginary numbers.

(b) Evaluating $A + A^*$ and $A - A^*$ gives $A_1 = B_1$ and $A_2 = B_2$, respectively.



Note. I asked for feedback in math discord, and eigentaylor gave me the [following tips](#) .

- Hermitian matrices ($A^* = A$) are like real numbers.
- Skew-Hermitian matrices ($A^* = -A$) are like imaginary numbers.
- Unitary matrices ($A^* = A^{-1}$) are complex numbers lying on the unit circle.

Also, the conjugate transpose is the ‘true’ matrix transpose — results that hold true for the transpose of real matrices hold true for the conjugate transpose of complex matrices. Here is my own example: For $A \in M_{n \times n}(\mathbb{K})$ and eigenvectors $x, y \in \mathbb{K}^n$ corresponding to λ and μ , such that $\lambda \neq \bar{\mu}$, notice that $\langle x, y \rangle = 0$ iff A is Hermitian.

Observation. Let $A \in M_{n \times n}(\mathbb{K})$.

A	iA
Hermitian	Skew-Hermitian
Skew-Hermitian	Hermitian
Unitary	Unitary

Exercise 6.1.23. Let $V = \mathbb{K}^n$ and $A \in M_{n \times n}(\mathbb{K})$.

- Prove that $\langle x, Ay \rangle = \langle A^*x, y \rangle$ for all $x, y \in V$.
- Suppose that for some $B \in M_{n \times n}(\mathbb{K})$, we have $\langle x, Ay \rangle = \langle Bx, y \rangle$ for all $x, y \in V$. Prove that $B = A^*$.

- (c) Let α be the standard ordered basis for V . For any orthonormal basis β for V , let Q be the $n \times n$ matrix whose columns are the vectors in β . Prove that $Q^* = Q^{-1}$.
- (d) Define linear operators T and U on V by $T(x) = Ax$ and $U(x) = A^*x$. Show that $[U]_\beta = [T]_\beta^*$ for any orthonormal basis β for V .

Note. V is equipped with the *standard inner product* throughout this exercise.

Proof.

- (a) Clearly, $\langle x, Ay \rangle := (Ay)^*x = y^*A^*x =: \langle A^*x, y \rangle$.
- (b) Since $\langle A^*x, y \rangle = \langle Bx, y \rangle$ by (a), $A^* = B$.
- (c) See the **proof** of exercise 6.2.11.
- (d) From (c) and the fact that $A = (A^*)^*$, it follows that

$$[I]_\beta^\alpha [A]_\beta [I]_\alpha^\beta = \left([I]_\beta^\alpha [A^*]_\beta [I]_\alpha^\beta \right)^* = [I]_\beta^\alpha [A^*]_\beta^* [I]_\alpha^\beta.$$

(Here, for clarity, we replace the induced linear transformations by the matrices that induce them.) Applying the appropriate inverses hence returns $[A]_\beta = [A^*]_\beta^*$.



Exercise 6.1.26. Prove that the following are norms on the vector space V .

- (a) $V = \mathbb{R}^2$; $\|(a, b)\| = |a| + |b|$ for all $(a, b) \in V$.
- (b) $V = C([0, 1])$; $\|f\| = \max_{t \in [0, 1]} |f(t)|$ for all $f \in V$.
- (c) $V = C([0, 1])$; $\|f\| = \int_0^1 |f(t)| dt$ for all $f \in V$.
- (d) $V = M_{m \times n}(\mathbb{K})$; $\|A\| = \max_{i,j} |A_{ij}|$ for all $A \in V$.

Proof.

- (a) (i) If $a, b \neq 0$, then $|a| > 0$ and $|b| > 0$ so $\|(a, b)\| > 0$.
- (ii) If $a, b = 0$, then $\|(a, b)\| = 0 + 0 = 0$.
- (iii) For any $c \in \mathbb{R}$, we have $\|c(a, b)\| = |ca| + |cb| = |c|(|a| + |b|) = |c|\|(a, b)\|$.
- (iv) $\|(a, b) + (c, d)\| = |a + c| + |b + d| \leq |a| + |c| + |b| + |d| = \|(a, b)\| + \|(c, d)\|$.
- (b) (i) If $f \neq 0$, then $f(t) \neq 0$ for some $t \in [0, 1]$ so $\|f\| \geq |f(t)| \geq 0$.
- (ii) If $f = 0$, then $\|f\| = \max\{0\} = 0$.
- (iii) For any $c \in \mathbb{R}$, we have $\|cf\| = \max|cf| = |c| \max|f| = |c|\|f\|$.
- (iv) $\|f + g\| = \max|f + g| \leq \max(|f| + |g|) = \max|f| + \max|g| = \|f\| + \|g\|$.
- (c) (i) Suppose $f(x) \neq 0$ for some x . Wlog, $x \geq 0$. Then, $y := \inf\{t : |f(t)| = \frac{1}{2}|f(x)|\} < x$. So, $\|f\| \geq \int_y^x \frac{1}{2}|f(x)| dt = \frac{1}{2}(x - y)|f(x)| > 0$.
- (ii) If $f = 0$, then $\|f\| = \int_0^1 0 dt = 0$.
- (iii) For any $c \in \mathbb{R}$, we have $\|cf\| = \int_0^1 |cf(t)| dt = \int_0^1 |c||f(t)| dt = |c|\|f\|$.
- (iv) As before, this follows from the absolute value satisfying the triangle inequality.
- (d) Similarly straightforward.



Exercise 6.1.27. Use Exercise 11 to show that there is no inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^2$, such that $\|x\|^2 = \langle x, x \rangle$ for all $x \in \mathbb{R}^2$ if the norm is defined as in exercise 26(a).

Proof. Since $\|e_1 + e_2\|^2 + \|e_1 - e_2\|^2 = 8$ while $2\|e_1\|^2 + 2\|e_2\|^2 = 4$, this norm is not induced by any inner product. 

Question. Let V be a vector space with basis $\beta := \{v_\alpha\}$ and norm $\|\cdot, \cdot\|$. Fix $p \geq 1$. Consider the β^p -norm $\|\sum c_\alpha v_\alpha\|_* := \sum |c_\alpha| \|v_\alpha\|^p$. When is it induced by an inner product?

Exercise 6.1.29. Let $\|\cdot, \cdot\|$ be a norm on a real vector space V satisfying the parallelogram law given in Exercise 11. Define

$$\langle x, y \rangle := \frac{1}{4} [\|x + y\|^2 - \|x - y\|^2].$$

Prove that $\langle \cdot, \cdot \rangle$ defines an inner product on V such that $\|x\|^2 = \langle x, x \rangle$ for all $x \in V$.

Exercise 6.1.30. Let $\|\cdot, \cdot\|$ be a norm (as defined on page 337) on a complex vector space V satisfying the parallelogram law was given in Exercise 11. Prove that there is an inner product $\langle \cdot, \cdot \rangle$ on V , such that $\|x\|^2 = \langle x, x \rangle$ for all $x \in V$.

Proof. Consider the inner product $\langle x, y \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2$. (Work in progress) 

Corollary. A norm is induced by an inner product iff it satisfies the parallelogram law.

6.2 Gram-Schmidt Orthogonalisation Process

6.2.1 Theorems

From the name itself, we can guess that this process allows us to constructively obtain an orthogonal basis for an inner product space V , from an arbitrary basis for V . This can be easily done with the idea of projections using the inner product associated with V , giving us the following guess on the *Gram-Schmidt Orthogonalisation Process*:

Theorem. Consider a basis $\beta = \{v_1, v_2, \dots, v_n\}$ for an inner product space V . Define $u_1 := v_1$ and

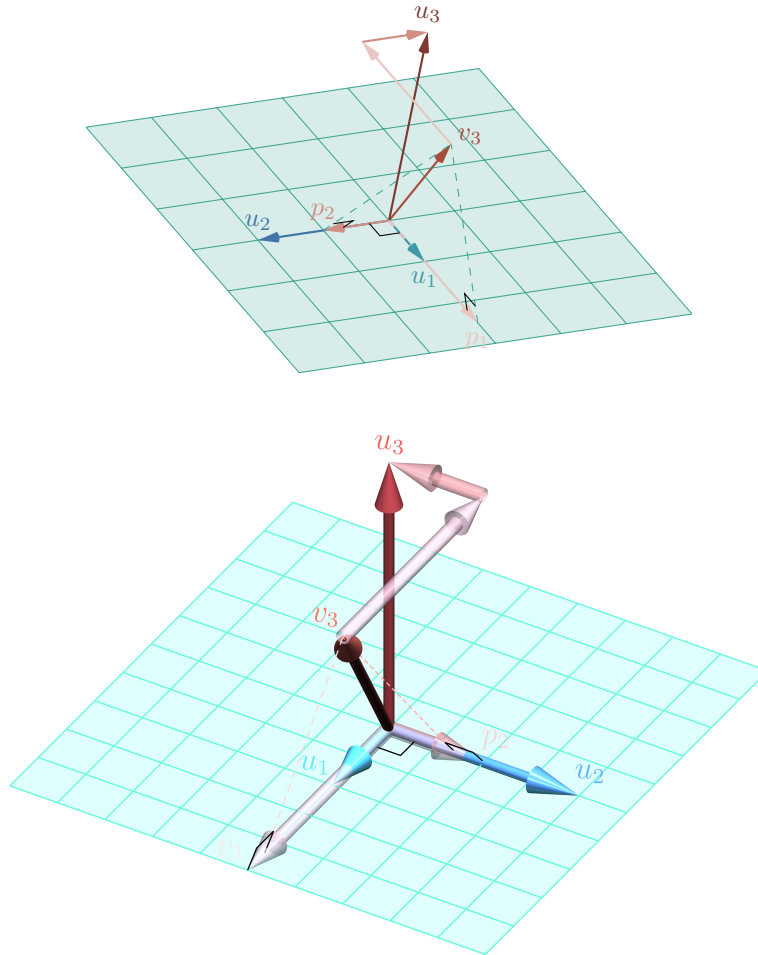
$$u_{j+1} := v_{j+1} - \sum_{k=1}^j \langle v_{j+1}, \hat{u}_k \rangle \hat{u}_k,$$

where $\hat{u}_k := \|u_k\|^{-1} u_k$. Then, $\beta_\perp := \{u_1, u_2, \dots, u_n\}$ is an orthogonal basis for V .

Proof. Clearly, $v - \langle v, \hat{u} \rangle \hat{u}$ is orthogonal to u , for any $u, v \in V$. Extending this idea, we have that $\beta_\perp$ is indeed orthogonal:

$$\langle u_i, u_j \rangle = \langle u_i, u_j - \langle u_j, \hat{u}_i \rangle \hat{u}_i \rangle - \sum_{\substack{k=1 \\ k \neq i}}^{j-1} \overline{\langle u_j, \hat{u}_k \rangle} \langle u_i, \hat{u}_k \rangle = 0 - \sum_{\substack{k=1 \\ k \neq i}}^{j-1} \overline{\langle u_j, \hat{u}_k \rangle} \cdot 0 = 0,$$

where $i < j$ wlog, and if $\{u_1, u_2, \dots, u_j\}$ is orthogonal. Furthermore, orthogonal sets are linearly independent, lest $u_j = \sum_{k=1}^n c_k u_k$ for some j and scalars c_k — (at least) one of which, c_ℓ , is nonzero. But this means u_j is not orthogonal to u_ℓ , a contradiction. So, $\beta_\perp$ must indeed be an orthogonal basis for V . 



Theorem 6.6. Let W be a finite-dimensional subspace of an inner product space V , and let $y \in V$. Then there exist unique vectors $u \in W$ and $z \in W^\perp$ such that $y = u + z$. Furthermore, if $\{v_1, v_2, \dots, v_k\}$ is an orthonormal basis for W , then $u = \sum_{i=1}^k \langle y, v_i \rangle v_i$.

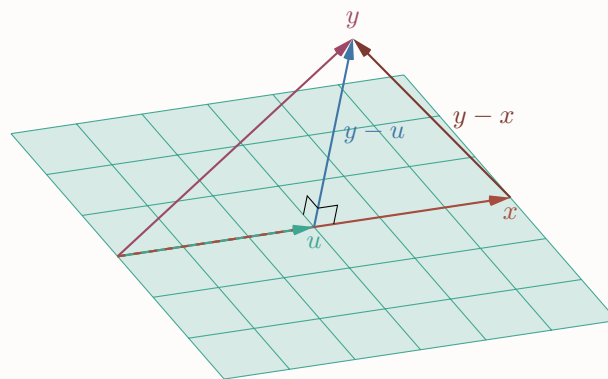
Proof. Since W is finite-dimensional, it has an orthonormal basis $\{v_1, v_2, \dots, v_k\}$. We see that $\pi_W(y) := \sum_{i=1}^k \langle y, v_i \rangle v_i \in W$ and $y - \pi_W(y) \in W^\perp$ sum to y . Moreover,

it is clear that $W \cap W^\perp = \{0\}$ so $V = W \oplus W^\perp$. Theorem 5.9 now implies the uniqueness of this representation. 

Corollary. In the notation of Theorem 6.6, the vector u is the unique vector in W that is “closest” to y ; that is, for any $x \in W$, it holds that $\|y - x\| \geq \|y - u\|$, and this inequality is an equality iff $x = u$.

Proof. Since $y - x$ can be uniquely written as the sum of the orthogonal vectors $u - x \in W$ and $y - u \in W^\perp$,

$$\|y - x\|^2 = \|u - x\|^2 + \|y - u\|^2 \geq \|y - u\|^2.$$



6.2.2 Exercises

Exercise 6.2.1. Label the following statements as true or false.

- (a) The Gram-Schmidt orthogonalisation process produces an orthonormal set from an arbitrary linearly independent set.
- (b) Every nonzero finite-dimensional inner product space has an orthonormal basis.
- (c) The orthogonal complement of any set is a subspace.
- (d) If $\{v_1, v_2, \dots, v_n\}$ is a basis for an inner product space V , then for any $x \in V$ the scalars $\langle x, v_i \rangle$ are the Fourier coefficients of x .
- (e) An orthonormal basis must be an ordered basis.
- (f) Every orthogonal set is linearly independent.
- (g) Every orthonormal set is linearly independent.

Proof.

- (a) False ✓; it produces an orthogonal set. Though, of course, this orthogonalisation process can be easily modified to become an orthonormalisation

process.

- (b) True ✓.
- (c) True ✓. Given any subset S of an inner product space V , we have $0 \in S^\perp$ and $\langle cu_1 + u_2, v \rangle = c\langle u_1, v \rangle + \langle u_2, v \rangle = 0$ — for all $u_1, u_2 \in S$ and $v \in V$ and scalars $c \in \mathbb{K}$.
- (d) No ✓. The basis $\{v_1, v_2, \dots, v_n\}$ must be orthonormal for the scalars $\langle x, v_i \rangle$ to be Fourier coefficients of x .
- (e) True ✓.
- (f) True. ✗ $\{0\}$ is not linearly independent.
- (g) True ✓.



Exercise 6.2.2. In each part, apply the Gram-Schmidt process to the given subset S of the inner product space V to obtain an orthonormal basis β for $\text{span}(S)$, and compute the Fourier coefficients of the given vector relative to β . Finally, use Theorem 6.5 to verify your result.

- (b) $V = \mathbb{R}^3$, $S = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$, and $x = (1, 0, 1)$.
- (k) $V = \mathbb{C}^4$, $S = \{(-4, 3 - 2i, i, 1 - 4i), (-1 - 5i, 5 - 4i, -3 + 5i, 7 - 2i), (-27 - i, -7 - 6i, -15 + 25i, -7 - 6i)\}$, and $x = (-13 - 7i, -12 + 3i, -39 - 11i, -26 + 5i)$.
- (m) $V = M_{2 \times 2}(\mathbb{C})$, $S = \left\{ \begin{pmatrix} -1 + i & -i \\ 2 - i & 1 + 3i \end{pmatrix}, \begin{pmatrix} -1 - 7i & -9 - 8i \\ 1 + 10i & -6 - 2i \end{pmatrix}, \begin{pmatrix} -11 - 132i & -34 - 31i \\ 7 - 126i & -71 - 5i \end{pmatrix} \right\}$, and $A = \begin{pmatrix} -7 + 5i & 3 + 18i \\ 9 - 6i & -3 + 7i \end{pmatrix}$.

Proof.

- (b) Using the Gram-Schmidt process with $v_1 := e_3$:

$$v_2 = (0, 1, 1) - \langle (0, 1, 1), e_3 \rangle e_3 = (0, 1, 1) - 1e_3 = e_2$$

and

$$v_3 = (1, 1, 1) - \langle (1, 1, 1), e_3 \rangle e_3 - \langle (1, 1, 1), e_2 \rangle e_2 = (1, 1, 1) - 1e_3 - 1e_2 = e_1.$$

So, $\beta := \{v_1, v_2, v_3\}$ is an orthonormal basis for $\text{span}(S)$. Hence, the Fourier coefficients of x relative to β are 1, 0, 1.

(k) Using the Gram-Schmidt process with $v_1 := (-4, 3 - 2i, i, 1 - 4i)$:

$$\begin{aligned}
 v_2 &= \begin{pmatrix} -1 - 5i \\ 5 - 4i \\ -3 + 5i \\ 7 - 2i \end{pmatrix} - \left\langle \begin{pmatrix} -1 - 5i \\ 5 - 4i \\ -3 + 5i \\ 7 - 2i \end{pmatrix}, \widehat{\begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix}} \\
 &= \begin{pmatrix} -1 - 5i \\ 5 - 4i \\ -3 + 5i \\ 7 - 2i \end{pmatrix} - \frac{(-1 - 5i)(\overline{-4}) + (5 - 4i)(\overline{3 - 2i}) + \dots}{|-4|^2 + |3 - 2i|^2 + |i|^2 + |1 - 4i|^2} \begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix} \\
 &= \begin{pmatrix} -1 - 5i \\ 5 - 4i \\ -3 + 5i \\ 7 - 2i \end{pmatrix} + (-1 - i) \begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix} \\
 &= \begin{pmatrix} 3 - i \\ -5i \\ -2 + 4i \\ 2 + i \end{pmatrix}
 \end{aligned}$$

and

$$\begin{aligned}
 v_3 &= \begin{pmatrix} -27 - i \\ -7 - 6i \\ -15 + 25i \\ -7 - 6i \end{pmatrix} - \left\langle \begin{pmatrix} -27 - i \\ -7 - 6i \\ -15 + 25i \\ -7 - 6i \end{pmatrix}, \widehat{\begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix}} \\
 &\quad - \left\langle \begin{pmatrix} -27 - i \\ -7 - 6i \\ -15 + 25i \\ -7 - 6i \end{pmatrix}, \widehat{\begin{pmatrix} 3 - i \\ -5i \\ -2 + 4i \\ 2 + i \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} 3 - i \\ -5i \\ -2 + 4i \\ 2 + i \end{pmatrix}} \\
 &= \begin{pmatrix} -27 - i \\ -7 - 6i \\ -15 + 25i \\ -7 - 6i \end{pmatrix} - \frac{(-27 - i)(\overline{-4}) + (-7 - 6i)(\overline{3 - 2i}) + \dots}{|-4|^2 + |3 - 2i|^2 + |i|^2 + |1 - 4i|^2} \begin{pmatrix} -4 \\ 3 - 2i \\ i \\ 1 - 4i \end{pmatrix} \\
 &\quad - \frac{(-27 - i)(\overline{3 - i}) + (-7 - 6i)(\overline{-5i}) + \dots}{|3 - i|^2 + |-5i|^2 + |-2 + 4i|^2 + |2 + i|^2} \begin{pmatrix} 3 - i \\ -5i \\ -2 + 4i \\ 2 + i \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
&= \begin{pmatrix} -27-i \\ -7-6i \\ -15+25i \\ -7-6i \end{pmatrix} - (3-i) \begin{pmatrix} -4 \\ 3-2i \\ i \\ 1-4i \end{pmatrix} - (1-i) \begin{pmatrix} 3-i \\ -5i \\ -2+4i \\ 2+i \end{pmatrix} \\
&= \begin{pmatrix} -17-i \\ -9+8i \\ -18+16i \\ -9+8i \end{pmatrix}.
\end{aligned}$$

So, we obtain the orthonormal basis

$$\begin{aligned}
\beta &= \{\hat{v}_1, \hat{v}_2, \hat{v}_3\} \\
&= \left\{ \frac{1}{\sqrt{47}} \begin{pmatrix} -4 \\ 3-2i \\ i \\ 1-4i \end{pmatrix}, \frac{1}{2\sqrt{15}} \begin{pmatrix} 3-i \\ -5i \\ -2+4i \\ 2+i \end{pmatrix}, \frac{1}{2\sqrt{290}} \begin{pmatrix} -17-i \\ -9+8i \\ -18+16i \\ -9+8i \end{pmatrix} \right\}
\end{aligned}$$

for $\text{span}(S)$. The Fourier coefficients of x relative to β are hence

$$\begin{aligned}
\langle x, \hat{v}_1 \rangle &= \frac{1}{\sqrt{47}} \left\langle \begin{pmatrix} -13-7i \\ -12+3i \\ -39-11i \\ -26+5i \end{pmatrix}, \begin{pmatrix} -4 \\ 3-2i \\ i \\ 1-4i \end{pmatrix} \right\rangle = -\sqrt{47}(1+i), \\
\langle x, \hat{v}_2 \rangle &= \frac{1}{2\sqrt{15}} \left\langle \begin{pmatrix} -13-7i \\ -12+3i \\ -39-11i \\ -26+5i \end{pmatrix}, \begin{pmatrix} 3-i \\ -5i \\ -2+4i \\ 2+i \end{pmatrix} \right\rangle = -\frac{30}{\sqrt{15}}(1-2i) \\
\langle x, \hat{v}_3 \rangle &= \frac{1}{2\sqrt{290}} \left\langle \begin{pmatrix} -13-7i \\ -12+3i \\ -39-11i \\ -26+5i \end{pmatrix}, \begin{pmatrix} -17-i \\ -9+8i \\ -18+16i \\ -9+8i \end{pmatrix} \right\rangle = \frac{580}{\sqrt{290}}(1+i).
\end{aligned}$$



Exercise 6.2.9. Let $W := \text{span}(i, 0, 1)$ in $\mathbb{C}^3$. Find orthonormal bases for W and $W^\perp$.

Proof. Clearly, $\beta := \{(i, 0, 1)\}$ is an orthonormal basis for W . We use the Gram-

Schmidt process on $\{v_1 = (i, 0, 1), e_1, e_2\}$:

$$v_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \widehat{\begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}} = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix},$$

$$v_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \left\langle \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \widehat{\begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}} - \left\langle \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \widehat{\begin{pmatrix} 1/2 \\ 0 \\ i/2 \end{pmatrix}} \right\rangle \widehat{\begin{pmatrix} 1/2 \\ 0 \\ i/2 \end{pmatrix}} = e_2.$$

Thus, an orthonormal basis for $W^\perp$ is

$$\gamma := \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$



Exercise 6.2.10. Let W be a finite-dimensional subspace of an inner product space V . Prove that $V = W \oplus W^\perp$. Using the definition on page 76, prove that there exists a projection T on W along $W^\perp$ that satisfies $N(T) = W^\perp$. In addition, prove that $\|T(x)\| \leq \|x\|$ for all $x \in V$.

Proof. See theorem 6.6 for a proof of $V = W \oplus W^\perp$. That is, each vector $x \in V$ has a unique decomposition $w + \mathfrak{s}$, for some $w \in W$ and $\mathfrak{s} \in W^\perp$. Hence, the projection of V on W along $W^\perp$ is the required projection T . Lastly, by orthogonality,

$$\|x\|^2 = \|w\|^2 + \|\mathfrak{s}\|^2 \geq \|w\|^2 = \|T(x)\|^2.$$



Exercise 6.2.11. Let A be an $n \times n$ matrix with complex entries. Prove that $AA^* = I$ iff the rows of A form an orthonormal basis for $\mathbb{C}^n$. Visit goo.gl/iKcC4S for a solution.

Proof. Let $x \in \mathbb{C}^n$ be the i th row of A , so $(AA^*)_{ij} = x_i x_j^* = \langle x_i, x_j \rangle$. The biconditional hence holds.

$$\begin{pmatrix} \begin{array}{cccc} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{array} \end{pmatrix} \begin{pmatrix} \begin{array}{cccc} \overline{A_{11}} & \overline{A_{21}} & \cdots & \overline{A_{n1}} \\ \overline{A_{12}} & \overline{A_{22}} & \cdots & \overline{A_{n2}} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{A_{1n}} & \overline{A_{2n}} & \cdots & \overline{A_{nn}} \end{array} \end{pmatrix} = \begin{pmatrix} \begin{array}{cccc} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{array} \end{pmatrix}$$



Exercise 6.2.12. Prove that for any matrix $A \in M_{m \times n}(\mathbb{K})$, it holds that $(R(L_{A^*}))^\perp = N(L_A)$.

Proof. From exercise 6.1.23, we see that $\langle A^*x, y \rangle = \langle x, Ay \rangle = \langle x, 0 \rangle = 0$ for all $x \in \mathbb{C}^m$ and $y \in N(L_A)$. 

Exercise 6.2.13. Let V be an inner product space, S and S_0 be subsets of V , and W be a finite dimensional subspace of V . Prove the following results.

- (a) $S_0 \subseteq S$ implies that $S^\perp \subseteq S_0^\perp$.
- (b) $S \subseteq (S^\perp)^\perp$; so $\text{span}(S) \subseteq (S^\perp)^\perp$.
- (c) $V = (W^\perp)^\perp$
- (d) $V = W \oplus W^\perp$.

Proof.

- (a) If $v \in S^\perp$ and $u_0 \in S_0$, then $u_0 \in S$ implies $\langle v, u_0 \rangle = 0$. That is, $v \in S_0^\perp$.
- (b) By definition, $S \subseteq (S^\perp)^\perp$.
- (c) Since $V = W \oplus W^\perp$ when W is finite dimensional, each $y \in (W^\perp)^\perp$ can be decomposed into the sum of some $w \in W$ and $x \in W^\perp$. So,

$$\begin{aligned} \langle y, x \rangle &= 0 \\ \langle w, x \rangle + \langle x, x \rangle &= 0 \\ \langle x, x \rangle &= 0 \quad \text{by orthogonality} \\ x &= 0. \end{aligned}$$

Thus, $y = w \in W$. Together with (b), we hence conclude $(W^\perp)^\perp = W$.

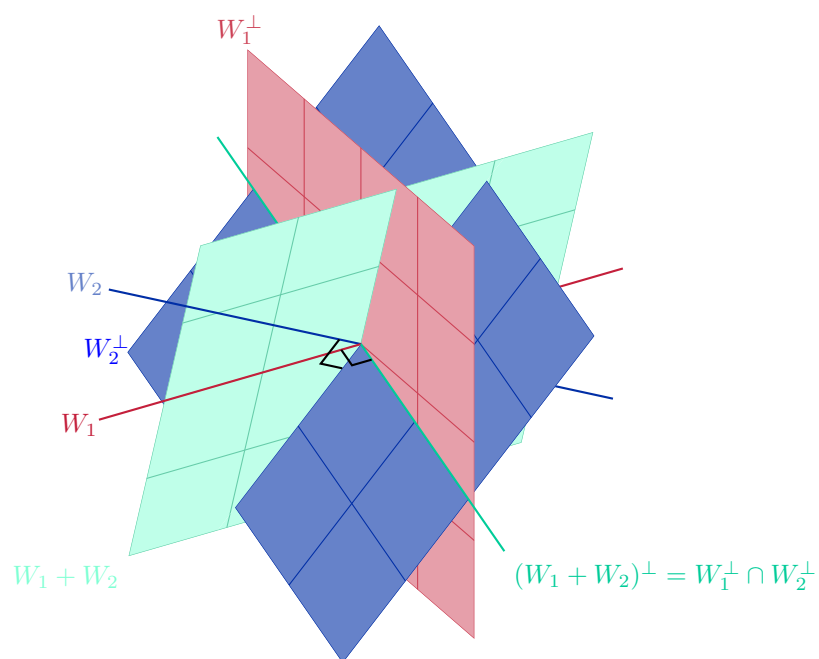
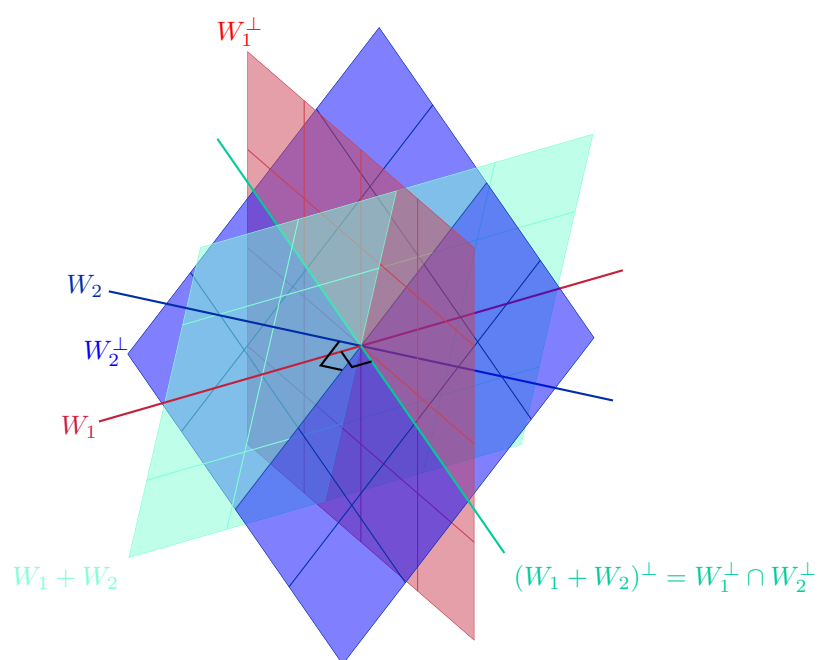
- (d) See the proof of theorem 6.6 

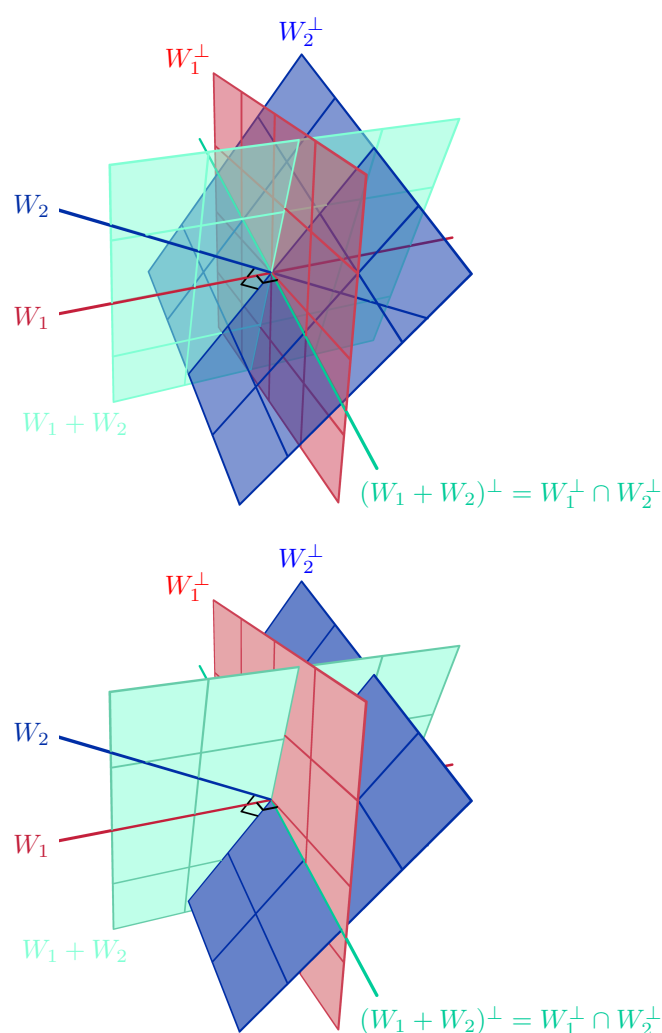
Exercise 6.2.14. Let W_1 and W_2 be subspaces of a finite-dimensional inner product space. Prove that $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$ and $(W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$.

Proof. Since $W_1, W_2 \subseteq W_1 + W_2$, the relation $(W_1 + W_2)^\perp \subseteq W_1^\perp \cap W_2^\perp$ follows straightforwardly. For the converse, notice that $\langle w_1 + w_2, v \rangle = \langle w_1, v \rangle + \langle w_2, v \rangle = 0 + 0 = 0$ for each $w_1 + w_2 \in W_1 + W_2$ and $v \in W_1^\perp \cap W_2^\perp$. Hence, $W_1^\perp \cap W_2^\perp \subseteq (W_1 + W_2)^\perp$; the equality holds. Furthermore, since the inner product space is finite-dimensional,

$$W_1^\perp + W_2^\perp = [(W_1^\perp + W_2^\perp)^\perp]^\perp = [(W_1^\perp)^\perp \cap (W_2^\perp)^\perp]^\perp = (W_1 \cap W_2)^\perp.$$







Note. The relation $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$ holds in all inner product spaces — finite or otherwise. But, $(W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$ does not hold generally in infinite inner product spaces. *insert example*

Exercise 6.2.15. Let V be a finite-dimensional inner product space over $\mathbb{K}$.

- (a) *Parseval's Identity.* Let $\{v_1, v_2, \dots, v_n\}$ be an orthonormal basis for V . For any $x, y \in V$ prove that

$$\langle x, y \rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

- (b) Use (a) to prove that

$$\langle \phi_\beta(x), \phi_\beta(y) \rangle' = \langle [x]_\beta, [y]_\beta \rangle' = \langle x, y \rangle,$$

for all $x, y \in V$ where $\langle \cdot, \cdot \rangle'$ is the standard inner product on $\mathbb{K}$.

Proof.

(a) By theorem 6.5,

$$\langle x, y \rangle = \left\langle x, \sum_{i=1}^n \langle y, v_i \rangle v_i \right\rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

(b) Since

$$[x]_{\beta} = \begin{pmatrix} \langle x, v_1 \rangle \\ \langle x, v_2 \rangle \\ \vdots \\ \langle x, v_n \rangle \end{pmatrix} \quad \text{and} \quad [y]_{\beta} = \begin{pmatrix} \langle y, v_1 \rangle \\ \langle y, v_2 \rangle \\ \vdots \\ \langle y, v_n \rangle \end{pmatrix},$$

the result follows immediately, by the standard inner product's definition.



Exercise 6.2.18. Let $V = C([-1, 1])$. Suppose that W_e and W_o denote the subspaces of V consisting of the even and odd functions, respectively. (See exercise 22 of section 1.3.) Prove that $W_e^{\perp} = W_o$, where the inner product on V is defined by

$$\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt.$$

Proof. For each nonzero $f \in W_e$, continuity allows us to pick an interval $[a, b] \subseteq [0, 1]$ over which f is positive. We hence obtain the function $1_{[-b, -a] \cup [a, b]}$ whose inner product with f is positive: $\langle f, 1_{[-b, -a] \cup [a, b]} \rangle = 2 \int_a^b f(t) dt > 0$. Since each function $h \in C([-1, 1])$ can be decomposed into the sum of the odd and even functions

$$g(t) := \frac{h(t) - h(-t)}{2} \quad \text{and} \quad f(t) := \frac{h(t) + h(-t)}{2},$$

we conclude that $W_e^{\perp} \subseteq W_o$. Finally, observe that, for each $f \in W_e$ and $f \in W_o$,

$$\langle f, g \rangle = \int_0^1 -f(x) dx + \int_0^1 f(t)g(t) dt$$

via the substitution $x = -t$. Hence, $W_o \subseteq W_e^{\perp}$ where equality holds.



Exercise 6.2.19. In each of the following parts, find the orthogonal projection of the given vector on the given subspace W of the inner product space V .

(b) $V = \mathbb{R}^3$, $u = (2, 1, 3)$, and $W = \{(x, y, z) \mid x + 3y - 2z = 0\}$.

Proof.

(b) ✓ We see that

$$W = \left\{ \begin{pmatrix} 2z - 3y \\ y \\ z \end{pmatrix} \mid x, y \in \mathbb{R} \right\} = \text{R} \begin{pmatrix} 0 & -3 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so it is clear that $\beta := \left\{ \begin{pmatrix} -3 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} \right\}$ is a basis for W . Its orthogonalisation is $\beta' = \left\{ \begin{pmatrix} 2 & 0 & 1 \end{pmatrix}, \begin{pmatrix} -3 & 5 & 6 \end{pmatrix} \right\}$. As such, the orthogonal projection of u on W is

$$\begin{aligned} \pi_W(u) &= \left\langle \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\rangle \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \left\langle \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix} \right\rangle \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix} \\ &= \frac{(2)(2) + (3)(1)}{2^2 + 1^2} \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \frac{(2)(-3) + (1)(5) + (3)(6)}{(-3)^2 + 5^2 + 6^2} \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix} \\ &= \frac{1}{14} \begin{pmatrix} 29 \\ 17 \\ 40 \end{pmatrix} \end{aligned}$$



Exercise 6.2.20. In each part of exercise 6.2.19, find the distance from the given vector to the subspace W .

Proof.

- (b) ✓ The required distance is $\|u - \pi_W(u)\| = 1/\sqrt{14}$.



Exercise 6.2.22. Let $V = C([0, 1])$ with the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. Let W be the subspace spanned by the linearly independent set $\{t, \sqrt{t}\}$.

- (a) Find an orthonormal basis for W .
 (b) Let $h(t) = t^2$. Use the orthonormal basis obtained in (a) to obtain the “best” (closest) approximation of h in W .

Proof.

- (a) Orthogonalising $\sqrt{t}$ gives

$$\sqrt{t} - \langle \sqrt{t}, \hat{t} \rangle \hat{t} = \sqrt{t} - \frac{1}{\int_0^1 x^2 dx} \left(\int_0^1 x^{3/2} dx \right) t = \sqrt{t} - \frac{6}{5}t.$$

Now, we normalise $\{t, \sqrt{t} - 6t/5\}$, giving us the orthonormal basis $\{t/\sqrt{3}, \sqrt{2}(5\sqrt{t} - 6t)\}$ for W .

- (b) The best approximation of h in W is simply its orthogonal projection on W , namely

$$\pi_W(h) = \frac{1}{3} \langle h, t \rangle t + 2 \langle h, 5\sqrt{t} - 6t \rangle (5\sqrt{t} - 6t) = \frac{45}{28}t - \frac{5}{7}\sqrt{t}.$$



Question. How does the inner product $\langle f, g \rangle := \int_0^1 f(t)g(t) dt$ give rise to a measure of the “distance” between two Riemann integrable functions $f, g: [0, 1] \rightarrow \mathbb{R}$ that is reasonable—with respect to the standard Euclidean view of $\mathbb{R}^2$?

Proof. We can define the distance between two Riemann integrable functions $f, g: [0, 1] \rightarrow \mathbb{R}$ as $d(f, g) := \|f - g\| = \sqrt{\int_0^1 (f - g)^2(t) dt}$. Being a norm-induced metric makes d a promising notion of distance between two Riemann integrable functions, and so does its parallel to the Euclidean distance $d(x, y) = \sqrt{(x - y)^2}$.



Exercise 6.2.23. Let V be the vector space defined in example 5 of section 1.2, the space of all sequences σ in $\mathbb{K}$ such that $\sigma(n) \neq 0$ for only finitely many positive integers n . For $\sigma, \mu \in V$, we define $\langle \sigma, \mu \rangle = \sum_{n=1}^{\infty} \sigma(n)\overline{\mu(n)}$. Since all but a finite number of terms of the series are zero, the series converges.

- (a) Prove that $\langle \cdot, \cdot \rangle$ is an inner product on V , and hence V is an inner product space.
- (b) For each positive integer n , let e_n be the sequence defined by $e_n(k) := \delta_{nk}$, where δ_{nk} is the Kronecker delta. Prove that $\{e_n\}_{n=1}^{\infty}$ is an orthonormal basis for V .
- (c) Let $\sigma_n = e_1 + e_n$ and $W = \text{span}\{\sigma_n \mid n \geq 2\}$.
 - (i) Prove that $e_1 \notin W$, so $W \neq V$.
 - (ii) Prove that $W^\perp = \{0\}$, and conclude that $W \neq (W^\perp)^\perp$.

Thus the assumption in exercise 6.2.13(c) that W is finite-dimensional is essential.

Proof.

- (a) (i) We see that

$$\begin{aligned} \langle z\sigma + \mu, \eta \rangle &= \sum (z\sigma(n) + \mu(n))\overline{\eta(n)} \\ &= z \sum \sigma(n)\overline{\eta(n)} + \sum \mu(n)\overline{\eta(n)} \\ &= z\langle \sigma, \eta \rangle + \langle \mu, \eta \rangle \end{aligned}$$

for each scalar $z \in \mathbb{K}$ and sequence $\sigma, \mu, \eta \in V$. So, linearity in the first coordinate holds.

- (ii) With equal simplicity, we verify the conjugate symmetry of $\langle \cdot, \cdot \rangle$:

$$\langle \sigma, \mu \rangle := \sum \sigma(n)\overline{\mu(n)} = \overline{\sum \mu(n)\overline{\sigma(n)}} = \overline{\langle \mu, \sigma \rangle}.$$

- (iii) Lastly, $\langle \cdot, \cdot \rangle$ is indeed positive-definite, since

$$\langle \sigma, \sigma \rangle = \sum \mu(n)\overline{\mu(n)} = \sum \|\mu(n)\|^2 > 0$$

for each nonzero $\sigma \in V$.

Hence, $\langle \cdot, \cdot \rangle$ is indeed an inner product on V .

- (b) Trivial.
- (c) (i) We see that any nonzero linear combination of $\{\sigma_n\}_{n=1}^\infty$ will have a nonzero entry in some ' n th coordinate', where $n \neq 1$. So, $e_1 \notin W$.
- (ii) Each sequence $\mu \in V$ can be decomposed into the form

$$z_1 e_{n_1} + z_2 e_{n_2} + \cdots + z_m e_{n_m},$$

for some nonzero $z_1, z_2, \dots, z_m \in \mathbb{K}$. Wlog $v \notin \text{span}\{e_1\}$, meaning $e_{n_k} \neq e_1$ for some $1 \leq k \leq m$. Hence

$$\langle \mu, z_k(\sigma_{n_k} - \sigma_{n_m+1}) \rangle = \langle \mu, z_k(e_{n_k} - e_{n_m+1}) \rangle = \|z_k\|^2 \neq 0.$$

That is, $\mu \notin W^\perp$. We conclude $W^\perp = \{0\}$, and consequently, $(W^\perp)^\perp = V \neq W$.



Question. If W is a subspace of the inner product space V , such that $\dim(W) = \dim(V)$, then must $W = \{0\}$?

Question. From [Wikipedia](#) , we see that the double orthogonal complement $(W^\perp)^\perp$ of a subspace W of a Hilbert space is just the closure $\overline{W}$ of W . Can we generalise this result to the following? If W is a subspace of the inner product space V , then $(W^\perp)^\perp$ is the completion $\mathcal{C}(W)$ of W .

6.3 The Adjoint of A Linear Operator

6.3.1 Theorems

Theorem 6.7. Let V be a finite-dimensional inner product space over $\mathbb{K}$, and let $g: V \rightarrow \mathbb{K}$ be a linear transformation. Then there exists a unique vector $y \in V$ such that $g(x) = \langle x, y \rangle$ for all $x \in V$.

Proof. Pick an orthonormal basis $\{v_i\}_{i=1}^n$ for V and let $y := \sum_{i=1}^n \overline{g(v_i)} v_i$. Since $g(v_i) = \langle v_i, y \rangle$, it holds that $g(x) = \langle x, y \rangle$ for all $x \in V$. Moreover, the uniqueness of this vector y is self-explanatory.



Theorem 6.8. For each linear operator T on the finite-dimensional inner product space V , there is a unique linear operator T^* on V , such that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in V$.

Proof. Let $\beta := \{v_i\}_{i=1}^n$ be an orthonormal basis for V , and hence, define the adjoint

$T^*: V \rightarrow V$ of T by

$$T^*(v_j) := \sum_{i=1}^n ([T]_\beta^*)_{ij} v_i = \sum_{i=1}^n \overline{\langle T(v_i), v_j \rangle} v_i. \quad (6.1)$$

It is clear that $\langle T(v_j), v_i \rangle = \overline{\langle T^*(v_i), v_j \rangle} = \langle v_j, T^*(v_i) \rangle$. That is, $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in V$. Lastly, to verify uniqueness, suppose $T_*: V \rightarrow V$ is a linear transformation satisfying $\langle T(x), y \rangle = \langle x, T_*(y) \rangle$. Then, $\langle x, T_*(y) \rangle = \langle x, T^*(y) \rangle$ for every $x, y \in V$ implies $T_*(y) = T^*(y)$ for each $y \in V$. Hence $T_* = T^*$ and uniqueness holds. 

Observation 6.9. The orthonormality of β cannot be replaced with the looser criterion of orthogonality. If β were merely orthogonal and T^* were defined as above, we would obtain the identity

$$\langle v_j, T^*(v_i) \rangle = \frac{\langle T(v_j), v_i \rangle}{\|v_i\|^2} \neq \langle T(v_j), v_i \rangle.$$

Theorem 6.10. Let β be an orthonormal basis for the finite-dimensional inner product space V . If T is a linear operator on V , then $[T^*]_\beta = [T]_\beta^*$.

Proof. This is clear from equation 6.1. 

Theorem 6.11. Let V be an inner product space, and let T and U be linear operators on V whose adjoints exist. Then

- (a) $(T + U)^* = T^* + U^*$.
- (b) $cT^* = \bar{c}T^*$ for any $c \in \mathbb{K}$.
- (c) $(TU)^* = U^*T^*$.
- (d) $T^{**} = T$.
- (e) $I^* = I$.

Proof.

- (c) A slightly different proof from that of the author's. Pick a basis $\beta = \{v_i\}_1^n$ for V ; observe that

$$\begin{aligned} (TU)^*(v_j) &= \sum_{i=1}^n ([TU]_\beta^*)_{ij} v_i \\ &= \sum_{i=1}^n ([U]_\beta^* [T]_\beta^*)_{ij} v_i \\ &= \sum_{i=1}^n ([U^*]_\beta [T^*]_\beta)_{ij} v_i \\ &= \sum_{i=1}^n ([U^* T^*]_\beta)_{ij} v_i \\ &= U^* T^*(v_j). \end{aligned}$$

Hence, $(TU)^* = U^*T^*$.

Note. Of course, this proof works only if V is finite-dimensional—I thought the author only defined adjoints for finite-dimensional spaces at the time of writing.



Lemma 6.3. For each $A \in M_{m \times n}(\mathbb{K})$, we have $\text{rank}(A^*A) = \text{rank}(A)$.

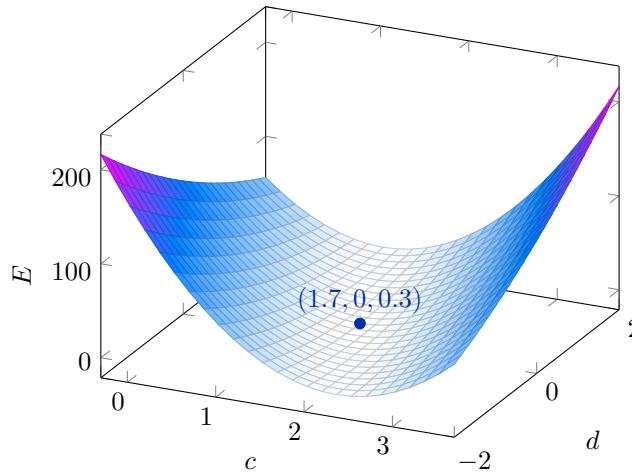
Proof. Pick an orthonormal basis $\{v_{k_i}\}_{i=1}^M$ for the row space of A and let

$$B := \begin{pmatrix} v_{k_1} \\ v_{k_2} \\ \vdots \\ v_{k_M} \end{pmatrix} \in M_{M \times n}(\mathbb{K}).$$

It follows, from (a slight generalisation of) exercise 6.2.11, that

$$\text{rank}(BB^*) = \text{rank}(I_M) = M = \text{rank}(A).$$

? In my handwritten brainstorming, I wrote $\text{rank}(AA^*) = \text{rank}(BB^*)$. But now I don't see a natural explanation; I'll leave this as something to revisit.



Theorem 6.13. Suppose $A \in M_{m \times n}(\mathbb{K})$ and $b \in \mathbb{K}^m$ are such that $Ax = b$ is consistent. Then the following results hold.

- (a) There exists a unique minimal solution s of $Ax = b$. In fact, $s \in R(L_{A^*})$.
- (b) The vector s is the only solution to $Ax = b$ that lies in $R(L_{A^*})$. That is, if $(AA^*)u = b$, then $s = A^*u$.

Proof.

- (a) The result is clear from $b = 0$, since $s = 0 \in R(L_{A^*})$. Now consider $b \neq 0$ and pick a solution $x \in \mathbb{K}^m$. Since $\mathbb{K}^m = N(L_A) \oplus R(L_{A^*})$ by exercise 6.2.12, the orthogonal projection $s = \pi_{R(L_{A^*})}(x)$ of x on $R(L_{A^*})$ is a solution.

Moreover, Theorem 3.9 implies the minimality and uniqueness of s .

- (b) Suppose that $AA^u = b$. We see that $(A(A^*u - s)) = 0$, so $A^*u - s \in N(L_A)$. Furthermore, $A^*u - s \in N(L_A)^\perp$ by linearity. Since $N(L_A) \cap N(L_A)^\perp$, it must be that $s = A^*u$.



6.3.2 Exercises

Exercise 6.3.1. Label the following statements as true or false. Assume that the underlying inner product spaces are finite-dimensional.

- (a) Every linear operator has an adjoint.
- (b) Every linear operator on V has the form $x \mapsto \langle x, y \rangle$ for some $y \in V$.
- (c) For every linear operator T on V and every ordered basis β for V , we have $[T^*]_\beta = ([T]_\beta)^*$.
- (d) The adjoint of a linear operator is unique.
- (e) For any operators T and U ; scalars a and b ,

$$(aT + bU)^* = aT^* + bU^*.$$

- (f) For any $n \times n$ matrix A , we have $(L_A)^* = L_A$.
- (g) For any linear operator T , we have $(T^*)^* = T$.

Proof.

- (a) True. ✓
- (b) False; it should be every linear functional (not operator). ✓
- (c) True. ✗ The basis β needs to be *orthonormal*, for the claim to hold in general.
- (d) False: $(aT + bU)^* = \bar{a}T^* + \bar{b}U^*$. ✓
- (e) True. ✓
- (f) True. ✓



Exercise 6.3.2. For each of the following inner product spaces V (over $\mathbb{K}$) and linear functional $\mathbf{g}: V \rightarrow \mathbb{K}$, find a vector y such that $\mathbf{g}(x) = \langle x, y \rangle$ for all $x \in V$.

- (b) $V = \mathbb{C}^2$ and $\mathbf{g}(z_1, z_2) = z_1 - 2z_2$.

Exercise 6.3.3. For each of the following inner product spaces V and linear operators T on V , evaluate T^* at the given vector in V .

- (c) $V = P_1(\mathbb{R})$ with

$$\langle f(x), g(x) \rangle := \int_0^1 f(t)g(t) dt, \quad T(f) := f' + 3f, \quad f(t) := 4 - 2t.$$

Exercise 6.3.7. Give an example of a linear operator T on an inner product space V such that $N(T) \neq N(T^*)$.

Proof. Let $T = L_{E^{12}}$ so

$$N(T) = \{(x \ 0) \mid x \in \mathbb{R}\} \quad \text{and} \quad N(T^*) = \{(x \ 0) \mid x \in \mathbb{R}\}.$$



Exercise 6.3.8. Let V be a finite-dimensional inner product space, and let T be a linear operator on V . Prove that if T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.

Proof. By the distributive property of the adjoint (theorem 6.11(c)),

$$T^*(T^{-1})^* = (T^{-1}T)^* = I_V^* = I_V$$

so $(T^*)^{-1} = (T^{-1})^*$.



Exercise 6.3.9. Prove that if $V = W \oplus W^\perp$ and T is the projection on W along $W^\perp$, then $T = T^*$.

Proof. Pick orthonormal bases $\alpha := \{w_k\}_1^n$ for W and $\beta := \{x_k\}_1^m$ for $W^\perp$; define $\gamma := \alpha \cup \beta$. Notice that

$$[T]_\gamma = \left(\begin{array}{ccc|ccc} & & & 0 & \cdots & 0 \\ & I_n & & \vdots & \ddots & \vdots \\ & & & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{array} \right).$$

So, $[T^*]_\gamma = [T]_\gamma^* = [T]_\gamma$. That is, $T^* = T$.



Exercise 6.3.10. Let T be a linear operator on an inner product space V . Prove that $\|T(x)\| = \|x\|$ for all $x \in V$ iff $\langle T(x), T(y) \rangle = \langle x, y \rangle$ for all $x, y \in V$.

Proof. If $\|T(x)\| = \|x\|$ for all $x \in V$ over $\mathbb{R}$, then by exercise 6.1.20(a)

$$\langle T(x), T(y) \rangle = \frac{1}{4}\|x+y\|^2 - \frac{1}{4}\|x-y\|^2 = \langle x, y \rangle.$$

The case of V being over $\mathbb{C}$ is similarly proven by using exercise 6.1.20(b).



Exercise 6.3.11. For a linear operator T on an inner product space V , prove that $T^*T = T_0$ implies $T = T_0$. Is the same result true if we assume that $TT^* = T_0$?

Proof. If $T^*T = T_0$, then

$$\langle T(x), T(x) \rangle = \langle x, T^*T(x) \rangle = \langle x, 0 \rangle = 0$$

for all $x \in V$. So, $T = T_0$; the same holds for $TT^* = T_0$, since $T^{**} = T$. 

Exercise 6.3.14. Let V be an inner product space and let $y, z \in V$. Define $T: V \rightarrow V$ by $T(x) := \langle x, y \rangle z$ for all $x \in V$. First prove that T is linear. Then show that T^* exists and find an explicit expression for it.

Proof. The linearity of T follows immediately from that of inner products in their first coordinate. Now define the linear operator U on V by $U(x) := \langle x, z \rangle y$. We see that

$$\langle x, U(w) \rangle = \langle z, w \rangle \langle x, y \rangle = \langle \langle x, y \rangle z, w \rangle = \langle T(x), w \rangle,$$

so $T^* = U$. 

Exercise 6.3.15. Let $T: V \rightarrow W$ be a linear transformation, where V and W are finite-dimensional inner product spaces with inner products $\langle \cdot, \cdot \rangle_V$ and $\langle \cdot, \cdot \rangle_W$, respectively. Prove the following results.

- (a) There is a unique adjoint T^* of T , which is linear.
- (b) If β and γ are orthonormal bases for V and W , respectively, then $[T^*]_\gamma^\beta = ([T]_\beta^\gamma)^*$.
- (c) $\text{rank}(T^*) = \text{rank}(T)$.
- (d) $\langle T^*(x), y \rangle_V = \langle x, T(y) \rangle_W$ for all $x \in W$ and $y \in V$.
- (e) For all $x \in V$, it holds that $T^*T = 0$ iff $T(x) = 0$.

Proof.

- (a) Pick orthonormal bases $\beta := \{v_k\}_1^n$ for V and $\gamma := \{w_k\}_1^m$ for W . Define $U: W \rightarrow V$ by

$$U(w_j) = \sum_{i=1}^n ([T]_\beta^\gamma)_{ij}^* v_i.$$

$$\text{So, } \langle v_i, U(w_j) \rangle_V = \overline{\langle U(w_j), v_i \rangle_V} = \overline{\langle T(v_i), w_j \rangle_W} = \langle T(v_i), w_j \rangle_W.$$

- (b)–(e) As done in (a), mirror the corresponding proofs of these facts done for $V = W$. 

Exercise 6.3.17. Let $T: V \rightarrow W$ be a linear transformation, where V and W are finite-dimensional inner product spaces. Prove that $R(T^*)^\perp = N(T)$, using the preceding definition.

Proof. We see that

$$\langle v, T^*w \rangle_V = \langle T(v), w \rangle_W = \langle 0, w \rangle_W = 0$$

for each $v \in N(T)$ and $w \in W$. That is, $N(T) \subseteq R(T^*)^\perp$. The converse follows similarly:

$$\langle u, T^*(w) \rangle_V = 0 \quad \text{implies} \quad \langle T(u), w \rangle_W = 0,$$

so $R(T^*)^\perp \subseteq N(T)$.



Exercise 6.3.21. In Physics, *Hooke's Law* states that (within certain limits) there is a linear relationship between the length x of a spring and the force y applied to (or exerted by) the spring. That is, $y = cx + d$, where c is called the *spring constant*. Use the following data to estimate the spring constant (the length is given in inches and the force is given in pounds).

Length	Force
x	y
3.5	1.0
4.0	2.2
4.5	2.8
5.0	4.3

Julia.

```

1  using LinearAlgebra
2
3  A = [7//2  1
4        ;4   1
5        ;9//2 1
6        ;5   1]
7
8  y = [1
9        ,11//5
10       ,14//5
11       ,43//10]
12
13 # The matrix division \ solves a system Ax = b, when A is
   ↪ invertible. In this case, it uses an LU decomposition (if A'A
   ↪ is not already diagonal/triangular) followed by forward and
   ↪ backward substitutions. So, we are only applying what has
   ↪ been proven in FIS; we aren't cheating.
14
15 x = (A'A)\A'y
16 println(x)
17 println(float(x))

```

Answer. The coefficient matrix that gives the least squares line is hence $x_0 = (2.1 \ -6.35)$. So, $c = 2.1$; the spring constant is approximately 2.1.



Exercise 6.3.22. Find the minimal solution to each of the following systems of linear equations.

$$\begin{aligned} x + y - z &= 0 \\ \text{(c) } 2x - y + z &= 3 \\ x - y + z &= 2 \end{aligned}$$

Julia.

```

1  using LinearAlgebra, LinearAlgebraX, Random
2
3  # Defining the system Ax = b
4      A = [1  1  -1
5           ;2  -1  1
6           ;1  -1  1]
7
8      b = [0
9           ,3
10          ,2]
11
12 # The rref of the augmented matrix (A*A'|b)
13     rref_augmented = rrefx([A*A' b])
14
15 # Defining M and N to be the number of rows and columns of A*A',
16   ↪ respectively
17     const M = size(A*A', 1)
18     const N = size(A*A', 2)
19
20 # The last column u of rref_augmented is a solution to the
21   ↪ corresponding system (A*A')u = b
22     u = rref_augmented[:, N+1]
23
24 # The minimal solution s to the system Ax = b
25     s = A'u

```

Answer. The minimal solution is $s = (1 \quad -1/2 \quad 1/2)$, which can be verified numerically. 

Julia.

```

1  # Correctness checks
2
3      # Check that (A * A')u = b
4      if (A * A')u == b
5          println("True: (A * A')u == b")
6      else println("False: (A * A')u == b")

```

```

7         end
8
9         # Computing the minimal solution s with random parameters to
10        ↪ verify its uniqueness
11        # Obtaining the basis {v_1,...,v_N} - {0} of the solution
12        ↪ space of A*A'u = 0
13        function v(n::Int)
14            if all(m == n || rref_augmented[m, n] == 0 for m
15            ↪ = 1 : M) # identify columns of leading ones
16                return zeros(Int, M)
17            else
18                return let v = zeros(eltype(rref_augmented),
19                ↪ N)
20
21                    v[n] = 1
22                    for m in 1:M
23                        let k = findfirst(isequal(1),
24                        ↪ rref_augmented[m, :])
25                        # If k === nothing, the
26                        ↪ mth row of
27                        ↪ rref_augmented must
28                        ↪ be zero.
29                        if k !== nothing && k <=
30                        ↪ n
31                            v[k] =
32                            ↪ -rref_augmented[m,
33                            ↪ n]
34                        end
35                    end
36                    v
37                end
38            end
39        end
40
41        end
42
43        end
44
45        for x in 1:20
46            # Defining random coefficients
47            random_numerator = rand(1:100, N)
48            random_denominator = rand(1:100, N)
49            random_vector = Vector{undef}(undef, N)
50
51            for m in 1:N

```

```

38         random_vector[m] = random_numerator[m] //
           ↪ random_denominator[m]
39     end
40
41     # The corresponding solution
42     general_solution = sum(random_vector[n]*v(n) for
           ↪ n in 1:N)+u
43
44     # Verify that {v_1,...,v_N} + u generates the
           ↪ solution set of A*A'u = b
45     if A*A'*general_solution == b
46         print("True: general_solution is correct for the
           ↪ randomly generated scalars ")
47         println(random_vector)
48     else print("False: general_solution is correct for
           ↪ the randomly generated scalars")
49         println(random_vector)
50     end
51
52     # Verify the uniqueness of s = A' * general_solution
53     let S = A' * general_solution
54     if s == S
55         print("True (uniqueness): s = A' * general
           ↪ solution for the randomly generated
           ↪ scalars ")
56         println(random_vector)
57     else print("False: s = A' * general solution for
           ↪ the randomly generated scalars ")
58         println(random_vector)
59     end
60
61     # Verify the minimality of s
62     if norm(s)<=norm(S)
63         print("True: s has smaller norm, based on the
           ↪ randomly generated scalars ")
64         println(random_vector)
65     else print("False: s has smaller norm, based on
           ↪ the randomly generated scalars ")
66         println(random_vector)
67     end
68     end
69 end
70

```

```

71 print("A solution u to (A * A')u = b is ")
72 println(u)
73 print("The minimal solution s to Ax = b is ")
74 println(s)

```

Exercise 6.3.23. Consider the problem of finding the least squares line $y = ct + d$ corresponding to the m observations $(t_1, y_1), (t_2, y_2), \dots, (t_m, y_m)$.

- (a) Show that the equation $(A^*A)x_0 = A^*y$ of theorem 6.12 takes the form of the *normal equations*:

$$\left(\sum_{i=1}^m t_i^2 \right) c + \left(\sum_{i=1}^m t_i \right) d = \sum_{i=1}^m t_i y_i$$

and

$$\left(\sum_{i=1}^m t_i \right) c + md = \sum_{i=1}^m y_i.$$

These equations may also be obtained from the error E by setting the partial derivatives of E with respect to both c and d equal to zero.

- (b) Use the second normal equation of (a) to show that the least squares line must pass through the *center of mass*, $(\bar{t}, \bar{y})$, where

$$\bar{t} = \frac{1}{m} \sum_{i=1}^m t_i \quad \text{and} \quad \bar{y} = \frac{1}{m} \sum_{i=1}^m y_i.$$

Proof.

- (a) At the vector $x_0 = (c_0 \ d_0)$ that minimises E , the partial derivatives of E with respect to c and d must be zero:

$$\begin{aligned} (\partial_1 E)(x_0) &= 0 & (\partial_2 E)(x_0) &= 0 \\ \sum_{i=1}^m -2t_i(y_i - c_0 t_i - d_0) &= 0 & \sum_{i=1}^m -2(y_i - c t_i - d_0) &= 0 \\ \left(\sum_{i=1}^m t_i^2 \right) c + \left(\sum_{i=1}^m t_i \right) d &= \sum_{i=1}^m t_i y_i & \left(\sum_{i=1}^m t_i \right) c + md &= \sum_{i=1}^m y_i. \end{aligned}$$

- (b) Dividing the second norm equation by m yields $\bar{y} = c_0 \bar{t} + d_0$. So, the least squares line must pass through the center of mass $(\bar{t}, \bar{y})$.



Exercise 6.3.24. Let V and $\{e_i\}_1^\infty$ be defined as in exercise 6.2.23. Define $T: V \rightarrow V$ by

$$T(\sigma)(k) := \sum_{i=k}^{\infty} \sigma(i) \quad \text{for every positive integer } k.$$

Notice that the infinite series in the definition of T converges because $\sigma(i) \neq 0$ for

only finitely many i . Prove the following facts:

- (a) T is a linear operator on V .
- (b) For any positive integer n , it holds that $T(e_n) = \sum_{i=1}^n e_i$.
- (c) T has no adjoint.

6.4 Normal and Self-Adjoint Operators

6.4.1 Theorems

Lemma. Let T be a linear operator on a finite-dimensional inner product space V . If T has an eigenvalue, then so does T^* .

Proof. Suppose T has an eigenvector, then $\chi_T(\lambda) = 0$ for some eigenvalue $\lambda \in \mathbb{K}$. So,

$$\chi_{T^*}(\bar{\lambda}) = \det(A^* - \bar{\lambda}I) = \det(A - \lambda I)^* = \overline{\det(A - \lambda I)} = \overline{\chi_T(\lambda)} = 0$$

where A is the matrix representation of T in some basis. Therefore, T^* has some eigenvector (corresponding to the eigenvalue $\bar{\lambda}$). 

Theorem 6.14 (Schur's Triangularisation Theorem). Let T be a linear operator on a finite-dimensional inner product space V . Suppose that the characteristic polynomial of T splits. Then there exists an orthonormal basis γ for V such that the matrix $[T]_\gamma$ is upper triangular.

Proof. Suppose this is true for all $\dim(V) \leq n-1$, and consider $\dim(V) = n$. First pick an orthonormal eigenbasis $\{\epsilon_j\}_1^m$ for the restriction of T to the direct sum E of all eigenspace of T . Since the characteristic polynomial of T splits, the same must hold for $\bar{T}: V/E \rightarrow V/E$; it also means $\dim(E) \geq 1$ so $\dim(V/E) \leq n-1$. Hence, with respect to the inner product $\langle u + E, v + E \rangle' := \langle \pi_{E^\perp}(u), \pi_{E^\perp}(v) \rangle$, choose an orthonormal basis $\{v_j + E\}_{m+1}^n$ that upper-triangularises $\bar{T}$. Consequently, the orthonormal basis $\gamma := \{u_j\}_1^n := \{\epsilon_j\}_1^m \cup \{\hat{\pi}_{E^\perp}(v_j)\}_{m+1}^n$ upper-triangularises T . 

Theorem 6.15. Let V be an inner product space, and let T be a normal operator on V . Then the following statements are true.

- (a) $\|T(x)\| = \|T^*(x)\|$ for all $x \in V$.
- (b) $T - cI$ is normal for every $c \in \mathbb{K}$.
- (c) If x is an eigenvector of T corresponding to the eigenvalue λ , then x is also an eigenvector of T^* corresponding to the eigenvalue $\bar{\lambda}$. That is, if $T(x) = \lambda x$, then $T^*(x) = \bar{\lambda}x$.
- (d) If λ_1 and λ_2 are distinct eigenvalues of T with corresponding eigenvectors x_1 and x_2 , then x_1 and x_2 are orthogonal.

Proof.

(a) This follows immediately from the definitions of the adjoint and normality:

$$\langle T(x), T(x) \rangle = \langle T^*T(x), x \rangle = \langle TT^*(x), x \rangle = \langle T^*(x), T^*(x) \rangle.$$

(b) Trivial:

$$\begin{aligned} (T - cI)(T - cI)^* &= (T - cI)(T^* - \bar{c}I) \\ &= (T - cI)T^* + (T - cI)(-\bar{c}I) \\ &= TT^* - cT^* - \bar{c}T + c\bar{c}I \\ &= T^*T - \bar{c}T - cT^* + \bar{c}cI \\ &= (T^* - \bar{c}I)T + (T^* - \bar{c}I)(-cI) \\ &= (T^* - \bar{c}I)(T - cI) \\ &= (T - cI)^*(T - cI). \end{aligned}$$

(c) For any $x, y \in E(\lambda, T)$, it is clear that $\langle x, T|_{E(\lambda, T)}^*(y) - \bar{\lambda}y \rangle = 0$, so $T|_{E(\lambda, T)}^*(y) = \bar{\lambda}y$. Furthermore, the uniqueness of adjoints implies $T|_{E(\lambda, T)}^*(y) = T^*|_{E(\lambda, T)}(y)$. Hence, $T^*(y) = \bar{\lambda}y$.

(d) We see that

$$\begin{aligned} \langle T(x), T(x) \rangle &= \langle \lambda x, \mu y \rangle \\ \langle x, T^*T(y) \rangle &= \langle \lambda x, \mu y \rangle \\ \langle x, \bar{\mu}\mu y \rangle &= \langle x, \bar{\lambda}\mu y \rangle \\ \langle x, \mu(\lambda - \mu)y \rangle &= 0 \\ \langle x, y \rangle &= 0. \end{aligned}$$



Theorem 6.16 (Complex Spectral Theorem). Let T be a linear operator on a finite-dimensional complex inner product space V . Then T is normal iff there exists an orthonormal basis for V consisting of eigenvectors of T .

Proof.

$\implies$ Since $\mathbb{C}$ is algebraically closed, [Schur's Triangularisation Theorem](#) holds; pick an orthonormal basis $\beta := \{v_k\}_1^n$ that upper-triangularises T — note $A_{mk} = 0$ for $k < m$, and $A_{km} = 0$ for $k > m$. To prove that this is indeed

an eigenbasis, first suppose $v_1, v_2, \dots, v_{m-1}$ are eigenvectors. Thus,

$$\begin{aligned} TT^*(v_m) &= T^*T(v_m) \\ \sum_{k=m}^n \overline{A_{mk}} T(v_k) &= \sum_{k=1}^m A_{km} T^*(v_k) \\ \sum_{k=m}^n \overline{A_{mk}} \lambda_k v_k &= \sum_{k=1}^m A_{km} \overline{\lambda_k} v_k. \end{aligned}$$

It follows from β being a basis, that $A_{km} = 0$ for all $1 \leq k \leq m-1$ — an eigenvector of T is v_m . Our induction is complete; β is an (orthonormal) eigenbasis for T .

$\Leftarrow$ Conversely, pick an orthonormal eigenbasis γ for T , where each eigenvector v_k corresponds to the eigenvalue λ_k . Since diagonal matrices commute, $[TT^*]_\gamma = [T^*T]_\gamma$. That is, T is normal.



After I spent many hours stuck on this proof — how to proceed after applying [Schur's Triangularisation Theorem](#), and whether that was the right move in the first place — I was skeptical at its simplicity. I might have unknowingly made some erroneous deduction. How should we proceed, with this skepticism? Let's implement a numerical verifier in [Julia](#)  to perform some sanity checks for us.

Julia.

```

1  # clears the terminal
2  print("\033c")
3
4  using LinearAlgebra, Random, Logging, Term
5
6  # Logging utilities
7      # Optionally write to a log file
8      # Open a log file
9      io = open("julia.log", "w")
10
11     # Create a simple logger
12     logger = SimpleLogger(io)
13
14     # Set the global logger to logger
15     global_logger(logger)
16
17     # Message generation
18     function Log_True_Up_To_Tolerance(statement, matrix,
19         ↪ storage_vector)
20         return let error = maximum(abs.(matrix))

```

```

20     @info "$statement up to \n error tolerance of
      ↳ magnitude $(floor(log10(error))) \n exact error
      ↳ = $error"
21     push!(storage_vector, error)
22     end
23 end
24
25 function Print_All_True_Up_To_Tolerance(statement, vector)
      ↳
26     return let vec_error_magnitude = floor.(log10.(vector))
27         tprintln("All {bold}$statement{/bold} up to \n
      ↳ maximum error tolerance of magnitude
      ↳ {bold}$(maximum(vec_error_magnitude)){/bold}")
28     @info "All $statement up to \n maximum error
      ↳ tolerance of magnitude
      ↳ $(maximum(vec_error_magnitude)) \n error vector =
      ↳ $vec_error_magnitude"
29     end
30 end
31
32 #=====#
33 #===== Start of actual programme =====#
34 #=====#
35
36
37 # Defining constants / programme parameters
38     # The maximum size of square matrices to be tested
39     const N = 100
40     # Number of randomly generated normal matrices T that are
      ↳ tested, for each n in 1 : N.
41     const number_of_repeitions_per_index_n = 100
42
43 # Defining empty vectors to which values are to be added
44 vec_error_normality = []
45 vec_error_diagonalness = []
46
47 # Test our proof of the Complex Spectral Theorem (CST) for square
      ↳ matrices of dimension 1 x 1 up to N x N
48 for n in 1 : N
49     for i in 1 : number_of_repeitions_per_index_n
50         @info "n = $n"
51
52         # Defining the normal matrix T

```

```

53      # The change of basis matrix Q that changes the from
    ↪ orthonormal basis to the standard ordered basis
54      # Since we used QR decomposition to define Q, it
    ↪ is surely orthonormal (unless the
    ↪ decomposition is done wrongly).
55      Q = Matrix(qr(rand(Complex{Float64}, (n, n))).Q)
56
57      # The matrix representation of T under a basis that
    ↪ triangularises it
58      D = Diagonal(rand(Complex{Float64}, n))
59
60      # Define the normal matrix T
61      T = Q * D * inv(Q)
62
63      # The normality of T
64      Log_True_Up_To_Tolerance("T is normal (T * T' ==
    ↪ T' * T)", T * T' - T' * T,
    ↪ vec_error_normality)
65
66      # Define the upper triangularised representation of T
67      triangularised = schur(T).Schur
68
69      # How close the upper triangular representation
    ↪ of T is to being diagonal
70      triangularised_errors = copy(triangularised)
71      triangularised_errors[diagind(
    ↪ triangularised_errors)] .= 0
72
73      Log_True_Up_To_Tolerance("'triangularised' is
    ↪ diagonal", triangularised_errors,
    ↪ vec_error_diagonalness)
74  end
75 end
76 Print_All_True_Up_To_Tolerance("T is normal (T * T' == T' * T)",
    ↪ vec_error_normality)
77 Print_All_True_Up_To_Tolerance("'triangularised' is diagonal",
    ↪ vec_error_diagonalness)
78
79 # Close the log file
80 close(io)

```

The output:

Output.

```

1 All T is normal (T * T' == T' * T) up to
2   maximum error tolerance of magnitude -15.0
3 All 'triangularised' is diagonal up to
4   maximum error tolerance of magnitude -14.0

```

So, we can be quite certain, through the above heuristics, that our proof of the forward direction ($\implies$) is correct.

In the upcoming lemma, we are to show that the characteristic polynomials of certain real linear operators split. Recall what makes the complex field so special — it is the algebraic closure of the real field. With that in mind, we might try *complexifying* a real vector space — extending it to have complex scalars.

The idea is simple: the real vector space is complexified to form a complex vector space the same way as how the real field is complexified to form the complex field.

Definition. The *complexification* $V_{\mathbb{C}}$ of a real vector space V is $V \times V$. For better semantics, we write $u + iv$ for $(u, v) \in V_{\mathbb{C}}$.

Furthermore, we now equip $V_{\mathbb{C}}$ with addition $+: V_{\mathbb{C}} \times V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ and scalar multiplication $\cdot: \mathbb{C} \times V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$, as follows:

$$(u_1 + iv_1) + (u_2 + iv_2) := (u_1 + u_2) + i(v_1 + v_2) \quad \text{and} \quad (a + bi)(u + iv) := (au - bv) + i(bu + av).$$

Of course, all variants of u and v here are vectors in $V_{\mathbb{C}}$, with $a, b \in \mathbb{R}$.

For such a construction to be meaningful, $V_{\mathbb{C}}$ must be a vector space.

Theorem. The complexification $V_{\mathbb{C}}$ of any real vector space V is a complex vector space, under the aforementioned binary operations of addition and scalar multiplication.

Proof. For all vectors $u, u_1, u_2 \in V$; $v, v_1, v_2 \in V$; complex scalars $a + bi$ and $c + di$, it holds that:

$$\begin{aligned}
 \text{(VS 1)} \quad (u_1 + iv_1) + (u_2 + iv_2) &= (u_1 + u_2) + i(v_1 + v_2) \\
 &= (u_2 + u_1) + i(v_2 + v_1) \\
 &= (u_2 + iv_2) + (u_1 + iv_1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(VS 2)} \quad ((u_1 + iv_1) + (u_2 + iv_2)) + (u_3 + iv_3) &= ((u_1 + u_2) + i(v_1 + v_2)) + (u_3 + iv_3) \\
 &= ((u_1 + u_2) + u_3) + i((v_1 + v_2) + v_3) \\
 &= (u_1 + (u_2 + u_3)) + i(v_1 + (v_2 + v_3)) \\
 &= (u_1 + iv_1) + ((u_2 + u_3) + i(v_2 + v_3)) \\
 &= (u_1 + iv_1) + ((u_2 + iv_2) + (u_3 + iv_3))
 \end{aligned}$$

$$(VS\ 3) \quad (0 + i0) + (u + iv) = (0 + u) + i(0 + v) = u + iv$$

$$(VS\ 4) \quad (u + iv) + ((-u) + i(-v)) = (u - u) + i(v - v) = 0 + i0 = 0.$$

$$(VS\ 5) \quad 1(u + iv) = (1u + i(1v)) = u + iv$$

$$\begin{aligned} (VS\ 6) \quad & ((a + bi)(c + di))(u + iv) \\ &= ((ac - bd) + i(ad + bc))(u + iv) \\ &= ((ac - bd)u - (ad + bc)v) + i((ad + bc)u + (ac - bd)v) \\ &= (a(cu - dv) - b(du + cv)) + i(b(cu - dv) + a(du + cv)) \\ &= (a + bi)((cu - dv) + i(du + cv)) \\ &= (a + bi)((c + di)(u + iv)) \end{aligned}$$

$$\begin{aligned} (VS\ 7) \quad & (a + bi)((u_1 + iv_1) + (u_2 + iv_2)) \\ &= (a + bi)((u_1 + u_2) + i(v_1 + v_2)) \\ &= (a(u_1 + u_2) - b(v_1 + v_2)) + i(b(u_1 + u_2) + a(v_1 + v_2)) \\ &= ((au_1 - bv_1) + (au_2 - bv_2)) + i((bu_1 + av_1) + (bu_2 + av_2)) \\ &= ((au_1 - bv_1) + i(bu_1 + av_1)) + ((au_2 - bv_2) + i(bu_2 + av_2)) \\ &= (a + bi)(u_1 + iv_1) + (a + bi)(u_2 + iv_2) \end{aligned}$$

$$\begin{aligned} (VS\ 8) \quad & ((a + bi) + (c + di))(u + iv) \\ &= ((a + c) + i(b + d))(u + iv) \\ &= ((a + c)u - (b + d)v) + i((b + d)u + (a + c)v) \\ &= ((au - bv) + (cu - dv)) + i((bu + av) + (du + cv)) \\ &= ((au - bv) + i(bu + av)) + ((cu - dv) + i(du + cv)) \\ &= (a + bi)(u + iv) + (c + di)(u + iv) \end{aligned}$$

Hence, $V_{\mathbb{C}}$ is a vector space. 

Definition. The *complexification* of a linear operator T on a real vector space V is the linear operator $T_{\mathbb{C}}: V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ defined by $T(u + iv) = T(u) + iT(v)$.

Theorem. The complexification of a real linear operator $T: V \rightarrow V$ is indeed linear.

Proof. Consider the vectors $u_1 + iv_1$ and $u_2 + iv_2$ of $V_{\mathbb{C}}$; the complex scalar $a + bi$.

Then,

$$\begin{aligned}
 & T_{\mathbb{C}}((a + bi)(u_1 + iv_1) + (u_2 + iv_2)) \\
 &= T_{\mathbb{C}}(((au_1 - bv_1) + i(bu_1 + av_1)) + (u_2 + iv_2)) \\
 &= T_{\mathbb{C}}((au_1 - bv_1 + u_2) + i(bu_1 + av_1 + v_2)) \\
 &= T(au_1 - bv_1 + u_2) + iT(bu_1 + av_1 + v_2) \\
 &= (aT(u_1) - bT(v_1)) + i(bT(u_1) + aT(v_1)) + (T(u_2) + iT(v_2)) \\
 &= (a + bi)(T(u_1) + iT(v_1)) + (T(u_2) + iT(v_2)) \\
 &= (a + bi)T_{\mathbb{C}}(u_1 + iv_1) + T_{\mathbb{C}}(u_2 + iv_2).
 \end{aligned}$$

This confirms our claim: $T_{\mathbb{C}}$ is linear. 

Lemma (FIS page 370). Let T be a self-adjoint operator on a finite-dimensional inner product space V . Then

- (a) Every eigenvalue of T is real.
- (b) Suppose that V is a real inner product space. Then the characteristic polynomial of T splits.

Proof.

- (a) If λ is an eigenvalue of $T = T^*$, then $\lambda = \bar{\lambda}$ by Theorem 6.15(c). So, $\lambda \in \mathbb{R}$.
- (b) Every eigenvalue λ of $T_{\mathbb{C}}$ is an eigenvalue of T : for $u + iv \in E(\lambda, T_{\mathbb{C}})$,

$$\begin{aligned}
 T_{\mathbb{C}}(u + iv) &= \lambda u + i(\lambda v) \\
 T(u) + iT(v) &= \lambda u + i(\lambda v)
 \end{aligned}$$

so $u, v \in E(\lambda, T)$. Furthermore, $T_{\mathbb{C}}$ is self-adjoint; all its eigenvalues are real, by (a). As such, $T_{\mathbb{C}}$, and hence T , has $\dim(V_{\mathbb{C}}) = \dim(V)$ real roots λ_j . Recall that the characteristic polynomial of $T: V \rightarrow V$ is of degree $\dim(V)$. Hence, the characteristic polynomial $\chi_T(\lambda)$ of T splits—into linear factors $\lambda - \lambda_j$. 

Theorem 6.17 (Real Spectral Theorem). Let T be a linear operator on a finite-dimensional real inner product space V . Then T is self-adjoint iff there is an orthonormal eigenbasis for T .

Proof.

$\Rightarrow$ Since $T_{\mathbb{C}}$ is self-adjoint—and hence normal—pick an orthonormal basis $\{u_j + iv_j\}_1^n$ for V , using the **Complex Spectral Theorem**. Consider an eigenvalue λ of T . Wlog, $\{u_j + iv_j\}_k^{k+m}$ is a basis for $E(\lambda, T_{\mathbb{C}})$, for some

$k \in \mathbb{N}$. Recall that $u_j, v_j \in E(\lambda, T)$. Since

$$\dim_{V_{\mathbb{C}}} \text{span}\{u_j + iv_j\}_k^{k+m} = m,$$

we must have

$$\dim_V \text{span}\{u_j, v_j\}_k^{k+m} = m.$$

× Counterexample: even though

$$\dim_{V_{\mathbb{C}}} \{u + iu, 2u + iu\} = 2,$$

clearly we have

$$\dim_V \{u, u, 2u, u\} = 1.$$

⇒ Using **Schur's Triangularisation Theorem**, pick an orthonormal basis $\beta := \{v_1, v_2, \dots, v_n\}$ that upper-triangularises T . In fact, since T is self-adjoint, β diagonalises T .

⇐ Let β be an orthonormal eigenbasis for T . Then, as $[T]_{\beta}$ is a real diagonal matrix, T must be self-adjoint.



	Real	Complex
Numbers	$x = \bar{x}$	$z\bar{z} = \bar{z}z$
Vector spaces	$T = T^*$	$TT^* = T^*T$

6.4.2 Exercises

Exercise 6.4.4. Let T and U be self-adjoint operators on an inner product space V . Prove that TU is self-adjoint iff $TU = UT$.

Proof. If TU is self-adjoint, then $TU = (TU)^* = U^*T^* = UT$. Conversely, when $TU = UT$, we obtain $(TU)^* = U^*T^* = UT$.



Exercise 6.4.8. Let T be a normal operator on a finite-dimensional complex inner product space V , and let W be a subspace of V . Prove that if W is T -invariant, then W is also T^* -invariant.

Proof. Pick an orthonormal eigenbasis $\{w_j\}_1^n$ for the normal operator T_W . If W is T -invariant,

$$\begin{aligned}\langle T(w_j), w_j \rangle &= \langle \lambda_j w_j, w_j \rangle \\ \langle w_j, T^*(w_j) \rangle &= \langle w_j, \bar{\lambda}_j w_j \rangle.\end{aligned}$$

So, $T^*(w_j) = \bar{\lambda}_j w_j \in W$, which means that W is T^* -invariant.



Exercise 6.4.10. Let T be a normal operator on a finite-dimensional inner product space V . Prove that for all $x \in V$

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2.$$

Deduce that $T - iI$ is invertible and that the adjoint of $(T - iI)^{-1}$ is $(T + iI)^{-1}$.

Proof. First, we see that $\langle T(x), x \rangle \in \mathbb{R}$, since $\langle T(x), x \rangle = \langle x, T(x) \rangle = \overline{\langle T(x), x \rangle}$. Now $\Re\langle T(x), ix \rangle = \Re(-i\langle T(x), x \rangle) = 0$;

$$\begin{aligned} \|T(x) \pm ix\|^2 &= \langle T(x), T(x) \rangle \pm 2\Re\langle T(x), ix \rangle \pm \overbrace{\langle ix, ix \rangle}^{i(-i)\cdot\langle x, x \rangle = \langle x, x \rangle} \\ &= \|T(x)\|^2 + \|x\|^2. \end{aligned}$$

Consequently, $\|(T - iI)(x)\| \geq \|x\|^2 > 0$ for all nonzero x —that is, $N(T - iI) = \{0\}$. We deduce that $T - iI$ is invertible. In fact, by exercise 6.3.8, $((T - iI)^{-1})^* = ((T - iI)^*)^{-1} = (T + iI)^{-1}$. 

Exercise 6.4.11. Assume that T is a linear operator on a complex (not necessarily finite-dimensional) inner product space V with an adjoint T^* . Prove the following results.

- (a) If T is self-adjoint, then $\langle T(x), x \rangle$ is real for all $x \in V$.
- (b) If T satisfies $\langle T(x), x \rangle = 0$ for all $x \in V$, then $T = T_0$.
- (c) If $\langle T(x), x \rangle$ is real for all $x \in V$, then T is self-adjoint.

Proof.

- (a) If T is self-adjoint,

$$\langle T(x), x \rangle = \langle x, T^*(x) \rangle = \langle x, T(x) \rangle = \overline{\langle T(x), x \rangle}$$

so $\langle T(x), x \rangle$ is real.

- (b) Suppose that $\langle T(x), x \rangle = 0$ for all $x \in V$. Pick any $x \in V$; notice that

$$\begin{aligned} \langle T(x + T(x)), x + T(x) \rangle &= 0 \\ \langle T(x), T(x) \rangle + \langle T^2(x), x \rangle &= 0 \\ \langle T(x), T(x) \rangle &= -\langle x, T^2(x) \rangle. \end{aligned}$$

Simultaneously, $\langle T(x), T(x) \rangle = \langle x, T^2(x) \rangle$; it must be that $\langle T(x), T(x) \rangle = 0$. That is, $T(x) = 0$.

× Ah I forgot that we are no longer assuming T to be self-adjoint.

- (c) Since $\langle T(x), x \rangle = \langle x, T^*(x) \rangle \in \mathbb{R}$, it follows that $\langle T(x), x \rangle = \langle T^*(x), x \rangle$. That is, $\langle (T - T^*)(x), x \rangle = 0$. By exercise 6.4.11(b), we have $T = T^*$. In other words, T is self-adjoint. 

Corollary. Any linear operator T on an inner product space V is self-adjoint ($T = T^*$) iff $\langle T(x), x \rangle \in \mathbb{R}$ for all $x \in V$.

Exercise 6.4.13. An $n \times n$ real matrix A is said to be a *Gramian* matrix if there exists a real (square) matrix B such that $A = B^t B$. Prove that A is a Gramian matrix iff A is symmetric and all of its eigenvalues are nonnegative.

Proof.

$\Rightarrow$ Gramian matrices $A \in M_{n \times n}(\mathbb{R})$ are clearly symmetric:

$$A^t = (B^t B)^t = B^t (B^t)^t = B^t B = A.$$

Hence, the **Real Spectral Theorem** is applicable. Paired with exercise 6.2.11, this entails the existence of $n \times n$ matrices Q and (diagonal) D with $A = Q D Q^t$. That is, $(BQ)^t BQ = D$, making it clear that any eigenvalue D_{nn} of A is the norm of the n th column of BQ — which is nonnegative.

$\Leftarrow$ Now consider symmetric A with all eigenvalues being nonnegative. Again, the **Real Spectral Theorem** applies; $A = Q D Q^t$ for some $n \times n$ real matrices Q and (diagonal) D . From here, let $\mathcal{D} \in M_{n \times n}(\mathbb{R})$ be defined by $\mathcal{D}_{mn} := \sqrt{D_{mn}}$. Since

$$(Q \mathcal{D} Q^t)^t = (Q^t)^t \mathcal{D} Q^t = Q \mathcal{D} Q^t,$$

it follows that A is Gramian:

$$A = Q D Q^t = (Q \mathcal{D} Q^t)(Q \mathcal{D} Q^t) = \underbrace{(Q \mathcal{D} Q^t)^t}_{B^t} \underbrace{(Q \mathcal{D} Q^t)}_B.$$



Exercise 6.4.14. *Simultaneous Diagonalisation.* Let V be a finite-dimensional real inner product space, and let U and T be self-adjoint linear operators on V such that $UT = TU$. Prove that there exists an orthonormal basis for V consisting of vectors that are eigenvectors of both U and T . (The complex version of this result appears as exercise 6.6.10.)

Proof. Let $\{\lambda_i\}_1^n$ be the set of all eigenvalues of T . Since $\lambda_i U(v) = TU(v)$ for any $v \in E(T, \lambda_i)$, it follows that $E(T, \lambda_i)$ is U -invariant; $U_{E(T, \lambda_i)}$ is a (self-adjoint) linear operator. Hence, the **Real Spectral Theorem** enables us to pick an orthonormal eigenbasis β_i for $U_{E(T, \lambda_i)}$. So, $\beta := \bigcup_1^n \beta_i$ is an orthonormal eigenbasis for both T and U .



Exercise 6.4.15. Let A and B be symmetric $n \times n$ matrices such that $AB = BA$. Use exercise 6.4.14 to prove that there exists an orthogonal matrix P such that $P^t A P$ and $P^t B P$ are both diagonal matrices.

Proof. By exercise 6.4.14, there is an orthonormal eigenbasis β that simultaneously diagonalises L_A and L_B . Note that, with respect to the standard ordered basis α , the change of basis matrix $P := [I_{\mathbb{R}^n}]$ is orthogonal by exercise 6.2.11. Hence, $P^t A P$ and $P^t B P$ are diagonal. 

Exercise 6.4.16. Prove the *Cayley-Hamilton-Frobenius theorem* for a complex $n \times n$ matrix A . That is, if $f(t)$ is the characteristic polynomial of A , prove that $f(A) = O$.

Lemma. Let T be a diagonalisable linear operator on some vector space V . Then, every eigenvector of T^2 is an eigenvector of T .

Proof. Let $\{\lambda_i\}_1^n$ be the set of all distinct eigenvalues of T . Pick an eigenvector x of T^2 , with corresponding eigenvalue some λ_i^2 . Since $V = \oplus_1^n E(\lambda_i, T)$ by the diagonalisability of T , there are unique scalars c_i and eigenvectors $x_i \in E(\lambda_i, T)$, such that $x = \sum_1^n c_i x_i$. So,

$$\lambda_i^2 x = T^2(x) = \sum_1^n c_i \lambda_i^2 x_i.$$

By the linear independence of eigenspaces (with distinct eigenvalues), $x = c_i x_i$ must follow. That is, $x \in E(\lambda_i, T)$; x is an eigenvector of T . 

Exercise 6.4.17. Let T and U be self-adjoint linear operators on an n -dimensional inner product space V , and let $A = [T]_\beta$, where β is an orthonormal basis for V . Prove the following results.

- (a) T is positive definite [semidefinite] iff all of its eigenvalues are positive [nonnegative].
- (b) T is positive definite iff

$$\sum_{i,j} A_{ij} a_j \overline{a_i} > 0 \text{ for all nonzero } n\text{-tuples } (a_1, a_2, \dots, a_n).$$

- (c) T is positive semidefinite iff $A = B^* B$ for some square matrix B .
- (d) If T and U are positive semidefinite operators such that $T^2 = U^2$, then $T = U$.
- (e) If T and U are positive definite operators such that $TU = UT$, then TU is positive definite.
- (f) T is positive definite [semidefinite] iff A is positive definite [semidefinite].

Proof.

- (a) First consider T being positive definite. Then, $\lambda \langle x, x \rangle = \langle T(x), x \rangle > 0$ for any eigenvector x of T corresponding to the eigenvalue λ . That is, $\lambda > 0$. For the converse, suppose all eigenvalues λ of T are positive. By the **Real/Complex Spectral Theorems**, there is an orthonormal basis $\beta := \{x_i\}_1^n$

for T ; each x_i corresponds to some eigenvalue λ_i . Hence, for any scalars c_i ,

$$\begin{aligned} \left\langle T \left(\sum_{i=1}^n c_i x_i \right), \sum_{j=1}^n c_j x_j \right\rangle &= \sum_{i=1}^n c_i \lambda_i \left\langle x_i, \sum_{j=1}^n c_j x_j \right\rangle \\ &= \sum_{i=1}^n c_i \lambda_i \sum_{j=1}^n \bar{c}_j \langle x_i, x_j \rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n c_i \bar{c}_j \lambda_i \langle x_i, x_j \rangle \\ &= \sum_{i=1}^n \underbrace{|c_i|^2}_{\geq 0} \underbrace{\lambda_i}_{>0} \underbrace{\langle x_i, x_i \rangle}_{>0}. \end{aligned}$$

Thus, T is positive definite. The corresponding statement for positive semidefinite T follows from a trivial modification of the above.

(b) The result follows trivially from this:

$$\begin{aligned} \sum_{i,j} A_{i,j} a_j \bar{a}_i &> 0 \\ \iff \sum_{i,j} \langle x_j, x_i \rangle a_j \bar{a}_i &> 0 \\ \iff \sum_{i,j} \lambda_j \langle x_j, x_i \rangle a_j \bar{a}_i &> 0 \\ \iff \sum_{i,j} \langle a_j \lambda_j x_j, a_i x_i \rangle &> 0 \\ \iff \left\langle T \left(\sum_{j=1}^n a_j x_j \right), \sum_{i=1}^n a_i x_i \right\rangle &> 0. \end{aligned}$$

(c) First consider T being positive semidefinite. By the **Real/Complex Spectral Theorems**, there is an orthonormal matrix Q and diagonal matrix D with $A = QDQ^*$. Since D is nonnegative by (a), we can define the diagonal matrix E by $E_{ii} := \sqrt{D_{ii}}$. Notice that

$$(QE Q^*)^* = (Q^*)^* (QE)^* = Q \bar{E} Q^* = QE Q^*$$

as E is a nonnegative (i.e. real) matrix. Consequently,

$$(QE Q^*)^* (QE Q^*) = QE Q^* QE Q^* = QDQ^* = A.$$

Conversely, suppose $A = B^* B$ for some square matrix B . Then, A is self-adjoint; there is an orthogonal matrix Q and diagonal matrix D with $A = QDQ^*$. So,

$$\begin{aligned} BB^* &= QDQ^* \\ (Q^* B)(B^* Q) &= D \\ (B^* Q)^*(B^* Q) &= D. \end{aligned}$$

Hence, for v_i being the i th column of B^*Q , we have $D_{ii} = \langle v_i, v_i \rangle \geq 0$. Recall that D is the diagonal matrix for all eigenvalues of T — which we have just shown to be nonnegative. Thus, T is positive semidefinite.

- (d) Pick an orthonormal eigenbasis $\{x_i\}_1^n$ for T , with corresponding eigenvalues $\{\lambda_i\}_1^n$. Then, each x_i is also an eigenvector of $T^2 = U^2$, with corresponding eigenvalue λ_i^2 . By the above lemma, every x_i is an eigenvector of U , with some eigenvalue μ_i . From $T^2 = U^2$, we deduce $\mu_i = \pm\lambda_i$. Since $\mu_i \langle x, x \rangle = \langle U(x), x \rangle \geq 0$ as U is positive semidefinite, $\mu_i \geq 0$; $\mu_i = \lambda_i$. Now, $T(x_i) = \lambda_i x_i = \mu_i x_i = U(x_i)$ for all $1 \leq i \leq n$. Hence, $T = U$.
- (e) If T and U are positive definite operators such that $TU = UT$, then exercise 6.4.14 implies that T and U can be simultaneously diagonalised by some orthonormal basis $\{v_i\}_1^n$. So, for λ_i and μ_i being the eigenvalues corresponding to the eigenvector v_i for T and U , respectively, the (self-adjoint) linear operator TU has eigenvalues $\lambda_i \mu_i$ for each $1 \leq i \leq n$. Since λ_i and μ_i are all positive by (a), so must $\lambda_i \mu_i$. Thus, TU is positive definite, again by (a).
- (f) Trivial by exercise 6.2.15(b).



Exercise 6.4.18. Let $T: V \rightarrow W$ be a linear transformation, where V and W are finite-dimensional inner product spaces. Prove the following results.

- (a) T^*T and TT^* are positive semidefinite.
 (b) $\text{rank}(T^*T) = \text{rank}(TT^*) = \text{rank}(T)$.

Proof.

- (a) Since $(T^*T)^* = T^*(T^*T)$ and

$$\langle T^*T(x), x \rangle_V = \langle T(x), T(x) \rangle_W \geq 0$$

for all $x \in V$, it follows that $T^*T: V \rightarrow V$ is positive semidefinite. The proof for TT^* is a trivial modification of the above.

- (b) Exercise 6.3.15?? tells us $N(T^*T) = N(T)$ so $R(T^*T) = R(T)$. Combining the above with 6.3.15?? now gives $\text{rank}(TT^*) = \text{rank}(T^*) = \text{rank}(T)$.



Question. If a linear operator on an inner product space is self-inverse and self-adjoint, does it have any special properties? Must it be the identity operator?

Exercise 6.4.20. Let V be an inner product space with inner product $\langle \cdot, \cdot \rangle$, and let T be a positive definite linear operator on V . Prove that $\langle x, y \rangle' = \langle T(x), y \rangle$ defines another inner product on V .

Proof. We verify the first coordinate linearity, conjugate symmetry and positive definiteness of $\langle \cdot, \cdot \rangle'$, in order. For any scalar $c \in \mathbb{K}$ and vectors $x, y, z \in V$:

- $\langle cx + y, z \rangle' := \langle T(cx + y), z \rangle$

$$= \langle cT(x) + T(y), z \rangle$$

$$= c\langle T(x), z \rangle + \langle T(y), z \rangle$$

$$= c\langle x, z \rangle' + \langle y, z \rangle'$$
- $\overline{\langle x, y \rangle'} := \overline{\langle T(x), y \rangle} = \overline{\langle T(x), y \rangle} \stackrel{T^* = T}{=} \langle T(y), x \rangle =: \langle y, x \rangle'$.
- If $x \neq 0$, then

$$\langle x, x \rangle' := \langle T(x), x \rangle \geq 0$$

by the positive definite nature of T .



Exercise 6.4.21. Let V be a finite-dimensional inner product space, and let T and U be self-adjoint operators on V such that T is positive definite. Prove that both TU and UT are diagonalisable linear operators that have only real eigenvalues.

Exercise 6.4.22. This exercise provides a converse to exercise 6.4.20. Let V be a finite-dimensional inner product space with inner product $\langle \cdot, \cdot \rangle$, and let $\langle \cdot, \cdot \rangle'$ be any other inner product on V .

- (a) Prove that there exists a unique linear operator T on V such that $\langle x, y \rangle' = \langle T(x), y \rangle$ for all x and y in V .
- (b) Prove that the operator T of (a) is positive definite with respect to both inner products.

Proof.

- (a) Pick an orthonormal basis $\beta := \{v_i\}_1^n$ for $(V, \langle \cdot, \cdot \rangle)$. Let $T: V \rightarrow V$ be defined by

$$T(v_i) := \sum_{j=1}^n \langle v_i, v_j \rangle' v_j.$$

We see that, as expected,

$$\langle T(v_i), v_j \rangle = \left\langle \sum_{k=1}^n \langle v_i, v_k \rangle' v_k, v_j \right\rangle = \sum_{k=1}^n \langle v_i, v_k \rangle' \overbrace{\langle v_k, v_j \rangle}^{\delta_{kj}} = \langle v_i, v_j \rangle'.$$

Furthermore, given any two linear operators T and U on V , such that $\langle T(x), y \rangle$ and $\langle U(x), y \rangle$ are both $\langle x, y \rangle'$ (for all $x, y \in V$), we have $\langle (T - U)(x), x \rangle = 0$. So, $T = U$ and uniqueness must hence hold.

- (b) We first see why T is self-adjoint:

$$([T^*]_{\beta})_{ij} = ([T]_{\beta})_{ij}^* = \overline{\langle v_j, v_i \rangle} = \langle v_i, v_j \rangle = ([T]_{\beta})_{ij}.$$

Hence, $\langle T(x), x \rangle' = \langle T^2(x), x \rangle = \langle T(x), T(x) \rangle > 0$ for nonzero vectors x . Also, $\langle T(x), x \rangle = \langle x, x' \rangle > 0$ by definition. That is, T is positive definite

with respect to both $\langle \cdot, \cdot \rangle$ and $\langle \cdot, \cdot \rangle'$.



Exercise 6.4.23. Let U be a diagonalisable linear operator on a finite-dimensional inner product space V such that all of the eigenvalues of U are real. Prove that there exist positive definite linear operators T_1 and T_1' ; self-adjoint linear operators T_2 and T_2' , such that $U = T_2 T_1 = T_1' T_2'$.

6.5 Unitary and Orthogonal Operators and Their Matrices

6.5.1 Theorems

Lemma (FIS page 377). Let U be a self-adjoint operator on an inner product space V and suppose that $\langle x, U(x) \rangle = 0$ for all $x \in V$. Then $U = T_0$.

Proof. Theorems 6.16 and 6.17 entails the existence of an orthonormal eigenbasis $\{v_k\}_1^n$ for U . If λ_k is the eigenvalue corresponding to v_k ,

$$\overline{\lambda_k} \langle v_k, v_k \rangle = \langle v_k, U(v_k) \rangle = 0$$

so $\lambda_k = 0$. That is, $U = T_0$.



Theorem 6.18. Let T be a linear operator on a finite-dimensional inner product space V . Then the following statements are equivalent.

- (a) $T^* T = I$.
- (b) $T T^* = I$.
- (c) $\langle T(x), T(y) \rangle = \langle x, y \rangle$ for all $x, y \in V$.
- (d) If β is an orthonormal basis for V , then $T[\beta]$ is an orthonormal basis for V .
- (e) There exists an orthonormal basis β for V , such that $T[\beta]$ is an orthonormal basis for V .
- (f) $\|T(x)\| = \|x\|$ for all $x \in V$.

Proof.

- (b) $\implies$ (a) If $T T^* = I$, then $T^{-1} = T^*$ so $T^* T = T^{-1} T = I$ follows immediately.
- (a) $\implies$ (c) We see that $\langle T(x), T(y) \rangle = \langle T^* T(x), y \rangle = \langle x, y \rangle$ for all $x, y \in V$, given $T^* T = I$.
- (c) $\implies$ (d) When $\beta := \{v_i\}_1^n$ is orthonormal and (c) holds,

$$\langle T(v_i), T(v_j) \rangle = \langle v_i, v_j \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

That is, $T[\beta]$ is orthonormal — and an orthonormal subset of V

with cardinality $n = \dim(V)$ must be a basis for V .

(d) $\implies$ (e) This is trivial since the corollary to theorem 1.13 says that all vector spaces have a basis, and by the Gram-Schmidt Orthonormalisation Process.

(e) $\implies$ (f) Assume there is an orthonormal basis $\beta := \{v_i\}_1^n$ for V such that $T[\beta]$ is also an orthonormal basis for V . Then, for any scalars c_i , let $x := \sum_1^n c_i v_i$. By the triangle inequality,

$$\|T(x)\| = \sqrt{\sum_1^n \|c_i T(v_i)\|^2} = \sqrt{\sum_1^n |c_i|^2} = \sqrt{\sum_1^n \|c_i v_i\|^2} = \|x\|.$$

(f) $\implies$ (a) Trivial by the above lemma, which when applied to

$$\begin{aligned} \|T^*(x)\| &= \|x\| \\ \langle T^*(x), T^*(x) \rangle &= \langle x, x \rangle \\ \langle x, (TT^* - I)(x) \rangle &= 0 \end{aligned}$$

yields $TT^* = I$.



Corollary 6.18.1. Let T be a linear operator on a finite-dimensional real inner product space V . Then V has an orthonormal basis of eigenvectors of T with corresponding eigenvalues of absolute value 1 iff T is both self-adjoint and orthogonal.

Proof. Suppose T has an orthonormal eigenbasis $\beta := \{v_i\}_1^n$ with corresponding eigenvalues all being ± 1 . By the Real Spectral Theorem, T is then self-adjoint. Moreover, $T[\beta]$ is clearly an orthonormal basis for V — each member of $T[\beta]$ or its additive inverse is in β — so theorem 6.18 implies the orthogonality of T .

Conversely, consider when T is self-adjoint and orthogonal. Again, the Real Spectral Theorem guarantees the existence of an orthonormal eigenbasis $\beta := \{v_i\}_1^n$ of T , with corresponding eigenvalues some $\{\lambda_i\}_1^n$. Furthermore, theorem 6.18 tells us

$$\begin{aligned} \|T(v_i)\| &= \|v_i\| \\ |\lambda_i| \cdot \|v_i\| &= \|v_i\| \\ |\lambda_i| &= 1. \end{aligned}$$



Corollary 6.18.2. Let T be a linear operator on a finite-dimensional complex inner product space V . Then V has an orthonormal basis of eigenvectors of T with corresponding eigenvalues of absolute value 1 iff T is unitary.

Proof. Similar to the proof of corollary 6.18.1.



Theorem 6.22. Let $f: V \rightarrow V$ be a rigid motion on a finite-dimensional real inner product space V . Then there exists a unique orthogonal operator on V and a unique translation g on V such that $f = g \circ T$.

Proof. Let the operator T on V be defined by $T(x) := f(x) - f(0)$. Below, we consider an arbitrary scalar $c \in \mathbb{K}$ and vectors $x, y \in V$; first notice that

$$\begin{aligned}\langle cf(x) \pm f(y), f(z) \rangle &= c\langle f(x), f(z) \rangle \pm \langle f(y), f(z) \rangle \\ &= c\langle x, z \rangle \pm \langle y, z \rangle \\ &= \langle cx \pm y, z \rangle.\end{aligned}$$

So, T is linear:

$$\begin{aligned}\|T(cx + y) - (cT(x) + T(y))\| &= \|f(cx + y) - (c(f(x) - f(0)) + f(y) - f(0))\| \\ &= \|f(cx + y) - cf(x) - f(y) + (c + 1)f(0)\| \\ &= \|cx + y - cx - y + (c - 1)(0)\| \\ &= \|0\| \\ &= 0.\end{aligned}$$

Furthermore, $\|T(x)\| = \|f(x) - f(0)\| = \|x\|$ so T is an orthogonal operator, by theorem 6.18. Thus, the corresponding translation $g: V \rightarrow V$ is the one defined by $g(x) := x + f(0)$.

Now, for uniqueness, suppose $f = h \circ U$ for some orthogonal operator U on V and translation h on V . Then, $(h \circ U)(0) = f(0)$ implies $h(0) = f(0)$; that is, $h: x \mapsto x + f(0)$ so $h = g$. Finally, we see that

$$\begin{aligned}f(x) &= U(x) + f(0) \\ U(x) &= f(x) - f(0) \\ U(x) &= T(x).\end{aligned}$$

Therefore, T and g are unique.



Theorem 6.23. Let T be an orthogonal operator on $\mathbb{R}^2$, and let $A = [T]_\beta$, where β is the standard ordered basis for $\mathbb{R}^2$. Then exactly one of the following conditions is satisfied:

- (a) T is a rotation, and $\det(A) = 1$.
- (b) T is a reflection about a line through the origin, and $\det(A) = -1$.

Proof. Let $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$. Since the columns of A are orthonormal,

$$\left\langle \begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix} \right\rangle = 0 \quad (\text{orthogonality})$$

$$ac + bd = 0$$

$$d = -\frac{ac}{b}$$

so

$$c^2 + d^2 = 1 \quad (\text{normality})$$

$$c^2 + \left(\frac{a^2}{b^2}\right)c^2 = 1$$

$$\left(\frac{a^2 + b^2}{b^2}\right)c^2 = 1 \quad (\text{normality})$$

$$\left(\frac{1}{b^2}\right)c^2 = 1$$

We hence obtain $c = \pm b$ and $d = \mp a$; we have

$$A = \begin{pmatrix} a & \pm b \\ b & \mp a \end{pmatrix}.$$

Since $a^2 + b^2 = 1$ also implies $a \in \text{dom}(\arccos)$, we see that

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \quad \text{or} \quad A = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix}$$

for $\theta := \arccos(a)$ and $d = a$, which represent rotations. As expected,

$$\det(A) = \cos^2(\pm\theta) + \sin^2(\pm\theta) = 1.$$

Now consider $d = -a$, where

$$A = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

First notice the eigenvalues of A are exactly ± 1 :

$$\det(A - \lambda I) = 0$$

$$(a - \lambda)(-a - \lambda) - b^2 = 0$$

$$(a - \lambda)(a + \lambda) + b^2 = 0$$

$$a^2 - \lambda^2 + b^2 = 0$$

$$\lambda^2 = 1$$

$$\lambda = \pm 1.$$

Therefore, A represents a reflection of $\mathbb{R}^2$ about $E(\lambda, 1)$. Finally, we note the trivial

fact

$$\det(A) = a(-a) - b^2 = -(a^2 + b^2) = -1.$$



6.5.2 Exercises

Exercise 6.5.4. For $z \in \mathbb{C}$, define $T_z: \mathbb{C} \rightarrow \mathbb{C}$ by $T_z(u) = zu$. Characterise those z for which T_z is normal, self-adjoint, or unitary.

Proof. Let $\beta := \{1\}$, the standard ordered basis for $\mathbb{C}$; then $[T_z]_\beta = \begin{pmatrix} z \end{pmatrix}$. So,

(a) T_z is normal for all complex z :

$$\begin{aligned} T_z^* &= T_z T_z^* \\ \iff (\bar{z})(z) &= (z)(\bar{z}) \\ \iff \bar{z}z &= z\bar{z}. \end{aligned}$$

(b) T_z is self-adjoint iff $z = \bar{z}$, that is, $z \in \mathbb{R}$.

(c) T_z is unitary iff $|z| = 1$:

$$\begin{aligned} T_z T_z^* &= I \\ \iff (z)(\bar{z}) &= 1 \\ \iff |z|^2 &= 1 \\ \iff |z| &= 1 \end{aligned}$$



Exercise 6.5.6. Let V be the inner product space of complex-valued continuous functions on $[0, 1]$ with the inner product

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt.$$

Let $h \in V$, and define $T: V \rightarrow V$ by $T(f) = hf$. Prove that T is a unitary operator iff $|h(t)| = 1$ for $0 \leq t \leq 1$.

Exercise 6.5.7. Prove that if T is a unitary operator on a finite-dimensional inner product space V , then T has a unitary *square root*; that is, there exists a unitary operator U such that $T = U^2$. Visit goo.gl/jADTaS for a solution.

Proof. Since U is unitary, $T^*T = T^{-1}T = I = TT^{-1} = TT^*$ so T is normal. We hence pick an orthonormal eigenbasis $\beta = \{v_i\}_1^n$ for T . Then, define $U: V \rightarrow V$ by

$$U(v_i) := \sqrt{A_{ii}} \cdot v_i,$$

where $A := [T]_\beta$ and some branch (e.g. the principal branch) of the complex square root is chosen. Now notice that, for the diagonal matrix $B := [U]_\beta$,

$$\begin{aligned} (BB^*)_{ii} &= B_{ii}\overline{B_{ii}} \\ &= |B_{ii}|^2 \\ &= |A_{ii}| \\ &= \sqrt{A_{ii}\overline{A_{ii}}} \\ &= \sqrt{(AA^*)_{ii}} \\ &= 1 \end{aligned}$$

so $BB^* = I$ — U is unitary. Moreover, $A = B^2$ is clear; $T = U^2$.



Exercise 6.5.20. Let V be a finite-dimensional inner product space. A linear operator U on V is called a *partial isometry* if there exists a subspace W of V such that $\|U(x)\| = \|x\|$ for all $x \in W$ and $U(x) = 0$ for all $x \in W^\perp$. Observe that W need *not* be U -invariant. Suppose that U is such an operator and $\{v_1, v_2, \dots, v_k\}$ is an orthonormal basis for W . Prove the following results.

- (a) $\langle U(x), U(y) \rangle = \langle x, y \rangle$ for all $x, y \in W$.
- (b) $\{U(v_1), U(v_2), \dots, U(v_k)\}$ is an orthonormal basis for $R(U)$.
- (c) There exists an orthonormal basis γ for V such that the first k columns of $[U]_\gamma$ form an orthonormal set and the remaining columns are zero.
- (d) Let $\{w_1, w_2, \dots, w_j\}$ be an orthonormal basis for $R(U)^\perp$ and $\beta = \{U(v_1), U(v_2), \dots, U(v_k), w_1, w_2, \dots, w_j\}$. Then β is an orthonormal basis for V .
- (e) Let T be the linear operator on V that satisfies $T(U(v_i)) = v_i$ ($1 \leq i \leq k$) and $T(w_i) = 0$ ($1 \leq i \leq j$). Then T is well-defined, and $T = U^*$.

Proof.

- (a) Since

$$\begin{aligned} \|U(v_i)\| &= \|v_i\| \\ \langle U(v_i), U(v_i) \rangle &= \langle v_i, v_i \rangle \\ \langle v_i, (U^*U - I)(x) \rangle &= 0, \end{aligned}$$

it follows from the lemma of theorem 6.18, that $U_W^*U_W = I$. So,

$$\langle Iv_i, v_j \rangle = \langle U(v_i), U(v_j) \rangle = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} = \langle v_i, v_j \rangle.$$

Clearly, this implies the more general result of

$$\langle U(x), U(y) \rangle = \langle x, y \rangle$$

for all $x, y \in V$ — by the linearity and conjugate linearity of inner products, in their first and second coordinates, respectively.

- (b) Trivial from (a).
- (c) Pick an extension $\gamma := \{v_i\}_1^n$ of $\alpha := \{v_i\}_1^k$ to a basis for V . Then the first k columns of $[U]_\gamma$ is just $U[\alpha]$, which is orthonormal. The remaining columns $U(v_i)$ ($k+1 \leq i \leq n$) are zero since $v_i \in W^\perp$ — that follows from α being a basis for W and $V = W \oplus W^\perp$, which implies $\gamma - \alpha = \{v_i\}_{k+1}^n$ is a basis for $W^\perp$.
- (d) Trivial from (d).
- (e) Well-definedness is trivial because β is a basis for V . We have four cases to evaluate; in each case we let i, j be arbitrary indices, for which the permissible range of values is easily inferred for each case.

- (i) Notice that $U^2(v_i)$ is a linear combination of $\{U(v_\ell)\}_1^k$ and must hence be orthogonal to w_j :

$$\langle U^2(v_i), w_j \rangle = 0 = \langle U(v_i), 0 \rangle = \langle U(v_i), T(w_j) \rangle.$$

- (ii) $\langle U(w_i), U(v_j) \rangle \stackrel{(a)}{=} \langle w_i, v_j \rangle = \langle w_i, TU(v_j) \rangle.$
- (iii) $\langle U^2(v_i), U(v_j) \rangle = \langle U(v_i), v_j \rangle = \langle U(v_i), TU(v_j) \rangle.$
- (iv) Since $U(w_i) \in R(U)$ and $w_j \in R(U)^\perp$,

$$\langle U(w_j), w_j \rangle = 0 = \langle w_i, 0 \rangle = \langle w_i, T(w_j) \rangle.$$

Hence, it is clear that $\langle U(x), y \rangle = \langle x, T(y) \rangle$ for all $x, y \in V$, by linearity.

In conclusion, $T = U^*$.



Exercise 6.5.28. Consider the expression $X^t A X$, where $X^t = (x, y, z)$ and A is as defined in exercise 2(e). Find a change of coordinates x', y', z' so that the preceding expression is of the form $\lambda_1(x')^2 + \lambda_2(y')^2 + \lambda_3(z')^2$

Proof. Throughout this exercise, we use Sage  to help with tedious computations and find careless mistakes; the corresponding code is displayed below this proof box. First, we diagonalise A . Its eigenvalues are 1 and 4:

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \det \begin{pmatrix} 2 - \lambda & 1 & 1 \\ 1 & 2 - \lambda & 1 \\ 1 & 1 & 2 - \lambda \end{pmatrix} &= 0 \\ -\lambda^3 + 6\lambda^2 - 9\lambda + 4 &= 0 \\ \boxed{\lambda = 4 \quad \text{or} \quad \lambda = 1} \end{aligned}$$

Now, we see that $\{p_1 = (1 \ 1 \ 1)^t\}$ and $\{p_2 = (1 \ -1 \ 0)^t, p_3 = (-1 \ 0 \ 1)^t\}$ are respective bases for the corresponding eigenspaces $E(4, T)$ and $E(1, T)$:

$$\begin{aligned}
 A - 4I &= \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} & A - I &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} & &\rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 1 & -2 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 3 & -3 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

So, $A = PDP^*$ for

$$P := (\hat{p}_1 \ \hat{p}_2 \ \hat{p}_3) \quad \text{and} \quad D := \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We recall that, where $X' := (x' \ y' \ z')^t := P^t(x \ y \ z)^t$, we have

$$X^t A X = X^t P D P^t X = (X')^t D X'.$$

Here, we notice that

$$X' = P^t X \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{pmatrix}^t \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} (1/\sqrt{3})(x + y + z) \\ (1/\sqrt{2})(-x + y) \\ (1/\sqrt{6})(x + y - 2z) \end{pmatrix}.$$

In other words, the change of coordinates is given by

$$\begin{aligned}
 x &\mapsto \frac{x + y + z}{\sqrt{3}} \\
 y &\mapsto \frac{-x + y}{\sqrt{2}} \\
 z &\mapsto \frac{x + y - 2z}{\sqrt{6}}.
 \end{aligned}$$



Jupyter.

```
[21]: var('t x y z')
```

```
[21]: (t, x, y, z)
```

```
[22]: A = Matrix( SR, [ [2, 1, 1], [1, 2, 1], [1, 1, 2] ] )
      B = A - t * identity_matrix(3)
      show( "A = " + latex(A) )
      show( "B =" + latex(B) )
```

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} -t+2 & 1 & 1 \\ 1 & -t+2 & 1 \\ 1 & 1 & -t+2 \end{pmatrix}$$

```
[23]: char_poly = det(B)
      char_poly_expanded = expand(char_poly)
      show(char_poly_expanded)
      if bool(char_poly == -t^3 + 6*t^2 - 9*t + 4):
          show( "✓:\\ " + LatexExpr(r"") + "\\text{char\\_poly} = "
                + latex(char_poly_expanded))
      else:
          show( "×:\\ " + LatexExpr(r"") + "\\text{char\\_poly} = "
                + latex(char_poly_expanded))
```

$$-t^3 + 6t^2 - 9t + 4$$

$$\checkmark: \text{char_poly} = -t^3 + 6t^2 - 9t + 4$$

```
[24]: show(solve(char_poly == 0, t))
```

$$[t = 4, t = 1]$$

```
[25]: expand(vector( SR, [ 1, 1, -2 ] ) ).norm(2)
```

```
[25]: sqrt(6)
```

```
[26]: A * vector( SR, [ 1, 1, -2 ] )
```

```
[26]: (1, 1, -2)
```

```
[81]: P = matrix( SR, [ [1/sqrt(3), -1/sqrt(2), 1/sqrt(6)], [1/
    ↪sqrt(3), 1/sqrt(2), 1/sqrt(6)], [1/sqrt(3), 0, -2/sqrt(6)]
] )
D = matrix( SR, [ [4, 0, 0], [0, 1, 0], [0, 0, 1] ] )
show( "P = " + latex(P) )
show( "D = " + latex(D) )
if A == P * D * P.conjugate_transpose():
    show( "✓:\\ " + LatexExpr("") + "A = P D P^{*}" )
else:
    show( "✗:\\ " + LatexExpr("") + "A = P D P^{*}" )
```

$$P = \begin{pmatrix} \frac{1}{3}\sqrt{3} & -\frac{1}{2}\sqrt{2} & \frac{1}{6}\sqrt{6} \\ \frac{1}{3}\sqrt{3} & \frac{1}{2}\sqrt{2} & \frac{1}{6}\sqrt{6} \\ \frac{1}{3}\sqrt{3} & 0 & -\frac{1}{3}\sqrt{6} \end{pmatrix}$$

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\checkmark: A = PDP^*$$

```
[80]: X = vector( SR, [x, y, z] )
X_prime = P.conjugate_transpose() * X
show( "X' = " + latex( vector( SR, [ factor(X_prime[i]) for i
in range(0, 3) ] ).column() ) )
quadratic_form = X * A * X
quadratic_form_diagonalised = X_prime * D * X_prime
show( "X^t A X = " + latex( expand( quadratic_form ) ) )
show( "(X')^t A X' = " + latex( expand(
quadratic_form_diagonalised ) ) )
if quadratic_form == quadratic_form_diagonalised:
    show( "✓:\\ " + LatexExpr("") + "X^t A X = (X')^t A X'"
" )
else:
    show( "✗:\\ " + LatexExpr("") + "X^t A X = (X')^t A X'"
" )
```

$$X' = \begin{pmatrix} \frac{1}{3}\sqrt{3}(x+y+z) \\ -\frac{1}{2}\sqrt{2}(x-y) \\ \frac{1}{6}\sqrt{6}(x+y-2z) \end{pmatrix}$$

$$X^t A X = 2x^2 + 2xy + 2y^2 + 2xz + 2yz + 2z^2$$

$$(X')^t A X' = 2x^2 + 2xy + 2y^2 + 2xz + 2yz + 2z^2$$

$$\checkmark: X^t A X = (X')^t A X'$$

Exercise 6.5.29. QR-Factorisation. Let $w_1, w_2, \dots, w_n$ be linearly independent vectors in $\mathbb{K}^n$, and let $v_1, v_2, \dots, v_n$ be the orthogonal vectors obtained from $w_1, w_2, \dots, w_n$ by the Gram-Schmidt process. Let $u_1, u_2, \dots, u_n$ be the orthonormal basis obtained by normalising the v_i 's.

- (a) Solving (1) in section 6.2 for w_k in terms of u_k , show that

$$w_k = \|v_k\|u_k + \sum_{j=1}^{k-1} \langle w_k, u_j \rangle u_j \quad (1 \leq k \leq n).$$

- (b) Let A and Q denote the $n \times n$ matrices in which the k th columns are w_k and u_k , respectively. Define $R \in M_{m \times n}(\mathbb{K})$ by

$$R_{jk} = \begin{cases} \|v_j\| & \text{if } j = k \\ \langle w_k, u_j \rangle & \text{if } j < k \\ 0 & \text{if } j > k, \end{cases}$$

Prove $A = QR$.

- (c) Compute Q and R as in (b) for the 3×3 matrix whose columns are the vectors $(1, 1, 0)$, $(2, 0, 1)$, and $(2, 2, 1)$.
- (d) Since Q is unitary [orthogonal] and R is upper triangular in (b), we have shown that every invertible matrix is the product of a unitary [orthogonal] matrix and an upper triangular matrix. Suppose that $A \in M_{m \times n}(\mathbb{K})$ is invertible and $A = Q_1 R_1 = Q_2 R_2$, where $Q_1, Q_2 \in M_{m \times n}(\mathbb{K})$ are unitary and $R_1, R_2 \in M_{m \times n}(\mathbb{K})$ are upper triangular. Prove that $D = R_2 R_1^{-1}$ is a unitary diagonal matrix.

Hint: Use exercise 6.5.17.

- (e) The QR factorisation described in (b) provides an orthogonalisation method for solving a linear system $Ax = b$ when A is invertible. Decompose A to QR , by the Gram-Schmidt process or other means, where Q is unitary and R is upper triangular. Then $QRx = b$, and hence $Rx = Q^*b$. This last system can be easily solved since R is upper triangular.

Use the orthonormalisation method and (c) to solve the system

$$\begin{aligned} x_1 + 2x_2 + 2x_3 &= 1 \\ x_1 &\quad + 2x_3 = 11 \\ x_2 + x_3 &= -1. \end{aligned}$$

Proof.

- (a) Trivially,

$$\begin{aligned} v_k &= w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j \\ \|v_k\|u_k &= w_k - \sum_{j=1}^{k-1} \langle w_k, u_j \rangle u_j \end{aligned} \tag{1}$$

since $u_k = v_k/\|v_k\|$ and $u_j = v_j/\|v_j\|$.

(b) Notice that the j th column of QR is

$$\sum_{i=1}^n R_{ij}u_i = \|v_j\|u_j + \sum_{i=1}^{j-1} \langle w_j, u_i \rangle u_i = w_j.$$

(c) We shall complete this with the help of [Sage](#) . First, notice that

$$\begin{aligned} v_1 &= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, & v_2 &= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \left\langle \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} \hat{1} \\ 1 \\ 0 \end{pmatrix} \right\rangle \begin{pmatrix} \hat{1} \\ 1 \\ 0 \end{pmatrix} \\ & & &= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}(2 + 0 + 0) \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ & & &= \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \end{aligned}$$

and

$$\begin{aligned} v_3 &= \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} - \left\langle \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} \hat{1} \\ -1 \\ 1 \end{pmatrix} \right\rangle \begin{pmatrix} \hat{1} \\ -1 \\ 1 \end{pmatrix} - \left\langle \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} \hat{1} \\ 1 \\ 0 \end{pmatrix} \right\rangle \begin{pmatrix} \hat{1} \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} - \frac{1}{3}(2 - 2 + 1) \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{1}{2}(2 + 2 + 0) \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}. \end{aligned}$$

Now,

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix},$$

so

$$Q = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{3} & \sqrt{2} & -1 \\ \sqrt{3} & -\sqrt{2} & -1 \\ 0 & \sqrt{2} & 2 \end{pmatrix}.$$

As for R , we note that

$$\begin{aligned} R_{12} &= \left\langle \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\rangle = \sqrt{2}, \\ R_{13} &= \left\langle \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\rangle = 2\sqrt{2}, \\ R_{23} &= \left\langle \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\rangle = \frac{1}{\sqrt{3}}. \end{aligned}$$

Therefore,

$$R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{3} & 1/\sqrt{3} \\ 0 & 0 & 2/\sqrt{6} \end{pmatrix}.$$

In sum, the requested QR decomposition is

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{3} & \sqrt{2} & -1 \\ \sqrt{3} & -\sqrt{2} & -1 \\ 0 & \sqrt{2} & 2 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} & \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{3} & 1/\sqrt{3} \\ 0 & 0 & 2/\sqrt{6} \end{pmatrix}.$$

- (d) Since $Q_2^* Q_1 = R_2 R_1^{-1}$ is a product of unitary [orthogonal] matrices, it must be a unitary [orthogonal] matrix itself. Now we show that R_1^{-1} is upper triangular; suppose, for contradiction, that the diagonal of R_1 has a zero entry. Then, pick the maximal $1 \leq k \leq n$ such that $(R_1)_{kk} = 0$. Then, for $R_1 = (r_1 \ r_2 \ \dots \ r_n)$, we see that

$$R_1 \rightarrow S \quad (r_i \rightarrow r_i - (R_{ij}/R_{jj}) \cdot r_j \text{ for all } k+1 \leq j \leq n \text{ and } 1 \leq i \leq j-1)$$

yields $\text{rank}(R_1) = \text{rank}(S) < n$, since the k th row of S is zero. A contradiction; the diagonal of R_1 must consist of only nonzero entries. As such, R_1 can be reduced to I through the following elementary row operations:

$$\begin{aligned} R_1 &\rightarrow R'_1 \quad (r_i \rightarrow r_i - (R_{ij}/R_{jj}) \cdot r_j \text{ for each } 1 \leq j \leq n \text{ and } 1 \leq i \leq j-1) \\ &\rightarrow I \quad (r'_i \rightarrow (1/R_{ii}) \cdot r'_i \text{ for each } 1 \leq i \leq n), \end{aligned}$$

where $R'_1 = (r'_1 \ r'_2 \ \dots \ r'_n)$. The first and second set of elementary row operations correspond to left multiplication on R_1 and R'_1 , respectively, by

the following products of elementary row matrices Π_1 and Π_2 :

$$\Pi_1 := \prod_{j=n}^1 \prod_{i=1}^{j-1} (I - (R_{ij}/R_{jj}) \cdot E^{ij}),$$

$$\Pi_2 := \begin{pmatrix} 1/R_{11} & 0 & \cdots & 0 \\ 0 & 1/R_{22} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1/R_{nn} \end{pmatrix},$$

where $\prod_{j=n}^1 M_j := (\prod_{j=n-1}^1 M_j)M_n$. Clearly, Π_1 , and hence $R^{-1} = \Pi_2\Pi_1$, being products of upper triangular matrices, must themselves be upper triangular. Now, $R_2R_1^{-1}$ is both unitary [orthogonal] and upper triangular, so it must be a unitary diagonal matrix.

(e) In (c), we saw that the QR decomposition of A is given by

$$Q = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{3} & \sqrt{2} & -1 \\ \sqrt{3} & -\sqrt{2} & -1 \\ 0 & \sqrt{2} & 2 \end{pmatrix} \quad \text{and} \quad R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{3} & 1/\sqrt{3} \\ 0 & 0 & 2/\sqrt{6} \end{pmatrix}.$$

This system $Ax = b \iff Rx = Q^*b$ has the augmented matrix

$$(R|Q^*b) = \left(\begin{array}{ccc|c} \sqrt{2} & \sqrt{2} & 2\sqrt{2} & 6\sqrt{2} \\ 0 & \sqrt{3} & 1/\sqrt{3} & -11\sqrt{3}/3 \\ 0 & 0 & 2/\sqrt{6} & 4\sqrt{6}/3 \end{array} \right).$$

So the solution to this system is $x = (4 \quad -5 \quad 3)$:

$$x_3 = \frac{4\sqrt{6}/3}{2/\sqrt{6}} = 4,$$

$$x_2 = \frac{1}{\sqrt{3}} \left(-\frac{11\sqrt{3}}{3} - \frac{1}{\sqrt{3}} \cdot x_3 \right) = -5,$$

$$x_1 = \frac{1}{\sqrt{2}} (6\sqrt{2} - \sqrt{2} \cdot x_2 - 2\sqrt{2} \cdot x_3) = 3.$$



Jupyter.

Trying to find my careless mistake

```
[1]: var('t')
```

```
[1]: t
```

```
[2]: A = matrix( [[1, 2, 2]
                  , [1, 0, 2]
                  , [0, 1, 1]] )
show( "A = " + latex(A) )
```

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

```
[3]: # Sage's .gram_schmidt() orthogonalises the rows of A, rather
      # than its columns, hence the need for transpose(A).
A_orthogonalised = transpose(A).gram_schmidt()[0].transpose()
Q = matrix( [ v.normalized() for v in A_orthogonalised.
             ↪ columns() ] ).transpose()
show( "A_\\text{orthogonalised} = " +
      latex(A_orthogonalised))
show( "Q = " + latex(Q) )
q_2 = Q.columns()[2]
my_q_2 = 1/sqrt(6) * vector( [-1, 1, 2] )
if q_2 == my_q_2:
    show( "✓:\\ " + latex(q_2.column()) + " =
    \\frac{1}{\\sqrt{6}} " + latex(vector( [-1, 1, 2] ).column())
    )
else:
    show( "✗:\\ " + latex(q_2.column()) + " =
    \\frac{1}{\\sqrt{6}} " + latex(vector( [-1, 1, 2] ).column())
    )
```

$$A_{\text{orthogonalised}} = \begin{pmatrix} 1 & 1 & -\frac{1}{3} \\ 1 & -1 & \frac{1}{3} \\ 0 & 1 & \frac{2}{3} \end{pmatrix}$$

$$Q = \begin{pmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{3}\sqrt{3} & -\frac{1}{2}\sqrt{\frac{2}{3}} \\ \frac{1}{2}\sqrt{2} & -\frac{1}{3}\sqrt{3} & \frac{1}{2}\sqrt{\frac{2}{3}} \\ 0 & \frac{1}{3}\sqrt{3} & \sqrt{\frac{2}{3}} \end{pmatrix}$$

$$\checkmark : \begin{pmatrix} -\frac{1}{2}\sqrt{\frac{2}{3}} \\ \frac{1}{2}\sqrt{\frac{2}{3}} \\ \sqrt{\frac{2}{3}} \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Well, we found it: I computed all the u_i 's correctly, but I accidentally wrote $u_2 = 1/\sqrt{3} \begin{pmatrix} 1 & -1 & 0 \end{pmatrix}$, instead of $u_2 = 1/\sqrt{3} \begin{pmatrix} 1 & -1 & 1 \end{pmatrix}$.

Verifying whether the QR decomposition I computed is correct

First, we check that $QR = A$

```
[4]: my_Q = 1/sqrt(6) * matrix( SR, [[sqrt(3), sqrt(2), -1]
                                     , [sqrt(3), -sqrt(2), 1]
                                     , [ 0, sqrt(2), 2]] )
my_R = matrix( SR, [[sqrt(2), sqrt(2), 2*sqrt(2)]
                    , [ 0, sqrt(3), 1/sqrt(3)]
                    , [ 0, 0, 2/sqrt(6)]] )

if A == my_Q * my_R:
    show("✓: A = my_Q * my_R")
else:
    show("✗: A = my_Q * my_R")
```

✓: A = my_Q * my_R

Now we check my QR decomposition by hand, against Sage's.

```
[5]: def R_entries(i, j):
    if i == j:
        return norm(A_orthogonalised.column(i))
    elif i < j:
        return A.column(j).dot_product(Q.column(i))
    elif i > j:
        return 0
    else:
        raise Exception("R_entries(i, j): greivous error:
somehow i != j, i !< j, and i !> j.")

R = matrix( SR, [ [ R_entries(i, j) for j in range(0, 3) ]
for i in range(0, 3) ] )

show( "Q = " + latex(Q) + "\\quad\\text{and}\\quad" + "R = "
+ latex(R) )

show( "\\texttt{my\\_Q} = " + latex(my_Q) +
"\\quad\\text{and}\\quad" + "\\texttt{my\\_R} = " +
latex(my_R) )

if Q == my_Q:
    show("✓: my_Q = Q")
else:
    show("✗: my_Q = Q")
if R == my_R:
    show("✓: my_R = R")
else:
    show("✗: my_R = R")
```

$$Q = \begin{pmatrix} \frac{1}{2}\sqrt{2} & \frac{1}{3}\sqrt{3} & -\frac{1}{2}\sqrt{\frac{2}{3}} \\ \frac{1}{2}\sqrt{2} & -\frac{1}{3}\sqrt{3} & \frac{1}{2}\sqrt{\frac{2}{3}} \\ 0 & \frac{1}{3}\sqrt{3} & \sqrt{\frac{2}{3}} \end{pmatrix} \quad \text{and} \quad R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{3} & \frac{1}{3}\sqrt{3} \\ 0 & 0 & \sqrt{\frac{2}{3}} \end{pmatrix}$$

$$\text{my_Q} = \begin{pmatrix} \frac{1}{6}\sqrt{6}\sqrt{3} & \frac{1}{6}\sqrt{6}\sqrt{2} & -\frac{1}{6}\sqrt{6} \\ \frac{1}{6}\sqrt{6}\sqrt{3} & -\frac{1}{6}\sqrt{6}\sqrt{2} & \frac{1}{6}\sqrt{6} \\ 0 & \frac{1}{6}\sqrt{6}\sqrt{2} & \frac{1}{3}\sqrt{6} \end{pmatrix} \quad \text{and} \quad \text{my_R} =$$

$$\begin{pmatrix} \sqrt{2} & \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{3} & \frac{1}{3}\sqrt{3} \\ 0 & 0 & \frac{1}{3}\sqrt{6} \end{pmatrix}$$

✓: my_Q = Q

✓: my_R = R

$$R = \begin{pmatrix} \|v_1\| & \langle w_2, u_1 \rangle & \langle w_3, u_1 \rangle & \cdots & \langle w_n, u_1 \rangle \\ 0 & \|v_2\| & \langle w_3, u_2 \rangle & \cdots & \langle w_n, u_2 \rangle \\ 0 & 0 & \|v_3\| & \cdots & \langle w_n, u_3 \rangle \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \|v_n\| \end{pmatrix}$$

Exercise 6.5.31. Let H_u be a Householder operator on a finite-dimensional inner product space V . Prove the following results.

- (a) H_u is linear.
- (b) $H_u(x) = x$ iff x is orthogonal to u .
- (c) $H_u(x) = -x$.
- (d) $H_u^* = H_u$ and $H_u^2 = I$, and hence H_u is a unitary [orthogonal] operator on V .

(Note: If V is a real inner product space, then in the language of section 6.11, H_u is a reflection.)

Proof.

- (a) Trivial by the first-coordinate linearity of inner products.
- (b) Trivial; we see that

$$\begin{aligned} H_u(x) = x &\iff x - 2\langle x, u \rangle u = x \\ &\iff \langle x, u \rangle u = 0 \\ &\iff \langle x, u \rangle = 0 \\ &\iff x \text{ is orthogonal to } u, \end{aligned}$$

since u , being a unit vector, is nonzero.

- (c) Trivial.
- (d) Since

$$\begin{aligned} \langle H_u(x), y \rangle &= \langle x - 2\langle x, u \rangle u, y \rangle \\ &= \langle x, y \rangle - 2\langle x, u \rangle \langle u, y \rangle \\ &= \overline{\langle y, x \rangle} - 2\overline{\langle y, u \rangle} \langle u, x \rangle \\ &= \overline{\langle y - 2\langle y, u \rangle u, x \rangle} \\ &= \overline{\langle H_u(y), x \rangle} \\ &= \langle x, H_u(y) \rangle, \end{aligned}$$

for any $x, y \in V$, it follows that H_u is self-adjoint. Further,

$$\begin{aligned}
 H_u^2(x) &= H_u(x) - 2\langle H_u(x), u \rangle u \\
 &= x - 2\langle x, u \rangle u - 2\langle x - 2\langle x, u \rangle u, u \rangle u \\
 &= x - 2\langle x, u \rangle u - 2\langle x, u \rangle u + 4\langle x, u \rangle \langle u, u \rangle u \\
 &= x - 4\langle x, u \rangle u + 5\langle x, u \rangle u \\
 &= x,
 \end{aligned}$$

for all $x \in V$. Thus, H_u is also self-inverse.



Exercise 6.5.32. Let V be a finite-dimensional inner product space over $\mathbb{K}$. Let x and y be linearly independent vectors in V such that $\|x\| = \|y\|$.

- (a) If $\mathbb{K} = \mathbb{C}$, prove that there exists a unit vector u in V and a complex number θ with $|\theta| = 1$ such that $H_u(x) = \theta u$.

Hint: Choose θ so that $\langle x, \theta y \rangle$ is real, and set $u = \frac{x - \theta y}{\|x - \theta y\|}$.

- (b) If $\mathbb{K} = \mathbb{R}$, prove that there exists a unit vector u in V such that $H_u(x) = y$.

Proof (without the hint).

- (a) I managed to derive the following, but was eventually stuck. Let $a, b \in \mathbb{C}$ and $u := \frac{x+y}{\|x+y\|}$:

$$\begin{aligned}
 H_u(x) &= x - \frac{2}{\|x+y\|^2} \cdot \langle x, x+y \rangle (x+y) \\
 &= x - \frac{2(\langle x, x \rangle + \langle x, y \rangle)}{\|x\|^2 + \|y\|^2 + 2\Re\langle x, y \rangle} (x+y) \\
 &= x - \frac{\langle x, x \rangle + \langle x, y \rangle}{\langle x, x \rangle + \Re\langle x, y \rangle} (x+y) \\
 &= x - \left(1 + \frac{\Im\langle x, y \rangle}{\langle x, x \rangle + \Re\langle x, y \rangle} \right) (x+y) \\
 &= \left(-\frac{\Im\langle x, y \rangle}{\langle x, x \rangle + \Re\langle x, y \rangle} \right) x + \left(-1 + \frac{\Im\langle x, y \rangle}{\langle x, x \rangle + \Re\langle x, y \rangle} \right) y
 \end{aligned}$$

Currently, we see that $H_u(x)$ has a possibly nonzero x -component, which we want to eliminate. So, let's try $u := \frac{ax+by}{\|ax+by\|}$:

$$\begin{aligned}
 H_u(x) &= x - \frac{2}{\|ax+by\|^2} \cdot \langle x, ax+by \rangle (ax+by) \\
 &= x - \frac{2(\bar{a}\langle x, x \rangle + \bar{b}\langle x, y \rangle)}{(|a|^2 + |b|^2)\|x\|^2 + 2\Re(\bar{a}b\langle x, y \rangle)} (ax+by) \\
 &=
 \end{aligned}$$

(b) We see that for $u := \frac{x-y}{\|x-y\|}$,

$$\begin{aligned}
 H_u(x) &= x - \frac{2}{\|x-y\|^2} \cdot \langle x, x-y \rangle (x-y) \\
 &= x - \frac{2(\langle x, x \rangle - \langle x, y \rangle)}{\|x\|^2 + \|y\|^2 - 2\Re\langle x, y \rangle} (x-y) \\
 &= x - \frac{2(\langle x, x \rangle - \langle x, y \rangle)}{2(\langle x, x \rangle - \Re\langle x, y \rangle)} (x-y) \\
 &= x - 1(x-y) \\
 &= y
 \end{aligned}$$

since $\mathbb{K} = \mathbb{R}$.



Proof (using the hint).

(a) Let $e^{i\alpha} = \frac{\langle x, y \rangle}{\langle y, x \rangle}$; pick $\theta := \exp(i\alpha/2)$ and $u := \frac{x-\theta y}{\|x-\theta y\|}$. We observe that

$$\begin{aligned}
 H_u(x) &= x - \frac{2}{\|x-\theta y\|^2} \cdot \langle x, x-\theta y \rangle (x-\theta y) \\
 &= x - \frac{2(\langle x, x \rangle - \langle x, \theta y \rangle)}{\|x\|^2 + \|\theta y\|^2 + 2\Re\langle x, \theta y \rangle}.
 \end{aligned}$$

Astute readers notice that $\|\theta y\|^2 = |\theta|^2 \|y\|^2 = \|x\|^2$ and $\langle x, \theta y \rangle \in \mathbb{R}$:

$$\begin{aligned}
 \theta/\bar{\theta} &= \frac{\langle x, y \rangle}{\langle y, x \rangle} \\
 \bar{\theta}\langle x, y \rangle &= \theta\langle y, x \rangle \\
 \langle x, \theta y \rangle &= \langle \theta y, x \rangle.
 \end{aligned}$$

Consequently,

$$H_u(x) = x - 1(x - \theta y) = \theta y.$$

(b) Now, it follows trivially from the above proof of (a), that $\theta = 1$ so $H_u(x) = \theta y = y$ for $u := \frac{x-\theta y}{\|x-\theta y\|}$.



6.6 (Self) Angles and isometries

The inner product is a generalisation of the dot product, and indeed has the properties we would expect it to hence possess, as exercise 15 demonstrates. Thus, we might attempt to define the angle between two vectors:

Definition. Let V be a real inner product space and $x, y \in V$. We define *the angle*

between x and y as

$$\theta(x, y) := \arccos \frac{\langle x, y \rangle}{\|x\| \|y\|}.$$

Naturally, we question our sanity¹:

Question. Let V and W be real inner product spaces. If $f: V \rightarrow W$ is an isometry, then must angles be preserved? That is, given $\|x - y\| = \|f(x) - f(y)\|$ for all $x, y \in V$, must $\theta(x, y) = \theta(f(x), f(y))$ for all $x, y \in V$?

Isometries preserve angles: given three points transformed under an isometry, the distances between them are invariant. i.e. the triangle before and after the isometry are congruent. As such, we expect that isometries preserve angles — the answer to the above is positive — if our definition of angle is reasonable.

Proof.



Question. Would extending the above definition to

$$\theta(x, y) := \arccos \frac{|\langle x, y \rangle|}{\|x\| \|y\|}$$

for complex inner product spaces be unreasonable?

Preliminary note: $\langle y - | \langle y, \hat{x} \rangle \hat{x}, x \rangle = \langle z, x \rangle - | \langle z, x \rangle |$.

Lemma. Let V be an inner product space. If $\|x + y\| = \|x\| + \|y\|$, then $x = cy$ for some $c \in \mathbb{C}$.

Proof. Let $y_{\parallel} = \langle y, \hat{x} \rangle \hat{x}$ and $y_{\perp} = y - y_{\parallel}$. So, $\langle x, y_{\perp} \rangle = 0$. Hence,

$$\|x + y\|^2 = \|x + y_{\parallel}\|^2 + \|y_{\perp}\|^2 = \|x\|^2 + 2\|x\|\|y_{\parallel}\| + \|y_{\parallel}\|^2 + \|y_{\perp}\|^2$$

and

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y_{\parallel}\|^2 + 2\|y_{\parallel}\|\|y_{\perp}\| + \|y_{\perp}\|^2.$$

When $\|x + y\| = \|x\| + \|y\|$,

$$\|x\|\|y_{\parallel}\| = \|x\|\|y\| + \|y_{\parallel}\|\|y_{\perp}\| \geq \|y_{\parallel}\|(\|x\| + \|y_{\perp}\|).$$

Therefore, $\|y_{\perp}\| = 0$. × Thanks to afqt who pointed out my implicit assumption that x is a real multiple of $y_{\parallel}$, and that the implication can be strengthened to $c \geq 0$.



Lemma. Let V be an inner product space and $x, y \in V$. If $\|x + y\| = \|x\| + \|y\|$, then $x = cy$ for some $c \geq 0$.

¹I phrased it this way for comedic purposes.

Proof. Let $y_{\parallel} = \langle y, \hat{x} \rangle \hat{x}$ and $y_{\perp} = y - y_{\parallel}$. So, $\langle x, y_{\perp} \rangle = 0$. Hence,

$$\|x + y\|^2 = \|x + y_{\parallel}\|^2 + \|y_{\perp}\|^2 = \|x\|^2 + 2\Re\langle x, y_{\parallel} \rangle + \|y_{\parallel}\|^2 + \|y_{\perp}\|^2$$

and

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = \|x\|^2 + 2\|y_{\parallel}\|\|x\| + \|y_{\parallel}\|^2 + 2\|y_{\parallel}\|\|y_{\perp}\| + \|y_{\perp}\|^2.$$

When $\|x + y\| = \|x\| + \|y\|$,

$$\Re\langle x, y_{\parallel} \rangle = \|x\|\|y\| + \|y_{\parallel}\|\|y_{\perp}\| \geq \|y_{\parallel}\|(\|x\| + \|y_{\perp}\|) \geq \Re\langle x, y_{\parallel} \rangle + \|y_{\parallel}\|\|y_{\perp}\|.$$

Therefore, $\|y_{\perp}\| = 0$ and $x = cy$, for some $c \in \mathbb{K}$. Since $|c + 1| = |c| + 1$, we have $c \geq 0$. 

Note. Afqt also mentioned this is just the equality case of Cauchy-Schwartz. But I'm still working on my proof of Cauchy-Schwartz; I don't have the required knowledge of how the proof goes, in order to do it that way. So, I used the above method, which applied Cauchy-Schwartz at the end.

Note. Let V be an inner product space and $x, y \in V$. While $\langle y, \hat{x} \rangle \hat{x}$ is always orthogonal to x , the vector $\langle \hat{x}, y \rangle \hat{x}$ is orthogonal to x iff $\langle x, y \rangle$ is real.

Question. If V is a normed space and $\|x + y\| = \|x\| + \|y\|$, must $x = cy$ for some $c \in \mathbb{C}$?

Question. Is an isometry T between two inner product spaces V and W linear?

Proof. Let $x, y \in V$ and $a \in \mathbb{K}$; notice that $\|T(ax) - T(x)\| = |a - 1|\|T(x)\|$. Since $\|aT(x)\| = \|T(ax)\|$, wlog $\|T(ax) - T(x)\| = \|T(ax)\| - \|T(x)\|$. By the above lemma, $T(ax) = cT(x)$ for some $c \geq 1$. In fact, $|a| = c$. 

If our definition of angle is sound, then we expect basic trigonometry to hold.

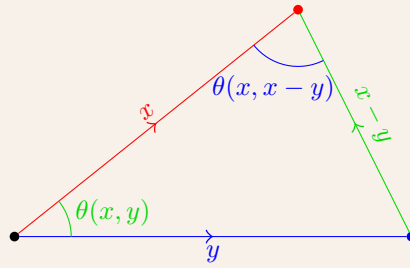
Claim. Let V be an inner product space. Then, for all $x, y \in V$, the following are true.

(a) The sine rule:

$$\frac{\sin \theta(x, y)}{\|x - y\|} = \frac{\sin \theta(x, x - y)}{\|y\|}.$$

(b) The cosine rule:

$$\cos \theta(x, y) = \frac{\|x\|^2 + \|y\|^2 - \|x - y\|^2}{2\|x\|\|y\|}.$$



Exercise. Find a metric on a vector space that does not induce a norm.

Proof. Consider the discrete metric d on $\mathbb{R}$. It clearly cannot be induced by any norm, since $d(1, 0) = d(2, 0)$. 

6.7 (Self) Linear transformations

Observation 6.24. Let V be a normed vector and $T: V \rightarrow V$ be linear. If $\text{nullity}(T) = 0$, we have the induced T -norm $\|v\|_T := \|T(v)\|$.

Question 6.7.1. Let V be a normed vector space. Does there exist a linear $T: V \rightarrow V$, such that $\|\cdot, \cdot\|_T$ is induced by some inner product on V ?

6.8 (Self) Continuity

In school, my teacher once handwaved why $T \lim T^n = \lim T^n$ (pointwise convergence). I asked on [math discord](#)  and found that any linear operator on a finite-dimensional normed space is continuous. So, let's try to prove it!

Observation 6.25. Let $\{x_n\}$ and $\{y_n\}$ be sequences in a normed space X with limits x and y , respectively; $c \in \mathbb{K}$. Then, $cx_n + y_n \rightarrow ax + y$.

Claim 6.26. Each inner product space X has an orthonormal basis.

Proof. Let $\mathcal{C}$ be a chain of orthonormal subsets of X and $x_i \in \beta_i \in \mathcal{C}$, where $\beta_1 \subseteq \dots \subseteq \beta_n$ wlog. As such, $x_i \in \beta_n$. So, $\mathcal{C}$ is orthonormal. There is hence a maximal orthonormal subset β of X , by Choice. Suppose, for contradiction, that there exists $x \in X - \text{span}(\beta)$. Then, $\hat{x} \cup \beta$ is an

Oh yep this doesn't work. 

Lemma 6.27. Let $\beta := \{x_1, x_2, \dots, x_m\}$ be a basis for the normed space X , and $t_n = \sum_{i=1}^m c_{in} x_i$. Then, $t_n \rightarrow x_1$ iff $c_{1n} \rightarrow 1$ and $c_{in} \rightarrow 0$ for each $i \neq 1$.

The proof is trivial if X is a finite-dimensional inner product space.

Claim 6.28. Let X and Y be finite-dimensional normed spaces. Any linear transformation $T: X \rightarrow Y$ is continuous.

Miscellaneous

Exercise (Leibniz's formula for determinants.). Let A be an $n \times n$ matrix. Prove that

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n A_{i\sigma(i)}.$$

An exercise by Ttera ([link to relevant discord messages](#)) ☞ .

Exercise. Suppose V is a real vector space of finite dimension which admits a linear operator $T: V \rightarrow V$ such that $T^2v = -v$ for all $v \in V$. Show that V is of even dimension.

A question asked in math discord ([link to relevant discord message](#)) ☞ .

Exercise. Let A and B be $n \times n$ real matrices. Also define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) = \det(A + tB)$. Compute $f'(0)$, and more generally, $f'(t)$ for any $t \in \mathbb{R}$.

An exercise by Daminark ([link to relevant discord messages](#)) ☞ .

Exercise.

- (a) Let A and B be commuting diagonalizable $n \times n$ matrices over $\mathbb{F}$. Then, they are simultaneously diagonalizable.
- (b) Commuting matrices $A, B \in M_{n \times n}(\mathbb{C})$ are simultaneously triangularizable. That is, there exists a basis β for which $[L_A] = [L_B]$ is upper triangular. (Think about what the $n = 1$ case means.)

Two exercises Neam sent, probably from LADR.

Exercise. Suppose $V_1, \dots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \dots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \dots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces, where $\times$ indicates the external direct sum.

Find a canonical isomorphism.

Proof. For each $v \in V_i$, let $v_i := (\overbrace{0, 0, \dots, 0}^{i-1}, v, 0, 0, \dots, 0)$. Now let the linear transformation.

$$T: \mathcal{L}(V_1 \times \dots \times V_m, W) \rightarrow \mathcal{L}(V_1, W) \times \dots \times \mathcal{L}(V_m, W)$$

be defined by $T(f) := (f_1, f_2, \dots, f_m)$, where $f_i(v) := f(v_i)$.

Suppose $T(f) = T(f')$, i.e. $f(v_i) = f'(v_i)$ for every $1 \leq i \leq m$ and $v \in V_i$. By linearity, $f = f'$. Hence injectivity holds.

Pick any $(f_1, f_2, \dots, f_m)$. We define $f(v_i) := f_i(v)$. Then, $T(f) = (f_1, f_2, \dots, f_m)$, as desired. Therefore, T is surjective.

Lastly, we see that

$$(af + f')_i(v) := (af + f')(v_i) = af(v_i) + f'(v_i) = af_i(v) + f'_i(v),$$

for any scalar $a \in \mathbb{F}$. So,

$$T(af + f') = (af_1 + f'_1, af_2 + f'_2, \dots, af_m + f'_m).$$

This implies T is linear. Consequently, T is an isomorphism. 

Remark. The generalised version for infinitely many vector spaces V_α , for $0 \leq \alpha \leq \kappa$, is as follows.

$$\mathcal{L}\left(\bigoplus_{\alpha} V_{\alpha}, W\right) \cong \bigotimes_{\alpha} \mathcal{L}(V_{\alpha}, W).$$

This should essentially follow from the above proof, except that each m -tuple is now replaced with a κ -sequence. i.e. $h: \kappa \rightarrow \bigcup_{\alpha} V_{\alpha}$, where $h(\alpha) \in V_{\alpha}$.

Note. For our isomorphism T to be injective, the (external) direct sum is necessary and can't be replaced with a direct product. Otherwise we cannot conclude $f = f'$ from $f(v_i) = f'(v_i)$. Similarly, since $T(f) := (f_1, f_2, \dots, f_m)$, its codomain must be the direct product and cannot be replaced with a direct sum.

Exercise. For each $0 \leq \alpha \leq \kappa$, let V_{α} be a vector space. Prove that the direct product of all V_{α} has strictly greater dimension than that of the direct sum. That is,

$$\dim\left(\bigotimes_{\alpha} V_{\alpha}\right) > \dim\left(\bigoplus_{\alpha} V_{\alpha}\right).$$

Proof. (Note. The letters κ , μ , and λ are taken to be cardinal numbers.)

Choose a basis

$$\beta_{\alpha} := \{v_1^{\alpha}, v_2^{\alpha}, \dots, v_{\mu_{\alpha}}^{\alpha}\}$$

for each V_{α} . It is clear that $\gamma^{\oplus} := \bigcup_{\alpha} \beta_{\alpha}$ is a basis for the direct sum $\bigoplus_{\alpha} V_{\alpha}$. Now

let $\mathcal{F}$ be the set of all $g_\lambda: \lambda \rightarrow \bigcup_\alpha^\kappa \beta_\alpha$, such that there is an injection $f_\lambda: \lambda \rightarrow \kappa$, for which $g_\lambda(\alpha) \in \beta_{f_\lambda(\alpha)}$ for all $1 \leq \alpha \leq \lambda$. Define $\delta^\otimes := \{\sum_\alpha^\lambda g_\lambda(\alpha) \mid g \in \mathcal{F}\}$.

$$c_i$$



Exercise. Suppose $W_1, \dots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \dots \times W_m)$ and $\mathcal{L}(V, W_1) \times \dots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

Find a canonical isomorphism.

Another exercise from LADR.

Exercise. Let V be a finite dimensional vector space and $T \in \mathcal{L}(V)$. Prove that T is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

Proof. If T is a scalar multiple of the identity, then the result is straightforward. Conversely, suppose $ST = TS$ for all linear operators S on V . We choose a basis $\beta := \{v_1, v_2, \dots, v_n\}$ for V . Let the linear operator S_i on V be defined by $S_i(v_i) := 0$, and $S_i(v_j) := v_j$ for $i \neq j$. It follows that

$$S_i T(v_i) = T S_i(v_i) = 0.$$

So, for each i we have $T(v_i) = c_i v_i$ for some scalar $c_i \in \mathbb{F}$. Now, define another linear operator U on V , by $U(v_i) = v_{i+1}$, where $v_{n+1} := 0$. For each i , it hence holds that $UT(v_i) = TU(v_i)$, i.e.

$$c_i v_{i+1} = c_{i+1} v_{i+1}.$$

Therefore, $c_1 = c_2 = \dots = c_n$ and T must be a scalar multiple of the identity. 

Question. Does the Cantor-Schröder-Bernstein theorem hold for linear functions? That is, if there exists the injective linear transformations $T: V \rightarrow W$ and $U: W \rightarrow V$, then there is a bijective linear transformation $S: V \rightarrow W$?

Proof. Suppose that such linear transformations T and U exist. Then, $\dim(V) = \dim(W)$ is clear. As such, $V \cong W$ implies such a bijective S exists. 

The following exercise is from Timothy Gowers' video, titled '[1. A strange determinant](#)'. The second part on trace is taken from [Wikipedia](#).

Exercise (Pascal's determinant.). The $n \times n$ (symmetric) Pascal matrix S_n is

$$\begin{pmatrix} 1 & 1 & 1 & \cdots \\ 1 & 2 & 3 & \cdots \\ 1 & 3 & 6 & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}.$$

Prove that the determinant of S_n is always 1 and

$$\text{tr}(S_n) = \sum_{k=0}^{n-1} \frac{(2k)!}{(k!)^2}.$$